

What efficacy and safety profiles can we expect

Mathematical and Computational Biology in Drug Discovery (MCBDD) Module IV

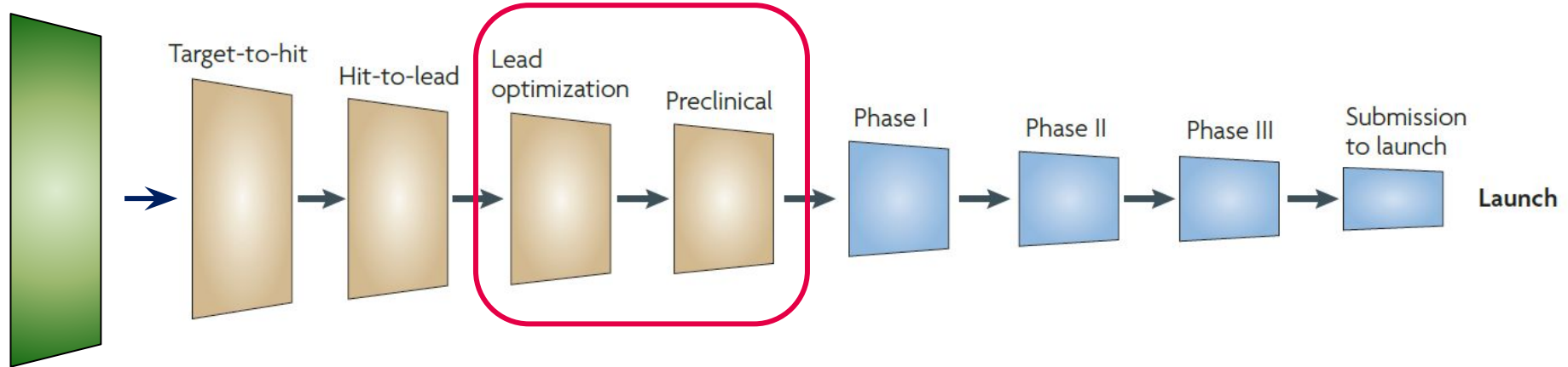
*PD Dr. Jitao David Zhang
April - May 2026*

Goals of the module

- ADME/PK prediction and PK/PD modelling
- Mechanistic models, machine learning, and causal models
- Protein turnover and its impact in drug discovery

Where are we now

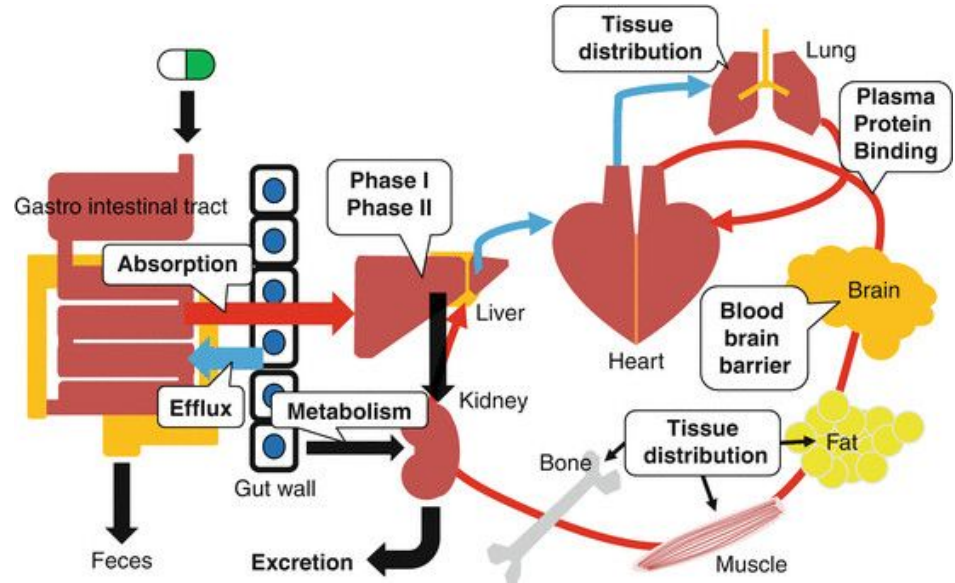
Target identification & assessment



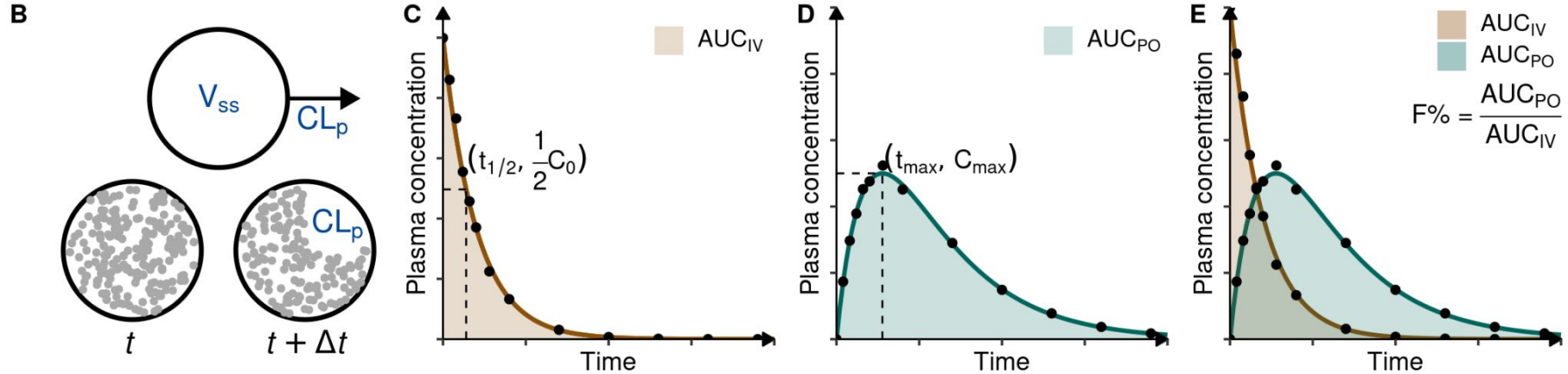
Goal: we want to select **one compound** from a few ($\sim 10^2$ - 10^0) for entry in human.

Key factors to consider in selecting compounds

- Pharmacokinetics (PK)
 - Absorption
 - Distribution
 - Metabolism
 - Excretion
- Potency, pharmacodynamics (PD), and efficacy
- Toxicology
- Biomarkers



Key PK parameters: V_{ss} , CL_p , $t_{1/2}$, t_{max} , C_{max} , and F



V_{ss}	Volume of distribution at steady state.	C_{max}	Maximum plasma concentration.
CL_p	Plasma clearance.	t_{max}	Time point in which the C_{max} is measured.
$t_{1/2}$	Half-life, time for a substance to reach the half concentration of the initial value (C_0).	$F\%$ (or F)	Bioavailability, the percentage of the administered compound reaching systemic circulation.

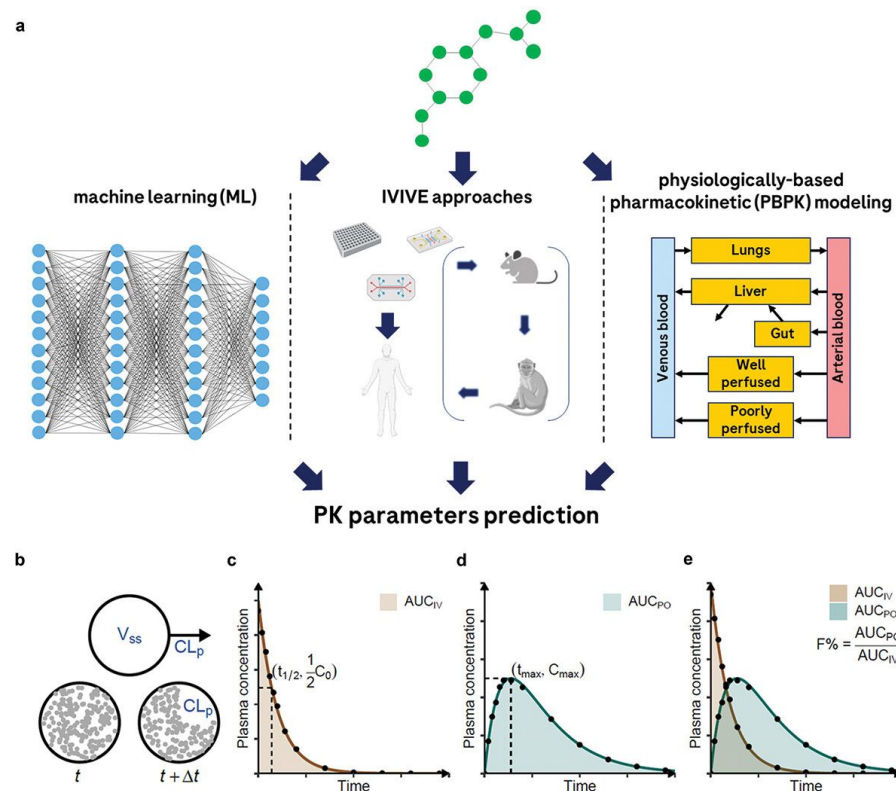
Three archetypal approaches to PK property prediction

Machine learning (ML): predict PK from molecule structure, using models trained from paired data of structures and PK parameters.

***In vitro-in vivo* extrapolation (IVIVE):** extrapolate *in vitro* measurements to *in vivo* predictions, e.g. predicting PK parameters from an *in vitro* measurement of intrinsic clearance (e.g. liver microsome assay).

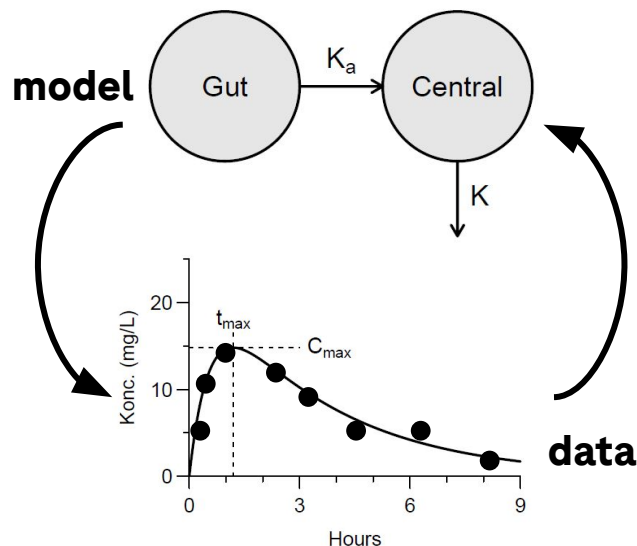
Physiology-based pharmacokinetic modeling (PBPK): predicting PK parameters combining molecule properties and mathematical models of human physiology.

Bassani, Davide, Neil John Parrott, Nenad Manevski, and Jitao David Zhang. 2024. “Another String to Your Bow: Machine Learning Prediction of the Pharmacokinetic Properties of Small Molecules.” *Expert Opinion on Drug Discovery* 19 (6): 683–98. For basics in PK/PD modeling, see [course material of AMIDD](#).



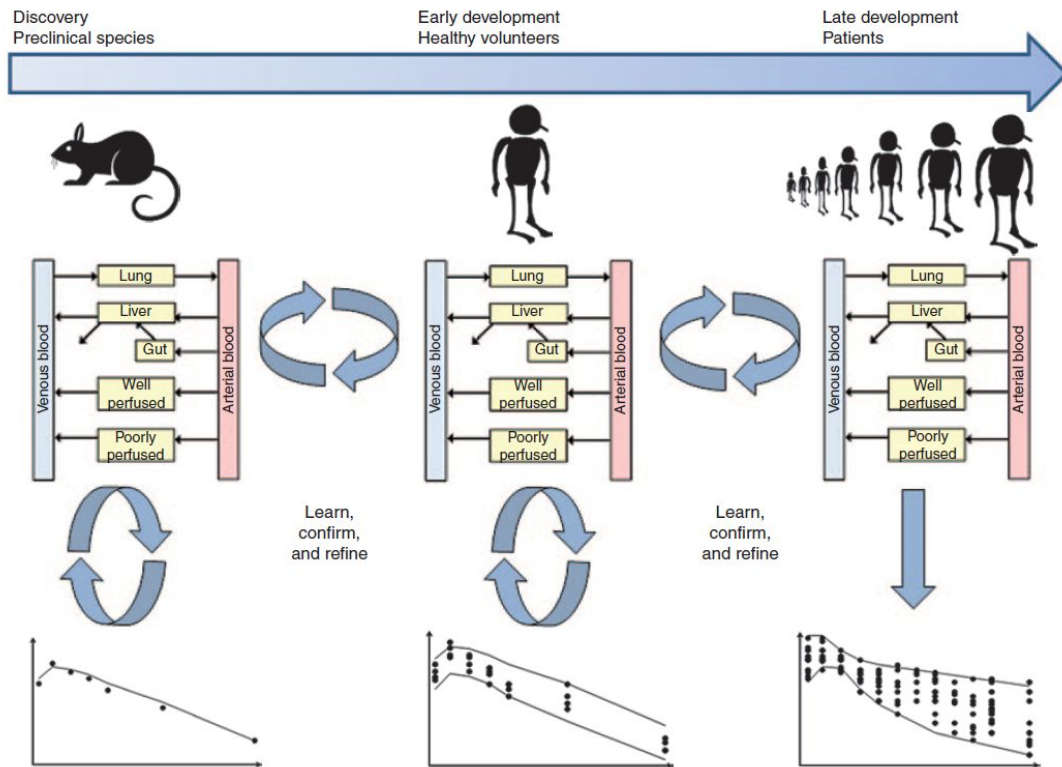
Physiology-based pharmacokinetic modeling (PBPK)

A mathematical modeling technique to predict pharmacokinetics (PK)

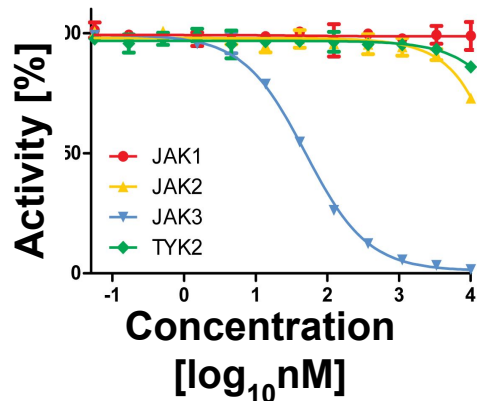


PBPK generates *compartment* models of PK (above) and implements a cycle of *Learn-Confirm-Refine* in translation (right).

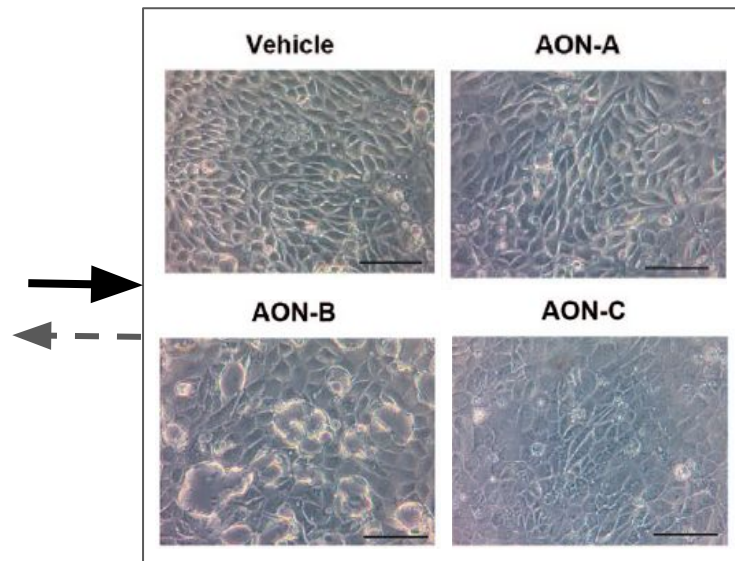
Jones, H. M., and K. Rowland-Yeo. 2013. "[Basic Concepts in Physiologically Based Pharmacokinetic Modeling in Drug Discovery and Development](#)." CPT: Pharmacometrics & Systems Pharmacology 2 (8): 63.



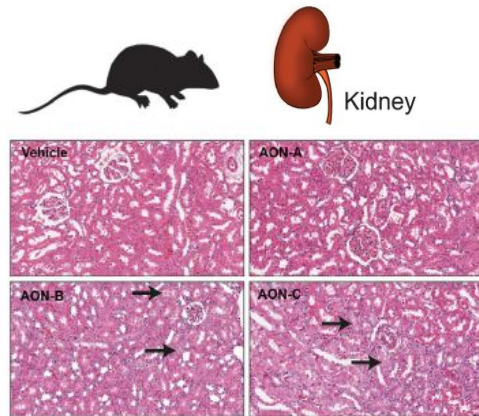
Classical workflow of efficacy and toxicity assessment



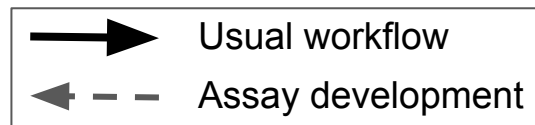
Biochemical & biophysical assays



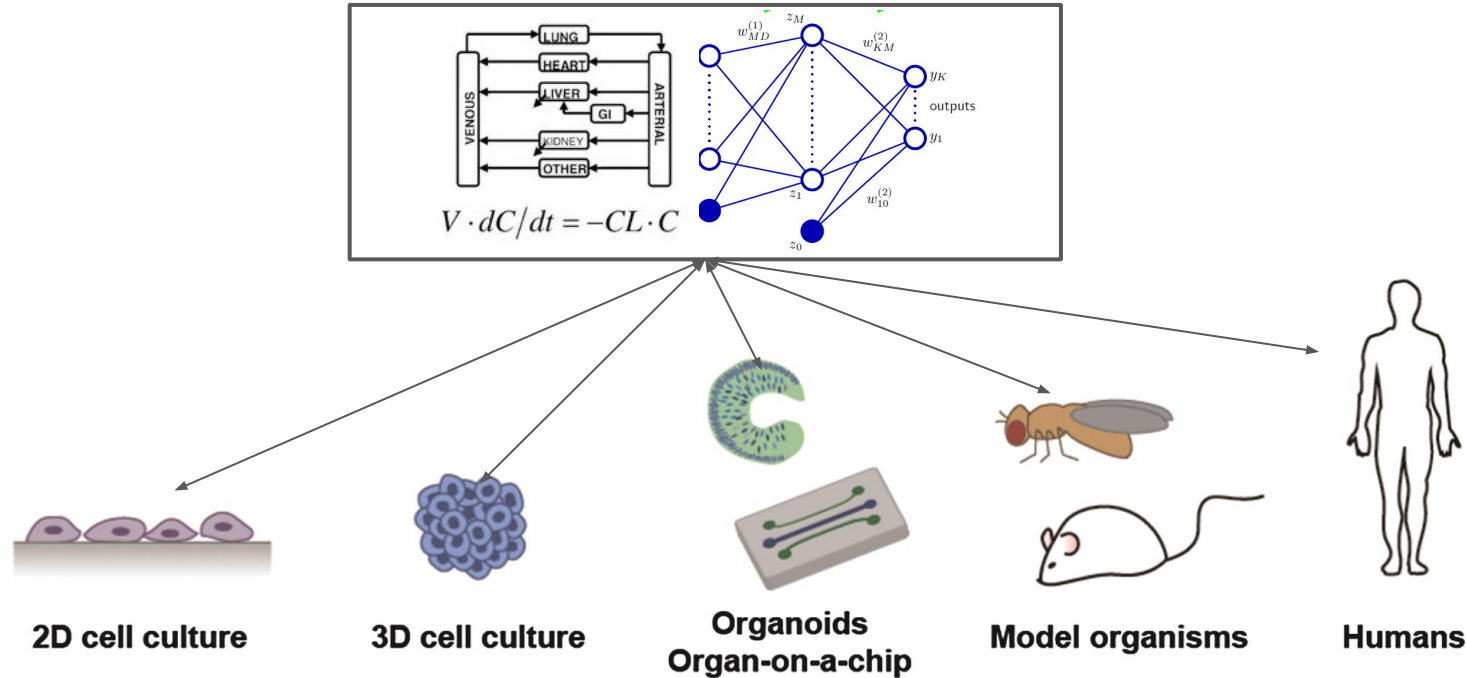
**Cellular assays
(*in vitro*)**



**Animal experiments
(*in vivo*)**



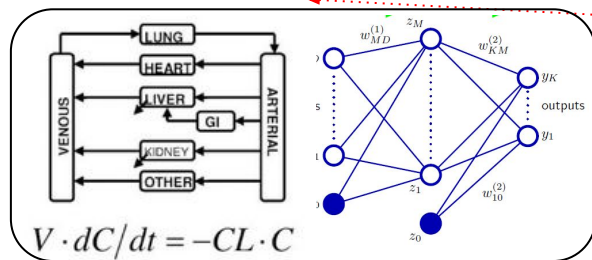
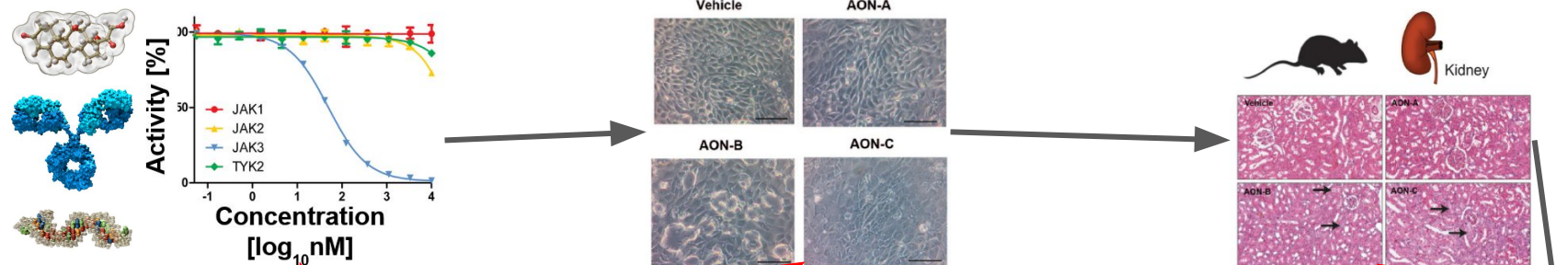
Biological and computational models of human diseases



Experimental Tractability

Physiological Relevance

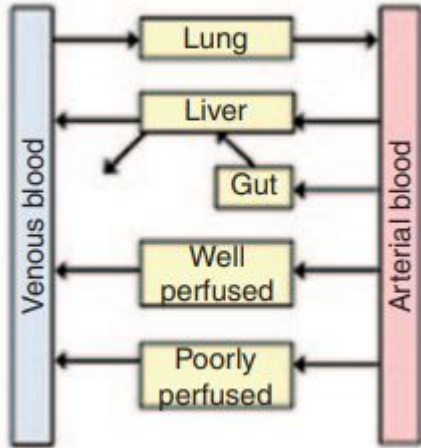
Computational methods empower efficacy and toxicity assessment



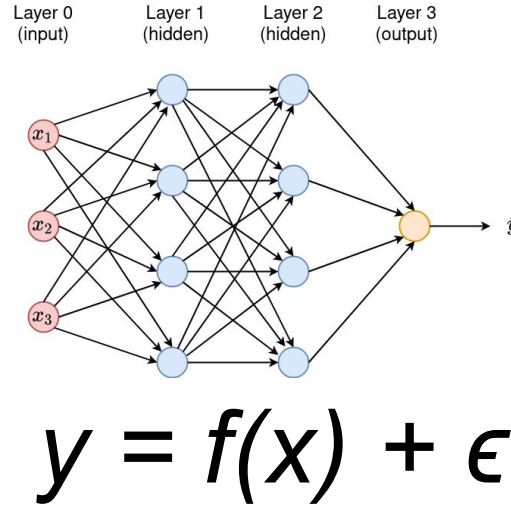
Computational models



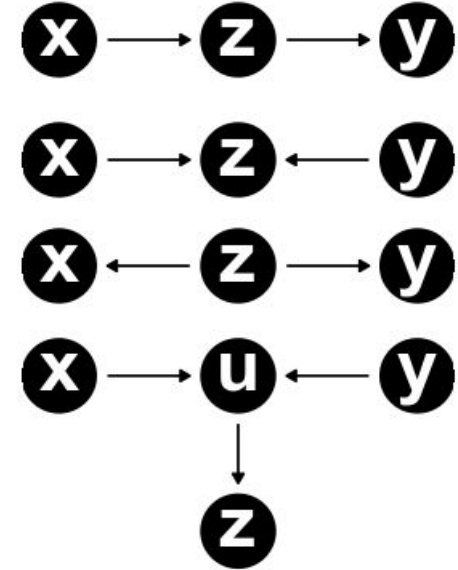
Three types of computational models



Mechanistic models



Statistical and machine-learning models

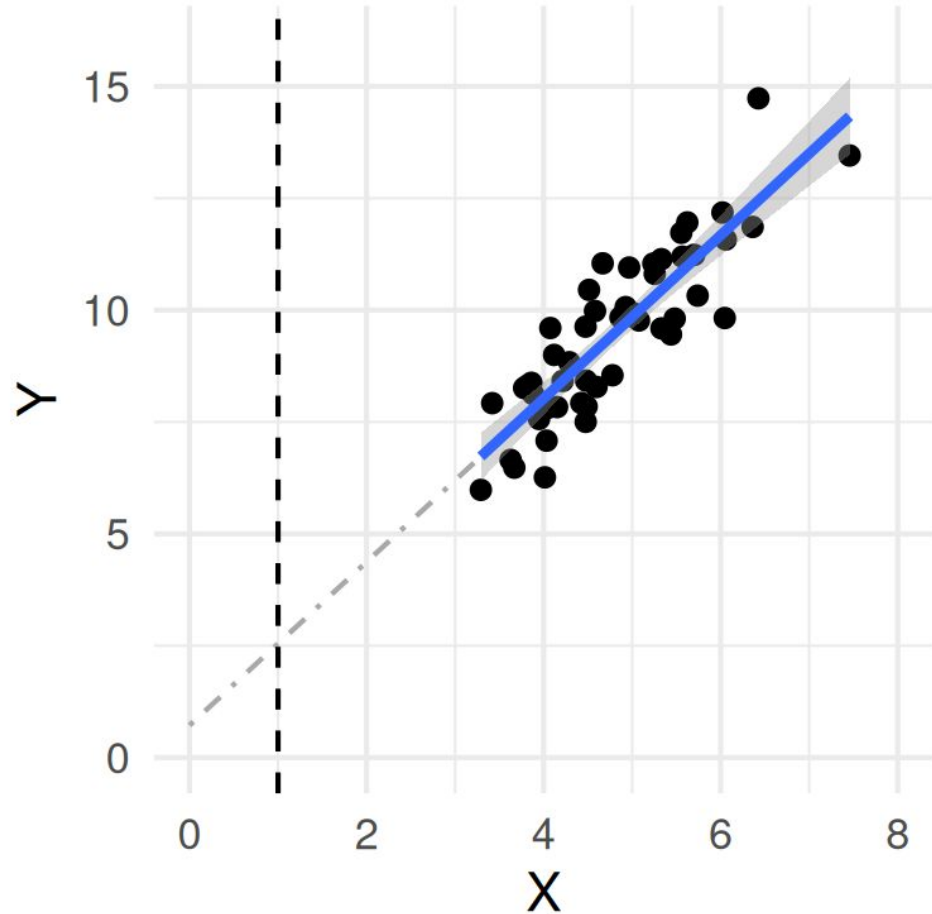


Causal models

We collected data from 50 independent observations, each consisting of paired measurements of X and Y . Assume that the correlation between X and Y is linear.

In a new observation, we manipulate the value of X by setting it to 1.

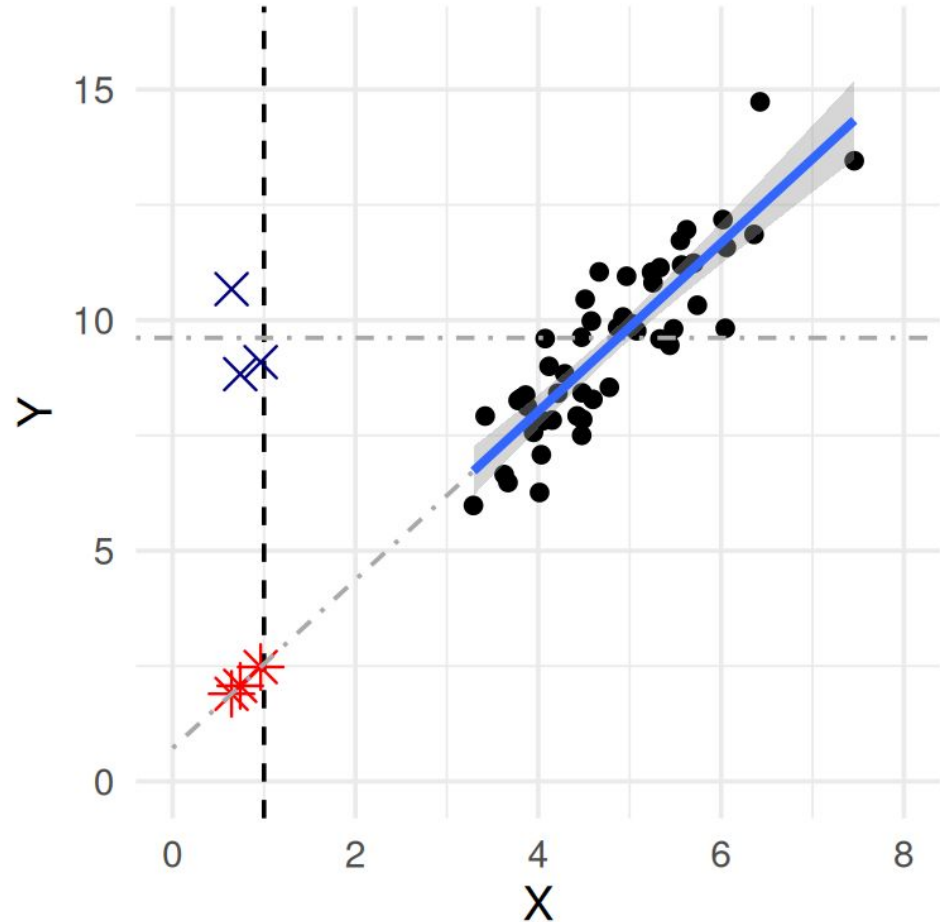
What value of Y do we expect?



Let's vote

We make three new observations where X is set to 1, the outcome would be:

- (A) Red stars;
- (B) Blue crosses;
- (C) Both red stars and blue crosses are possible;
- (D) I am not sure.

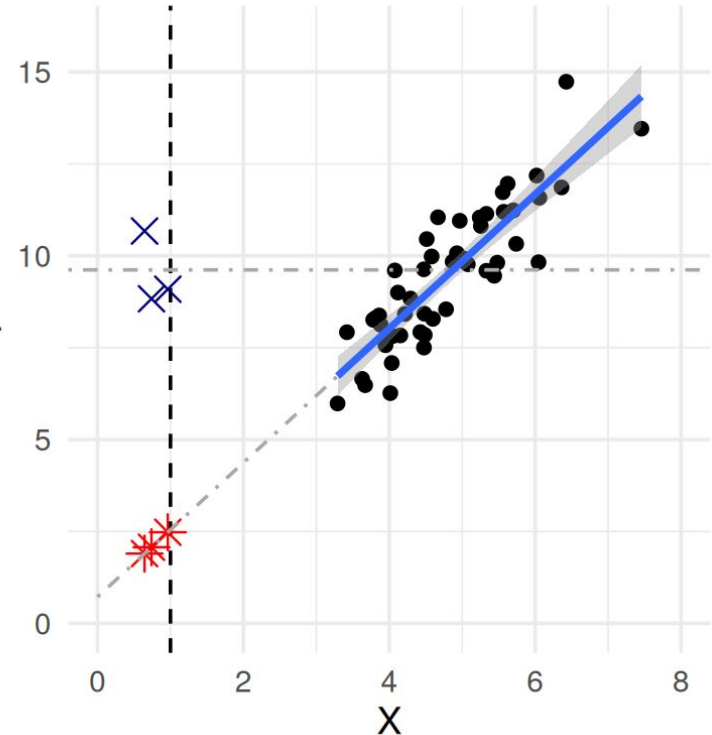


Predicting intervention outcome requires causal understanding

- Scenario 1:
 - X: highest temperature minus 30°C
 - Y: incidence of heat shock
 - Likely outcome of the intervention: **red stars**
- Scenario 2:
 - X: sales of ice-cream
 - Y: highest temperature minus 20°C
 - Likely outcome of the intervention: **blue crosses**
- Scenario 3:
 - X: sales of ice-cream
 - Y: incidence of drowning
 - Likely outcome of the intervention: **blue crosses**

Y

Both causation and confounding produce correlation. Predicting intervention outcome requires *causal understanding*, which data and statistical models alone cannot provide.

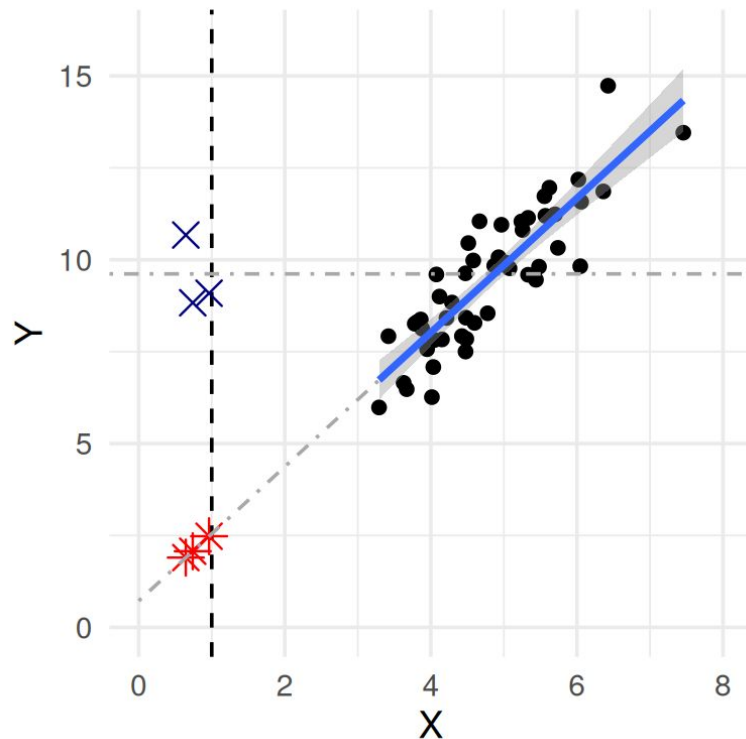
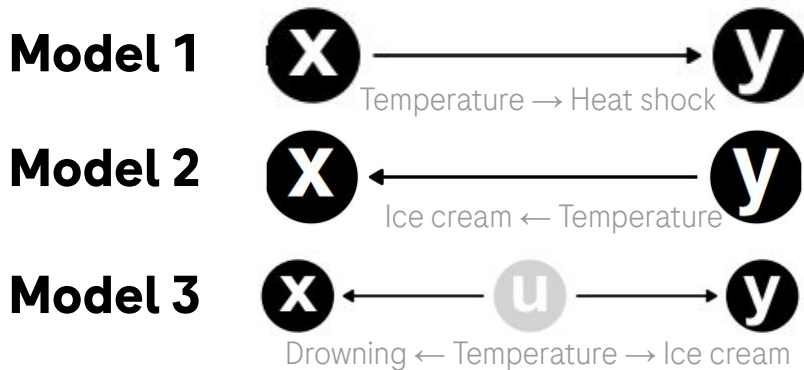


Causal models are critical when we intend to intervene

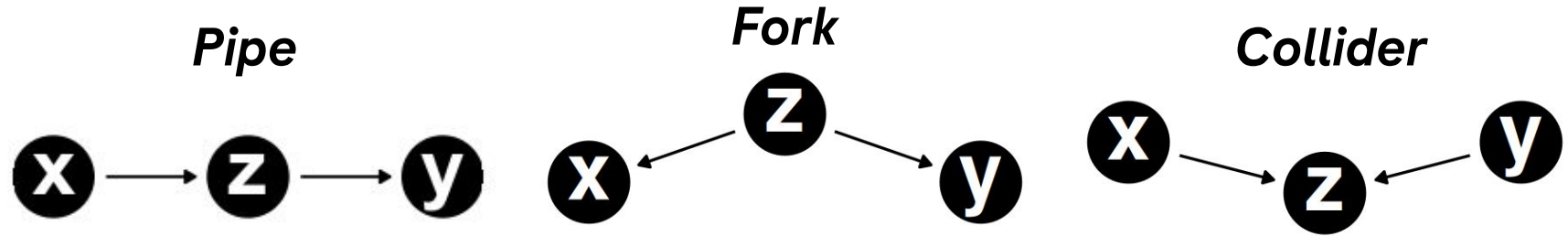
Imagine that X is the amount of a particular protein in the blood plasma of a patient of a disease, and Y is the disease severity of that patient.

A central question: *should we develop a drug to reduce the levels of this protein?*

We can use Directed Acyclic Graphs (DAGs) as causal models to identify causality.



Common structures of Directed Acyclic Graphs (DAGs)



X: The quality of a movie
Z: Review of an influencer
Y: Follower watching the movie

X and Y are marginally correlated. Intervening Z changes Y (but not X), removing the correlation between X and Y.

Z: Dog ownership
X: Number of daily walks
Y: Poop bags used

X and Y are correlated. Intervening Z changes both X and Y. Within each subgroup of Z, X and Y are not correlated.

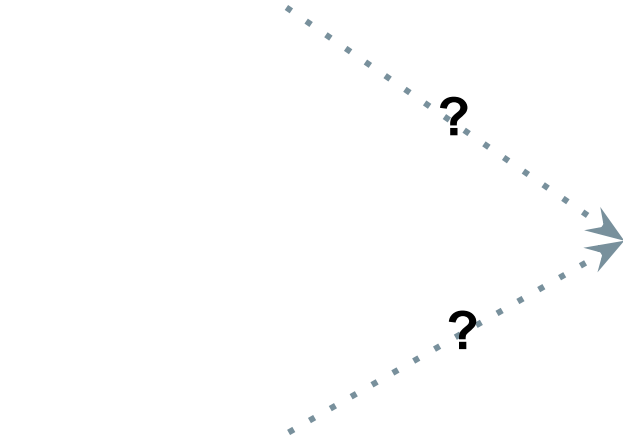
Z: Late arrival at work
X: Traffic congestion status
Y: Amount of oversleeping

X and Y are not correlated. Intervening Z has no effect on X or Y. Within each subgroup of Z, X and Y are (anti-)correlated.

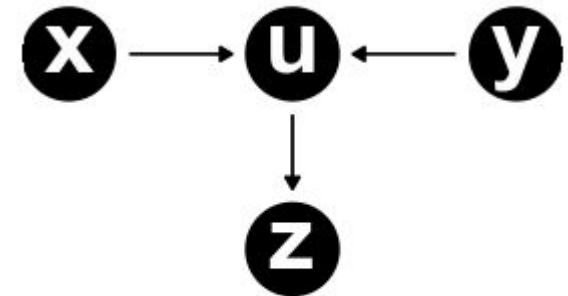
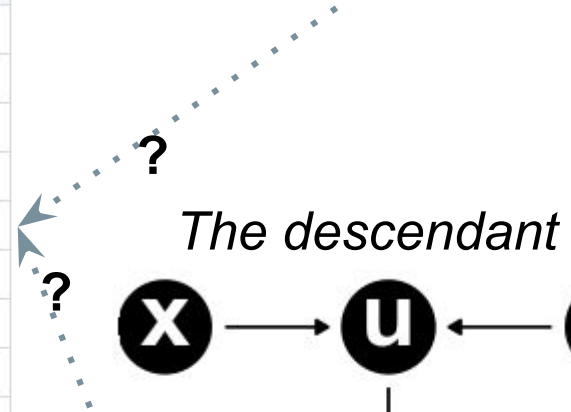
Causal structures and interventions can create or abolish correlations

Causality is crucial for drug discovery

Biomarker, tox study, pathology,
omics data, real-world data, ...



	x	z	y
1	0.835386320	1	-0.73897252
2	-0.005354014	-1	-0.82972315
3	0.058788286	1	0.76213369
4	-1.015602246	-1	-0.05951719
5	-0.339569780	-1	-0.11745910
6	-0.041077979	-1	-1.28243716
7	0.363740407	1	-0.30570762
8	0.119496314	-1	-1.19932461
9	0.257108454	-1	-1.06044066
10	0.304537158	-1	-0.43396492



We need both models (knowledge + assumptions) and data to infer causality.

Accurate predictions on small data with a tabular foundation model

Almost too good to be true?



"I knew the indoor pool was too good to be true."

Tabular Prior-data Fitted Network (TabPFN)

Article

Accurate predictions on small data with a tabular foundation model

<https://doi.org/10.1038/s41586-024-08328-6>

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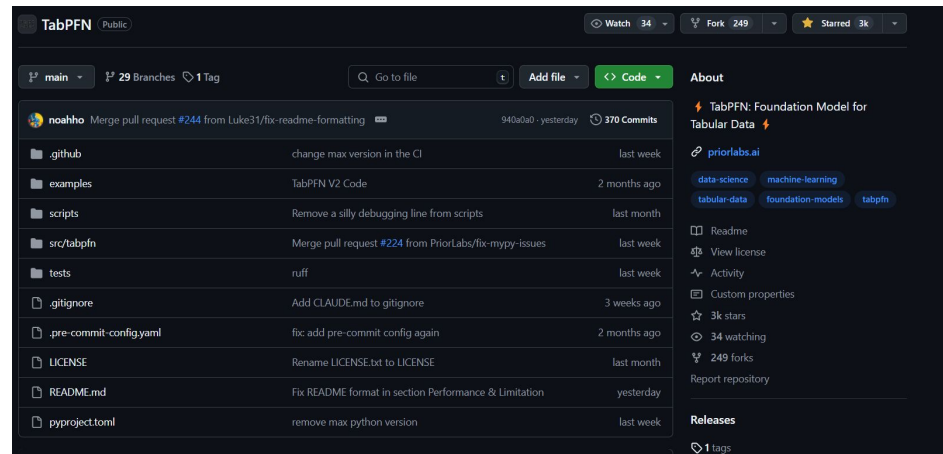
Published online: 8 January 2025

Open access

 Check for updates

Noah Hollmann^{1,2,3,7}, Samuel Müller^{1,7}, Lennart Purucker⁴, Arjun Krishnakumar⁴, Max Körfer¹, Shi Bin Hoo¹, Robin Tibor Schirmer^{1,5} & Frank Hutter^{1,5,8}

Tabular data, spreadsheets organized in rows and columns, are ubiquitous across scientific fields, from biomedicine to particle physics to economics and climate science^{1,2}. The fundamental prediction task of filling in missing values of a label column based on the rest of the columns is essential for various applications as diverse as biomedical risk models, drug discovery and materials science. Although deep learning has revolutionized learning from raw data and led to numerous high-profile success stories^{3–5}, gradient-boosted decision trees^{6–9} have dominated tabular data for the past 20 years. Here we present the Tabular Prior-data Fitted Network (TabPFN), a tabular foundation model that outperforms all previous methods on datasets with up to 10,000 samples by a wide margin, using substantially less training time. In 2.8 s, TabPFN outperforms an ensemble of the strongest baselines tuned for 4 h in a classification setting. As a generative transformer-based foundation model, this model also allows fine-tuning, data generation, density estimation and learning reusable embeddings. TabPFN is a learning algorithm that is itself learned across millions of synthetic datasets, demonstrating the power of this approach for algorithm development. By improving modelling abilities across diverse fields, TabPFN has the potential to accelerate scientific discovery and enhance important decision-making in various domains.



Source code at <https://github.com/PriorLabs/tabpfm>, released in a license analogous to Apache 2.0 (commercial use friendly).

In-Context Learning (ICL)

The model isn't explicitly trained on a particular task ahead of time. Instead, the prompt itself provides examples that enable the model to perform the task on the fly. It is one of the intriguing abilities of deep transformer models.

Who won the final game of World Cup 2014?

Reasoned for a few seconds

Germany won the final of the 2014 FIFA World Cup, defeating Argentina 1–0 in extra time.

2018 World Cup · 15 Jul 18

Full-time



4

-

2



France

Croatia

Final

Mario Mandžukić 18' (OG)

Antoine Griezmann 38' (P)

Paul Pogba 59'

Kylian Mbappé 65'



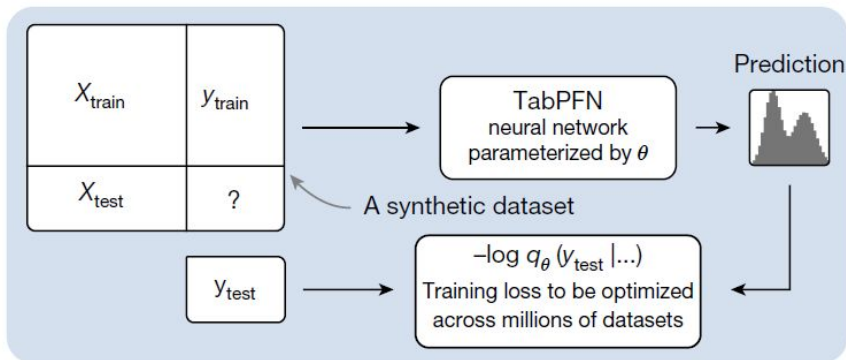
Ivan Perišić 28'

Mario Mandžukić 69'

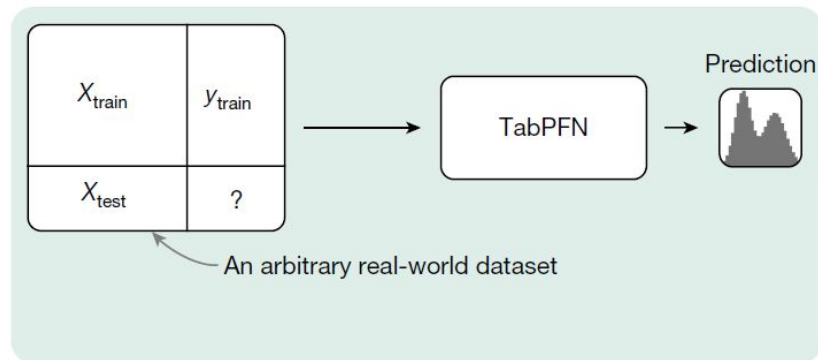
Left: chat history (ChatGPT o1 model, tested on 18.03.2025). Right: The ground truth.

Tabular Prior-data Fitted Network (TabPFN) is trained with synthesized data and predicts missing value in user data

TabPFN is trained on synthetic data to take entire datasets as inputs and predict in a forward pass



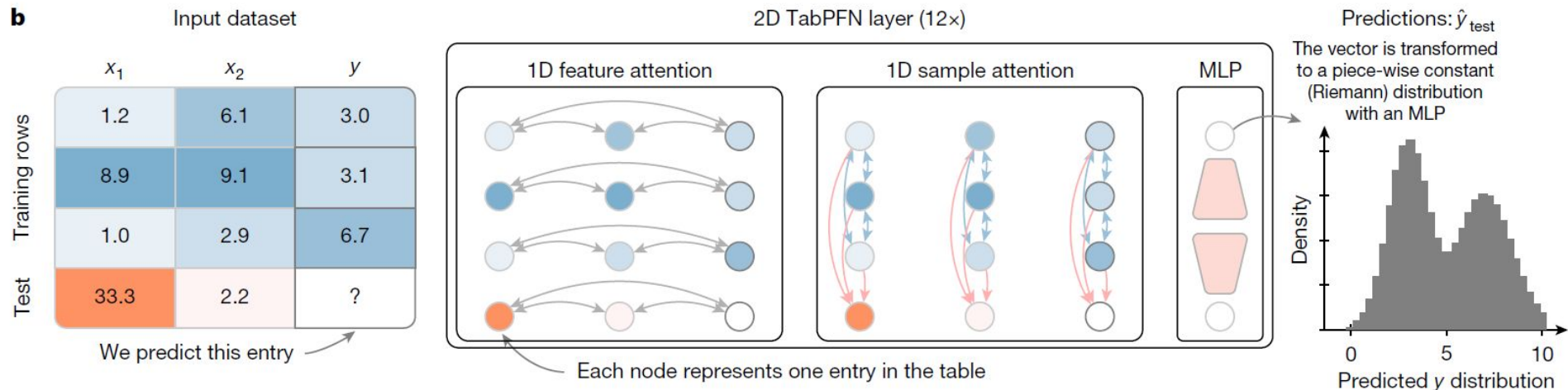
TabPFN can now be applied to arbitrary unseen real-world datasets



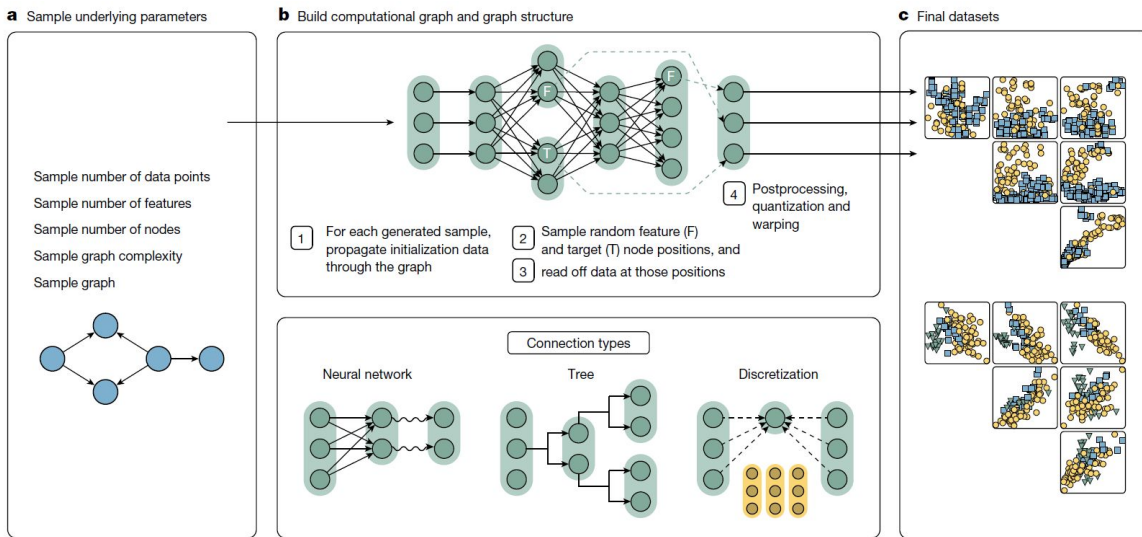
A metaphor: Imagine a lab where millions of billiard games are played simultaneously: in each game, different numbers of balls are placed randomly, and a white ball starts with a random velocity at a random position. By learning the trajectory of all balls of all games, one may learn to predict the trajectory of any real billiard game, as long as the positions of all balls and the initial velocity of the white ball is known.

Architecture of TabPFN uses both row-wise sample and column-wise feature information to predict a distribution

b

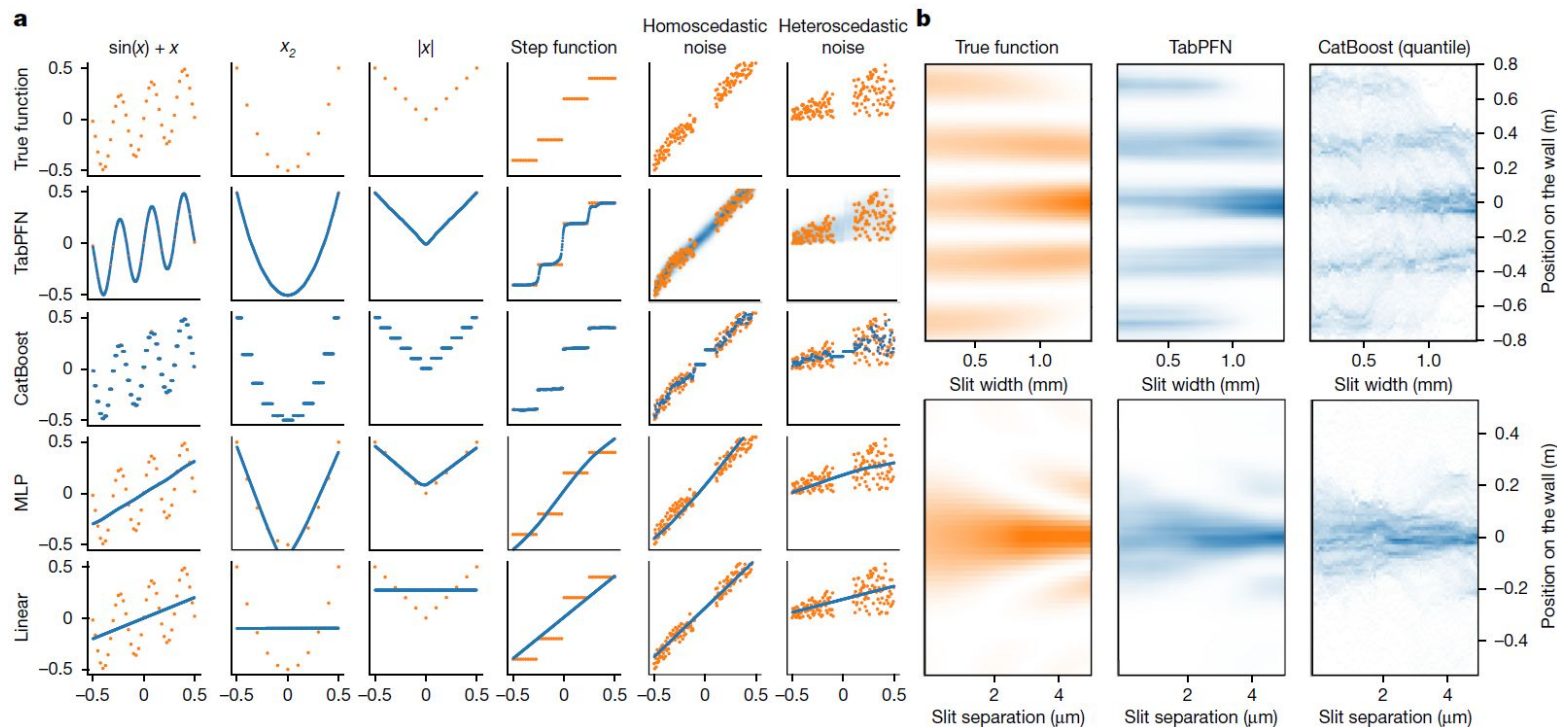


Generating synthesized data with causal models

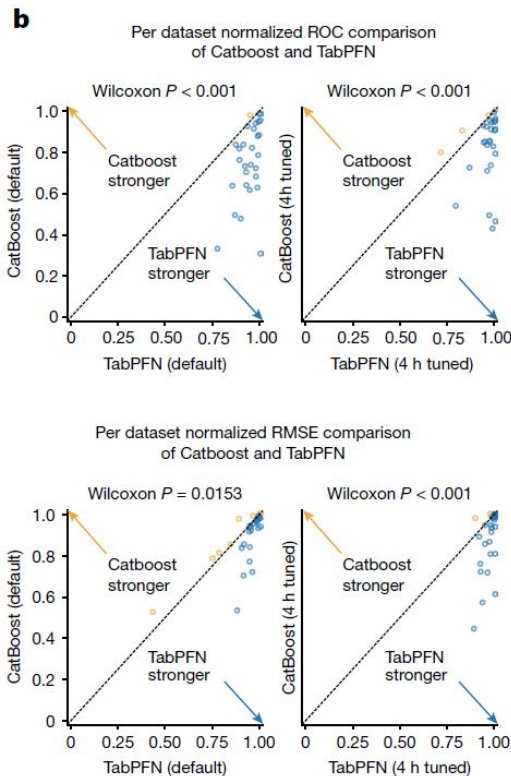
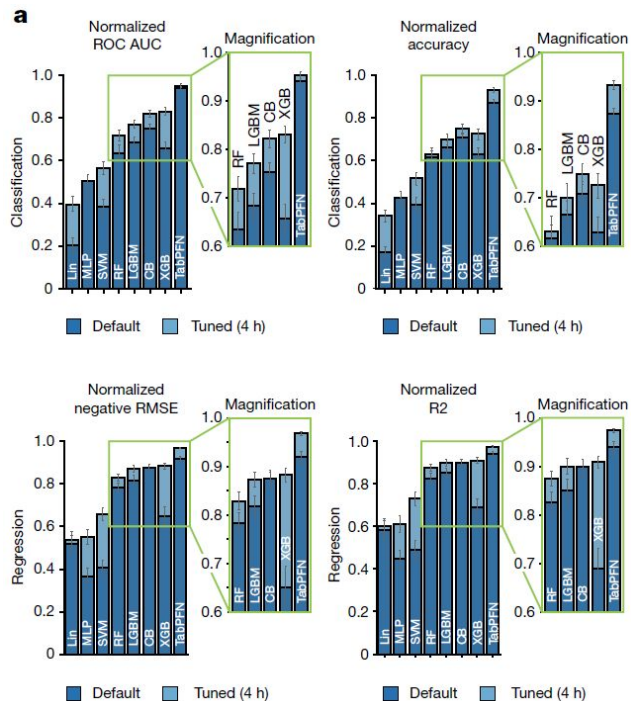


1. We sample 100-million structurally unique directed acyclic graphs (DAGs) as causal graphs that generate data.
2. Random data is assigned to the root node. Other nodes are propagated by rules specified by the edges, plus a Gaussian noise.
3. The values of sampled features (input to the model) and targets (output of the model) are extracted.
4. Data are post-processed (non-linear distortion, binning, random missing) to reflect real-world data processing.
5. Synthesized data from 10^8 causal experiments are used to train *TabPFN*.

TabPFN's prediction on data generated by simple functions



Performance benchmark against popular methods



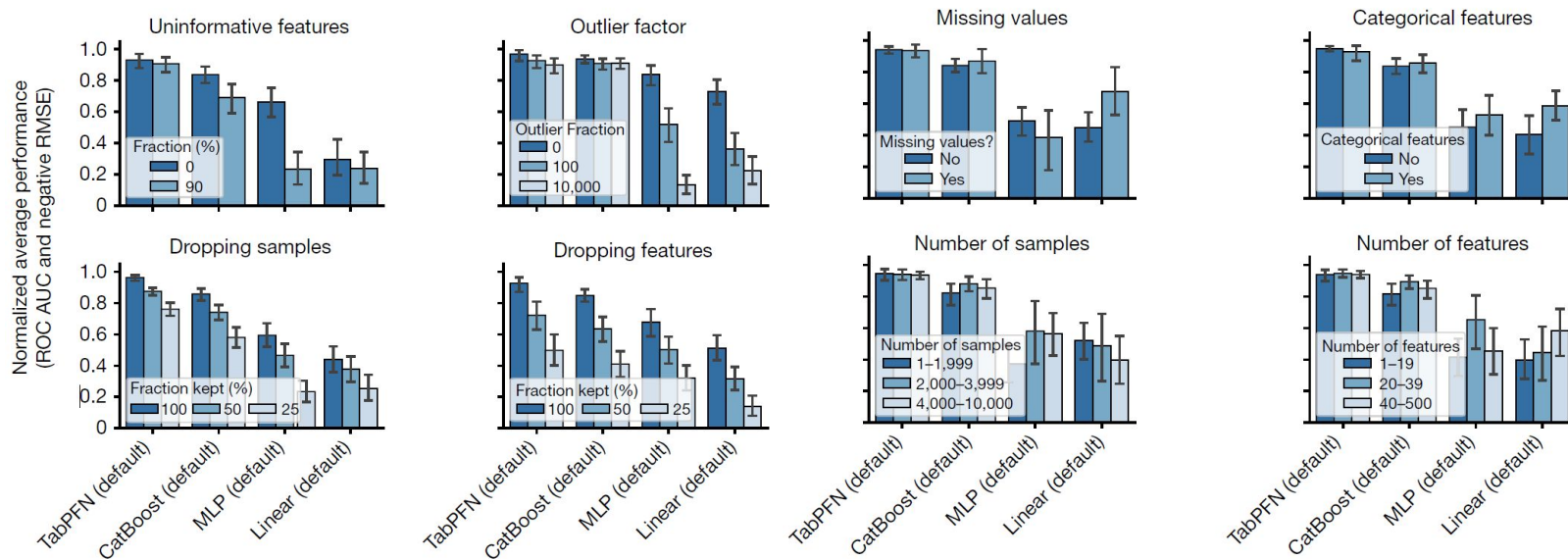
Left: Performance for classification (top) and regression (bottom) tasks.

Right: Comparing performance of TabPFN and CatBoost, the top contender, for classification (top) and regression (bottom) tasks.

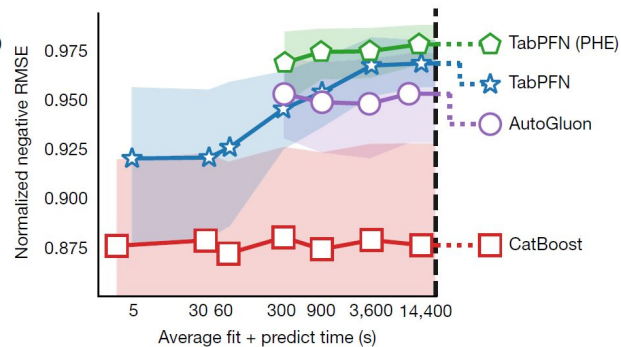
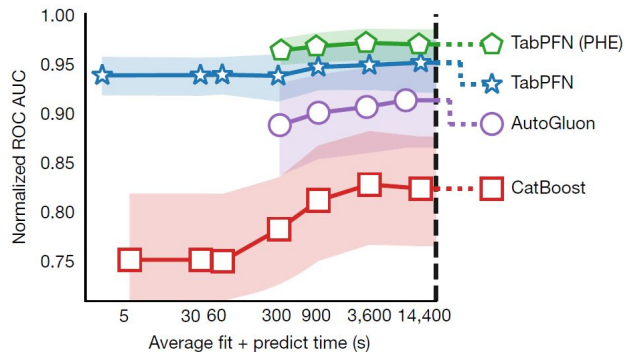
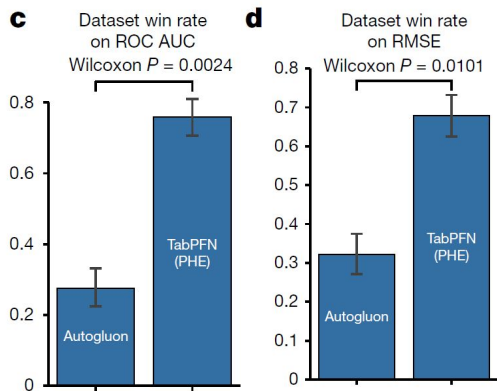
Hollmann, Noah, Samuel Müller, ..., Frank Hutter. 2025. "Accurate Predictions on Small Data with a Tabular Foundation Model." *Nature* 637 (8045): 319–26.

Lin: linear; MLP: multi-layer perceptrons; SVM: support vector machines; RF: random forest; LGBM: Light Gradient Boosting Machine; CB: CatBoost; XGB: XGBoost; ROC: receiver operating characteristic; RMSE: root mean square error.

Robustness against common caveats and problems



Benchmark against AutoML and CatBoost



Datasets that the authors tested for classification (left) and regression (right) tasks

Article

Extended Data Table 3 | List of test datasets used for primary evaluation of classification tasks

Name	OpenML ID	Domain	Features	Samples	Targets	Categorical Feats.
ada	41156	Census	48	4147	2	0
Australian	40981	Finance	14	690	2	8
blood-transfusion-service-center	1464	Healthcare	4	748	2	0
car	40975	Automotive	6	1728	4	6
churn	40701	Telecommunication	20	5000	2	4
cmc	23	Public Health	9	1473	3	7
credit-g	31	Finance	20	1000	2	13
dna	40670	Biology	180	3186	3	180
eucalyptus	188	Agriculture	19	736	5	5
first-order-theorem-proving	1475	Computational Logic	51	6118	6	0
GesturePhase Segmentation Processed	4538	Human-Computer Interaction	32	9873	5	0
jasmine	41143	Natural Language Processing	144	2984	2	136
kc1	1067	Software Engineering	21	2109	2	0
kr-vs-kp	3	Game Strategy	36	3196	2	36
madeline	41144	Artificial	259	3140	2	0
mfeat-factors	12	Handwriting Recognition	216	2000	10	0
ozone-level-8hr	1487	Environmental	72	2534	2	0
pc4	1049	Software Engineering	37	1458	2	0
philippine	41145	Bioinformatics	308	5832	2	0
phoneme	1489	Audio	5	5404	2	0
qsar-biodeg	1494	Environmental	41	1055	2	0
Satellite	40900	Environmental Science	36	5100	2	0
segment	40984	Computer Vision	16	2310	7	0
steel-plates-fault	40982	Industrial	27	1941	7	0
sylvine	41146	Environmental Science	20	5124	2	0
vehicle	54	Image Classification	18	846	4	0
wilt	40983	Environmental	5	4839	2	0
wine-quality-white	40498	Food and Beverage	11	4898	7	0
yeast	181	Biology	8	1484	10	0

All classification tasks from the AutoML Benchmark²⁶ with fewer 10,000 samples and 500 features. The benchmark comprises diverse real-world tabular datasets, curated for complexity, relevance, and domain diversity.

Extended Data Table 4 | List of test datasets used for primary evaluation of regression tasks

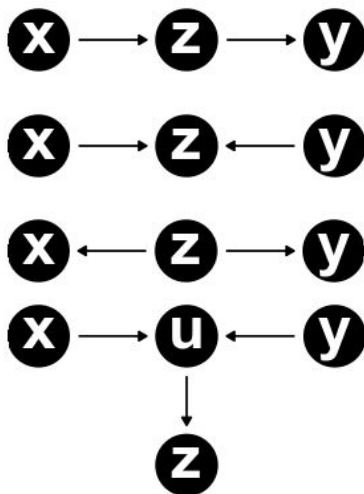
Name	OpenML ID	Domain	Features	Samples	Categorical Features
abalone	42726	Marine Biology	8	4177	1
airfoil_self_noise	44957	Aerospace Engineering	5	1503	0
auction_verification	44958	Economics	7	2043	2
boston	531	Real Estate	13	506	2
cars	44994	Automotive Engineering	17	804	0
colleges	42727	Education	44	7063	12
concrete_compressive_strength	44959	Materials Science	8	1030	0
cpu_activity	44978	Computer Engineering	21	8192	0
energy_efficiency	44960	Architectural Engineering	8	768	0
geographical_origin_of_music	44965	Music Information Retrieval	116	1059	0
grid_stability	44973	Power Systems Engineering	12	10000	0
house_prices_nominal	42563	Real Estate	79	1460	43
kin8nm	44980	Robotics	8	8192	0
Mercedes_Benz_Greener_Manufacturing	42570	Manufacturing	376	4209	8
MIP-2016-regression	43071	Operations Research	144	1090	1
Moneyball	41021	Sports Analytics	14	1232	6
pumadyn32nh	44981	Robotics	32	8192	0
QSAR_fish_toxicity	44970	Toxicology	6	908	0
quake	550	Geophysics	3	2178	0
SAT11-HAND-runtime-regression	41980	Computational Logic	116	4440	1
sensory	546	Food Science	11	576	11
socmob	541	Sociology	5	1156	4
space_ga	507	Political Science	6	3107	0
student_performance	44967	Education	30	649	17
tecator	505	Food Science	124	240	0
topo_2_1	422	Cheminformatics	266	8885	0
us_crime	42730	Criminology	126	1994	0
yprop_4_1	416	Cheminformatics	251	8885	0

All regression tasks from the AutoML²⁶ and OpenML-CTR23²⁷ Benchmarks with fewer 10,000 samples and 500 features. The benchmark comprises diverse real-world tabular datasets, curated for complexity, relevance, and domain diversity.

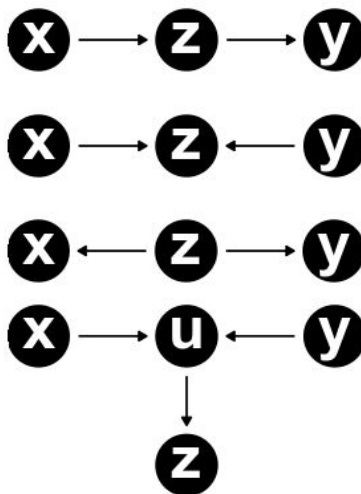
We witness a paradigm shift in analysing tabular data: from *causal model selection* to *causality-assisted prediction*

Until now: Which model(s) could have generated this piece of data? Use them for prediction.

	x	z	y
1	0.835386320	1	-0.73897252
2	-0.005354014	-1	-0.82972315
3	0.058788286	1	0.76213369
4	-1.015602246	-1	-0.05951719
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8	0.119496314	-1	-1.19932461
9	0.257108454	-1	-1.06044066
10	0.304537158	-1	-0.43396492



From now on: Sample a large number of causal models, synthesize data from them, and train ML models trained on synthesized



	x	z	y
1	0.835386320	1	-0.73897252
2	-0.005354014	-1	-0.82972315
3	0.058788286	1	0.76213369
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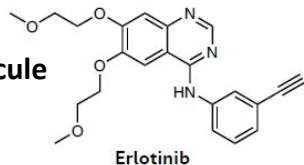
The difficulty of asserting causality remains. However, thanks to causal models and transformers, we now have better tools for prediction.

End of the first lecture

Many modalities are now available: most of them target proteins. Why?



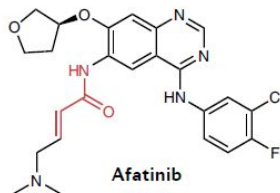
Reversible
small-molecule
enzyme
inhibitors



Monoclonal
antibodies



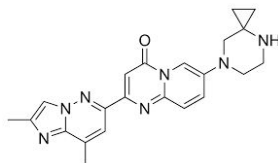
Irreversible
covalent
enzyme
inhibitors



Bispecific
antibodies



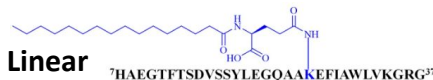
Small-
molecule
splicing
modifier



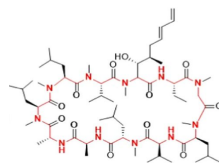
Antisense
oligonucleotides



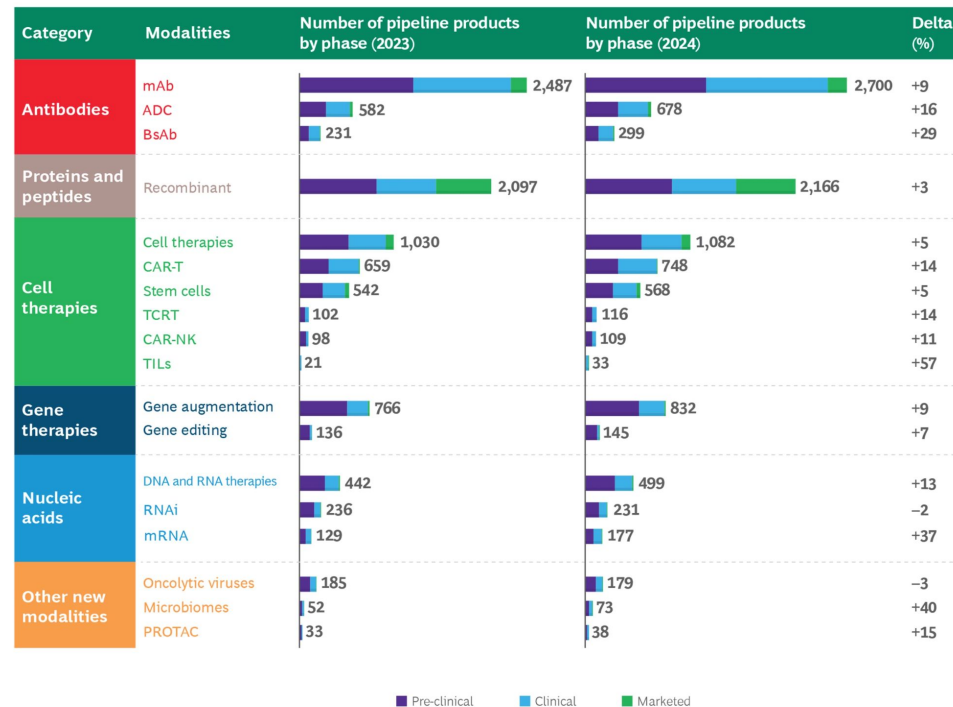
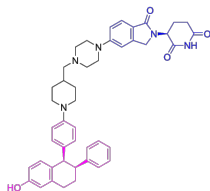
Linear
peptides



MacroCyclic
Peptides
(MCP)



PROteolysis
Targeting
Chimera
(PROTAC)

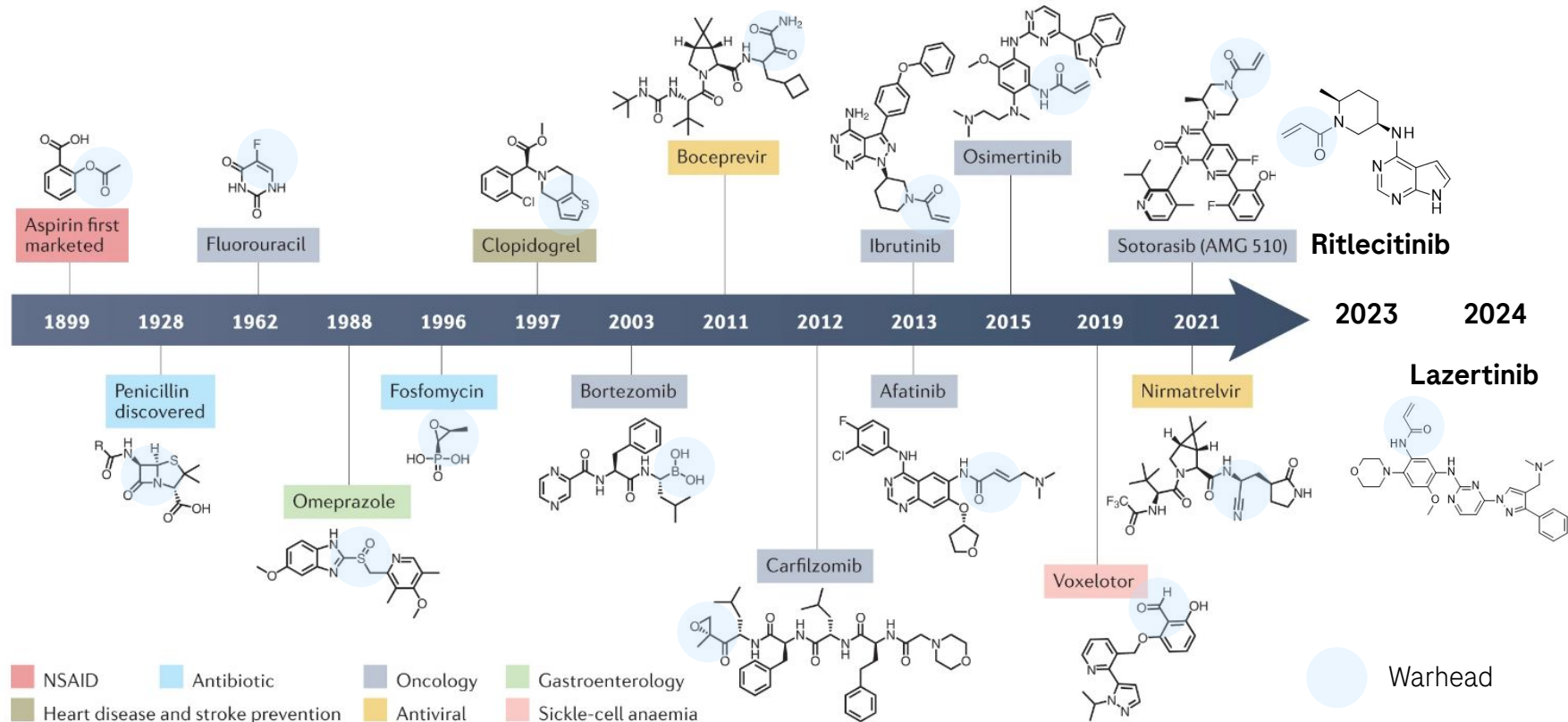


Sources: Evaluate Pharma; BCG analysis.

Notes: ADC = antibody-drug conjugate; BsAb = bispecific antibody; CAR-NK = chimeric antigen receptor-transduced natural killer cell; CAR-T = chimeric antigen receptor T cell; mAb = monoclonal antibody; PROTAC = proteolysis-targeting chimera; TCRT = T-cell receptor therapy; TIL = tumor-infiltrating lymphocyte.

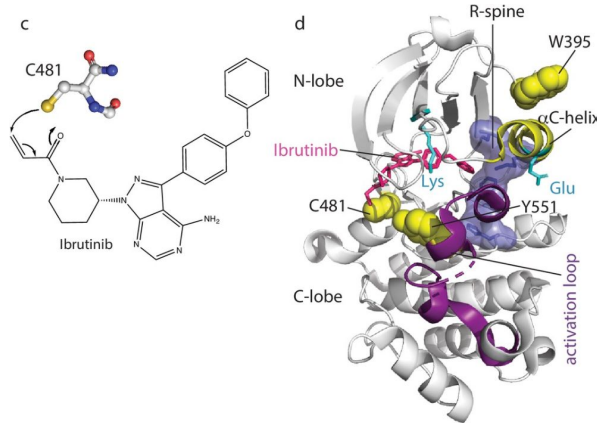
Left: various publications. Right: Chen, et al., [New Drug Modalities 2024](#), Boston Consulting Group.

Covalent drugs have gained renewed interests



Ibrutinib, a first-in-class inhibitor of BTK (Bruton's Tyrosine Kinase)

Ibrutinib	
Approval	AbbVie (2013)
Binding type	Irreversible covalent binding
Binding site	C481, ATP-binding domain
Warhead	Acrylamide
Half-life	~4-6 hours
Indication	CLL, MCL, MZL, WM, GVHD, BN
Dosage	420 mg, qd (CLL/SLL, WM); 560 mg, qd (MCL, MZL)
Administration	Oral

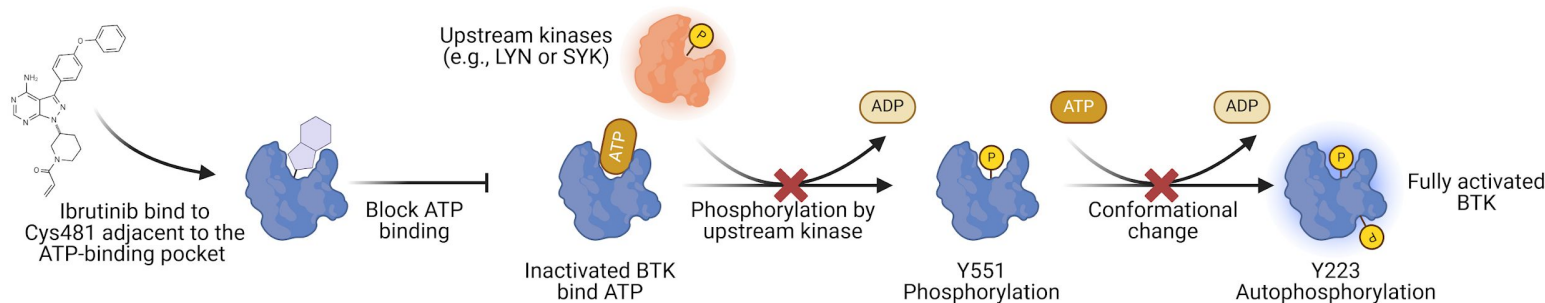


The ATP-binding pocket:

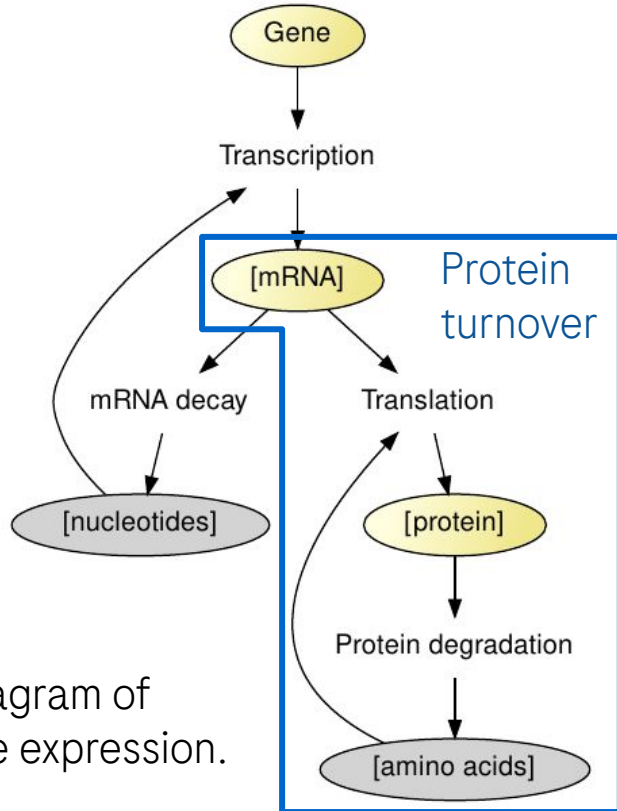
- A highly conserved region within the kinase domain.

Cysteine residue (Cys481):

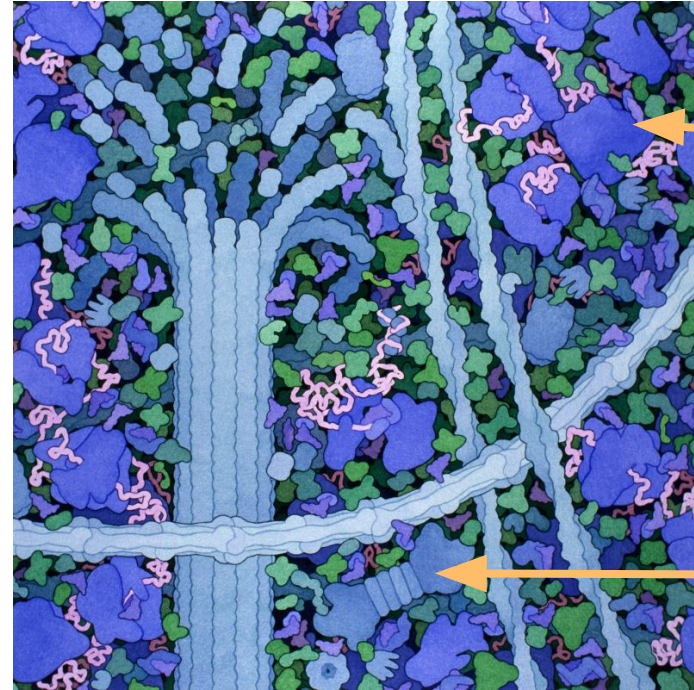
- Located adjacent to the ATP-binding pocket
- Cys481 is a relatively unique cysteine, enabling selective covalent inhibition.



Protein turnover consists of synthesis and degradation



A diagram of gene expression.



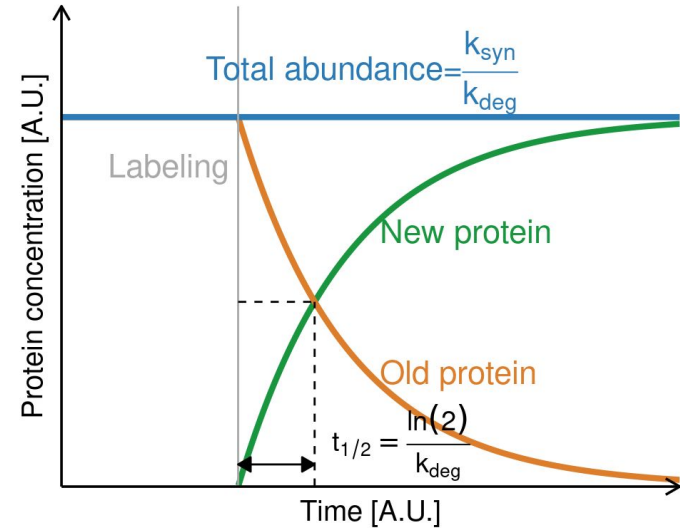
ribosome

proteasome

Illustration by David S. Goodsell. doi:
10.2210/rcsb_pdb/goodsell-gallery-006

Protein turnover is critical for drug discovery & development

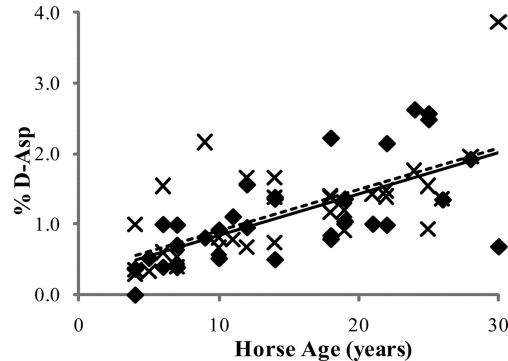
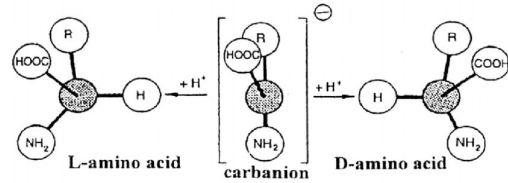
- Protein turnover affects efficacy, potency, ADME properties, and safety profiles of drug candidates.
- Protein turnover is essential for target prioritization and modality selection, for instance covalent binders and/or targeted protein degraders.
- Understanding protein turnover helps to translate pharmacokinetic and pharmacodynamic (PK/PD) relationships between systems.



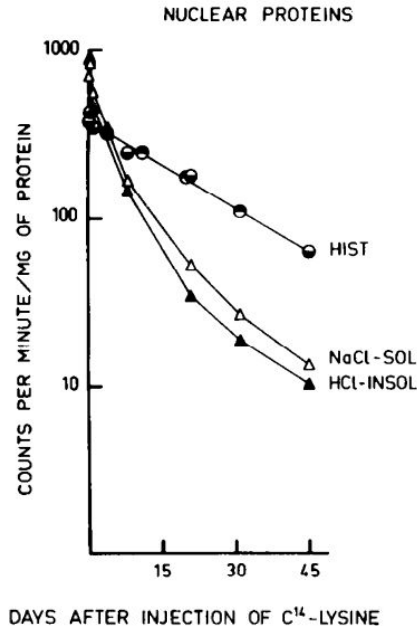
Assumptions: zero-order synthesis (rate k_{syn}), first-order degradation (rate k_{deg}), and steady state (i.e. no expression changes).

Quantifying protein turnover and long-living proteins

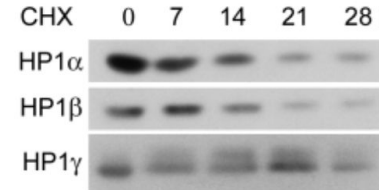
L- and D-Asp racemization



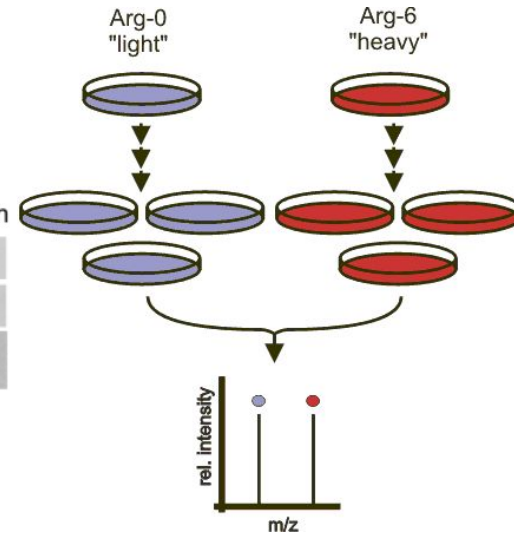
Radio isotope pulse-labelling



Western blotting following cycloheximide (CHX) treatment

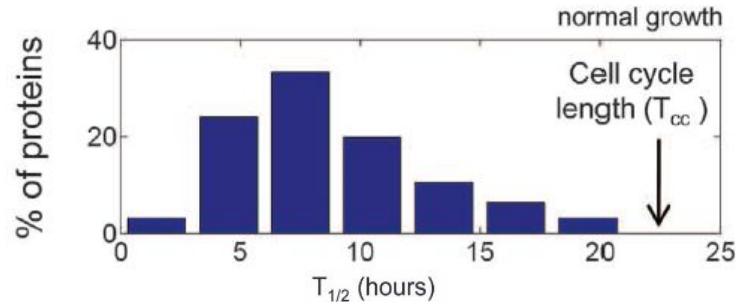


Stable isotope with amino acids in cell culture (SILAC) and mass spectrometry (MS)

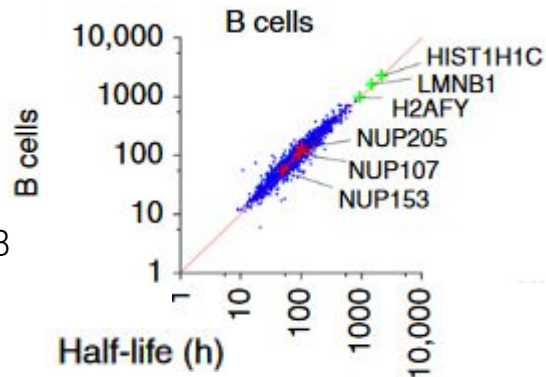


Source: various publications.

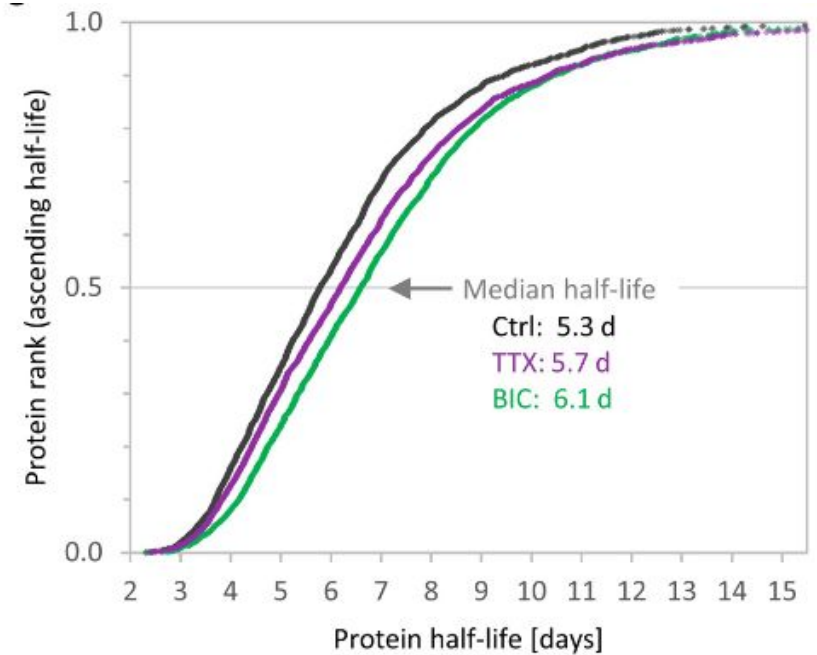
Protein half-life *in vitro* ranges between hours and days



Data from a human non-small cell lung cancer cell line (Eden *et al.*, 2011)

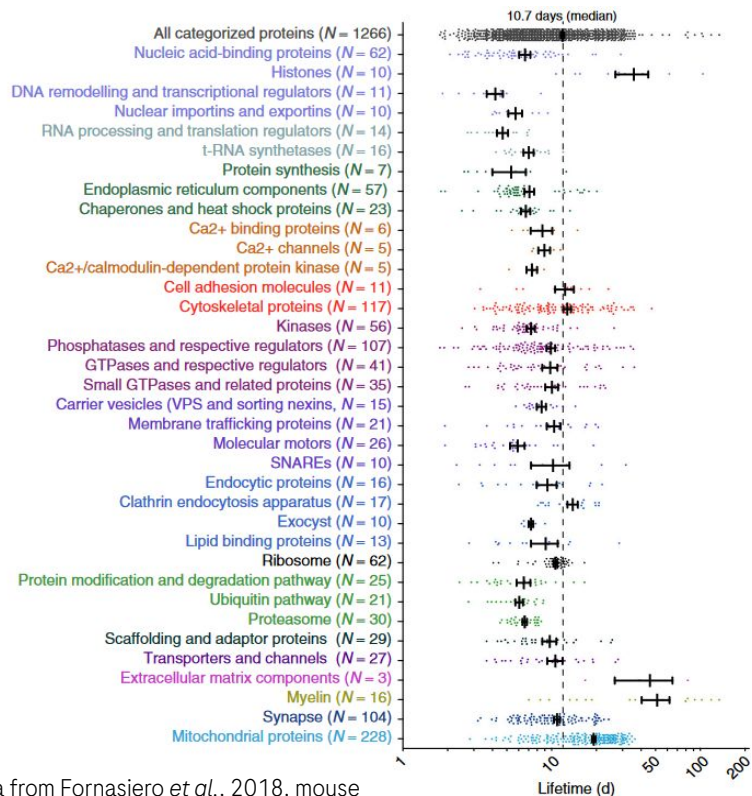


Data from primary human B cells (Mathieson *et al.*, 2018)



Data from primary hippocampal neuronal cells from rat (Dörrbaum *et al.*, 2020)

Protein half-life *in vivo* ranges between days and years



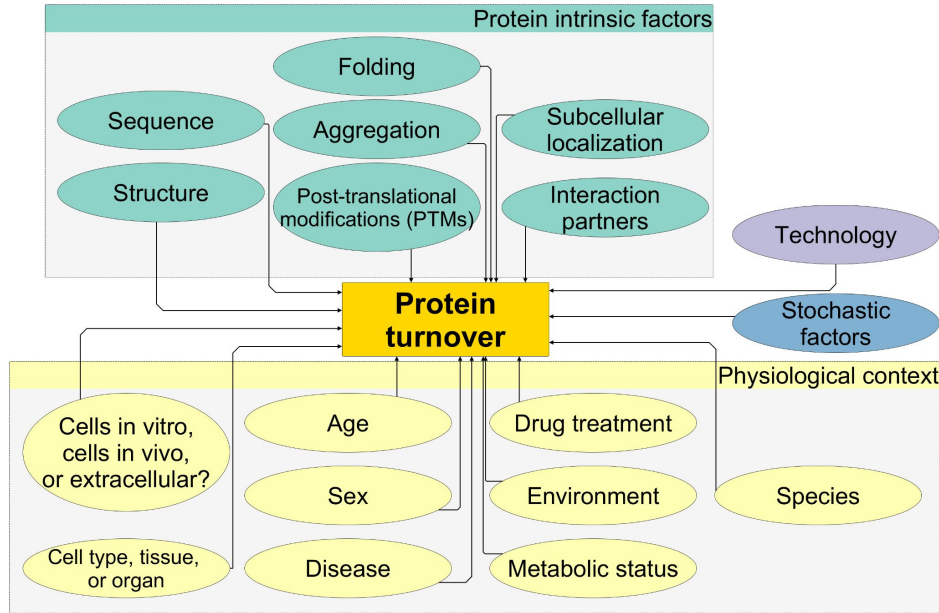
Data from Fornasiero *et al.*, 2018, mouse brain

Table 1 | Known long-lived proteins and molecules

Protein or molecule*	Age [‡]	Measure	Organism
Eye lens crystallin	>70 years	Lifetime	Human
Collagen	117 years	Half-life	Human
Elastin	>78 years	Lifetime	Human
Enamel and dentine	>70 years	Lifetime	Human
Histones	223 days	Half-life	Mouse
	117 days	Half-life	Mouse
	218 days	Half-life	Rat
Nuclear pore proteins	>1 month	Lifetime	Worm
	>1 year	Lifetime	Rat
Myelin	95 days	Half-life	Rat
	>100 days	Half-life	Mouse
Myelin proteolipid protein	>100 days	Half-life	Mouse
REC8	Weeks	Lifetime	Mouse
mRNA	Possibly indefinite	Lifetime	Plant seed
	>2 years	Half-life	Frog oocyte
Cholesterol	>18 months	Lifetime	Rabbit
Phospholipids	>192 days	Lifetime	Rabbit

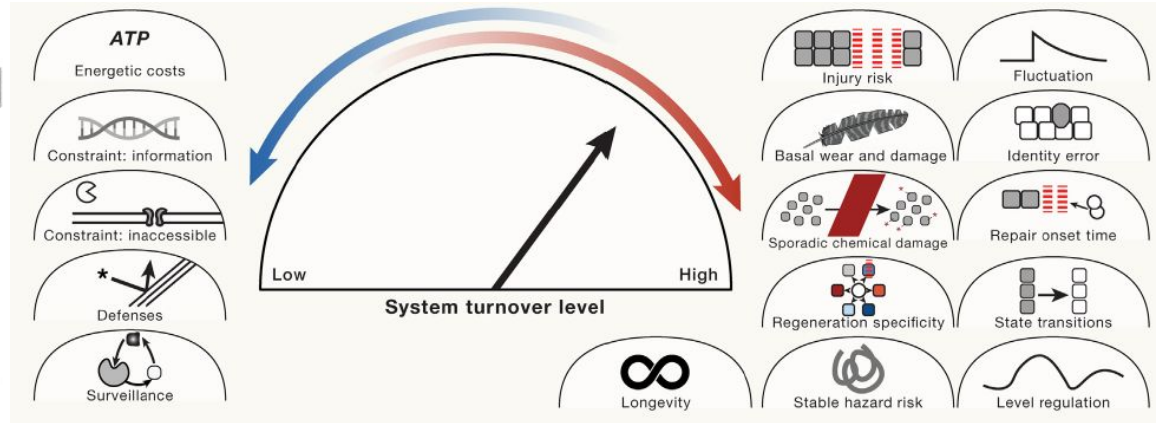
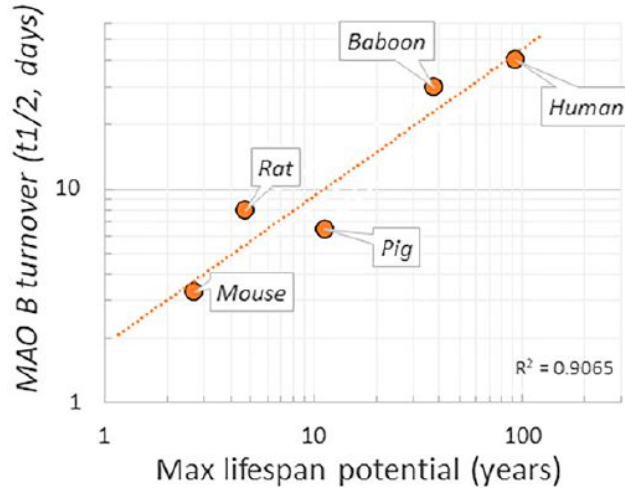
Toyama, Brandon H., and Martin W. Hetzer. "[Protein Homeostasis: Live Long, Won't Prosper.](#)" *Nature Reviews Molecular Cell Biology*, 2013

Half-life varies between proteins and contexts: influencing factors and an example



Condition	Half-life of protein X	Source
Human neurons <i>in vitro</i>	38.6h	Roche in-house data
Mouse neurons in vitro	34.1h (standard error:3.9h)	Fornasiero et al. , Nature Communications, 2018
Mouse cortex <i>in vivo</i>	619.2h, or 25.8d	Kluever et al. , Science Advances, 2022

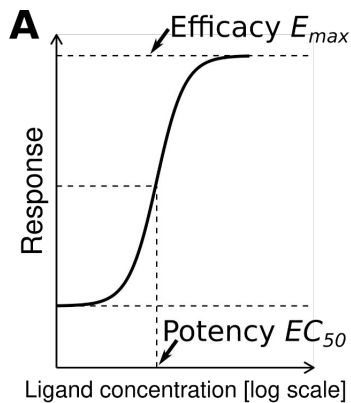
Nothing in Biology Makes Sense Except in the Light of Evolution: the purpose and ubiquity of turnover



Left: Gabrielsson, J., and S. Hjorth. 2023. "Turn On, Tune In, Turnover! Target Biology Impacts In Vivo Potency, Efficacy, and Clearance." *Pharmacological Reviews* 75 (3): 416–62. <https://doi.org/10.1124/pharmrev.121.000524>. Right: Reddien, Peter W. 2024. "The Purpose and Ubiquity of Turnover." *Cell* 187 (11): 2657–81. <https://doi.org/10.1016/j.cell.2024.04.034>. Quote: Theodosius Dobzhansky

Open models integrate protein turnover into pharmacological modeling

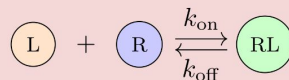
According to open models (see the comprehensive review by [Gabrielsson and Hjorth](#)), target turnover impacts *in vivo* potency, efficacy, and clearance.



B

Reversible binding

Closed pharmacological systems



$$E_{max} = B_{max}$$

$$EC_{50} \propto K_d = \frac{k_{off}}{k_{on}}$$

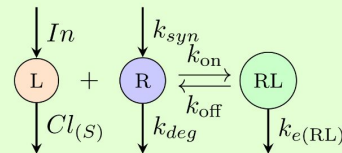
Irreversible binding



$$E_{max} = B_{max}$$

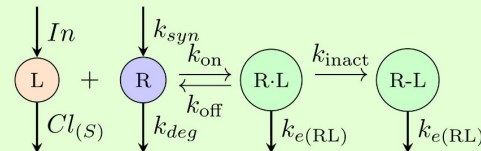
$$EC_{50} \propto \frac{k_{inact}}{k_{off}/k_{on}}$$

Open pharmacological systems



$$E_{max} = \rho \cdot \frac{k_{syn}}{k_{e(RL)}}$$

$$EC_{50} = \frac{k_{deg}}{k_{e(RL)}} \cdot \frac{k_{off} + k_{e(RL)}}{k_{on}}$$



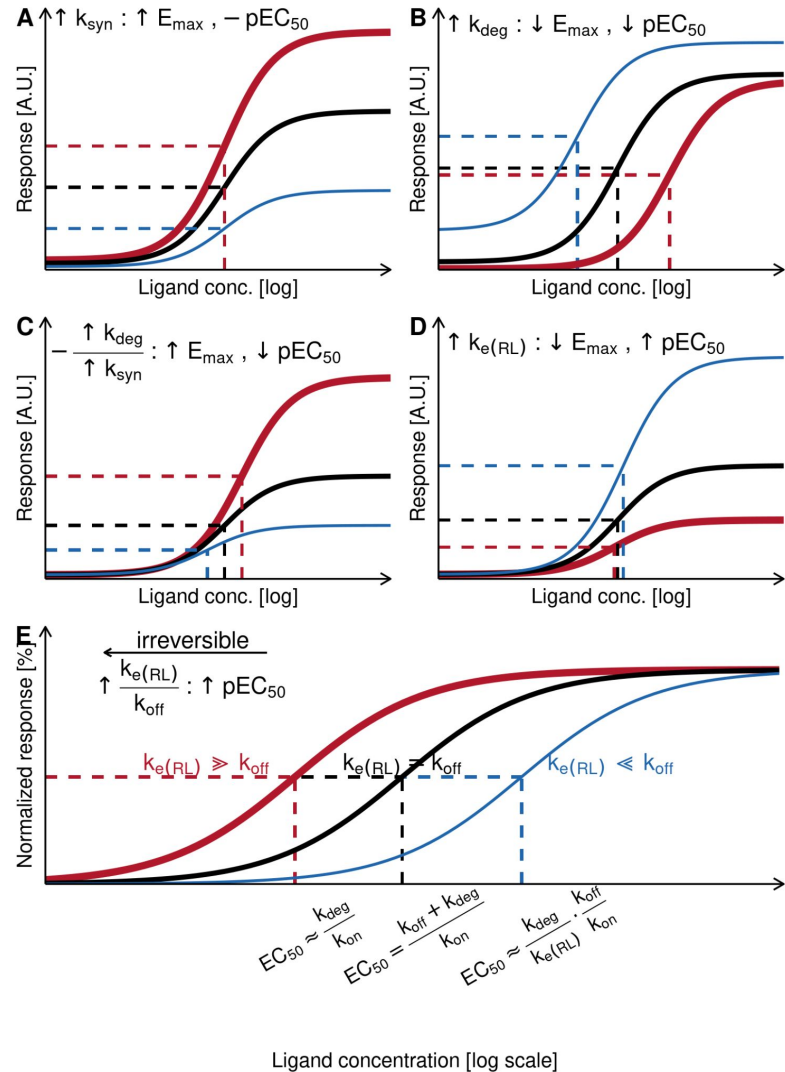
$$E_{max} = \rho \cdot \frac{k_{syn}}{k_{e(RL)}}$$

$$EC_{50} \propto \frac{k_{deg}}{k_{on}}$$

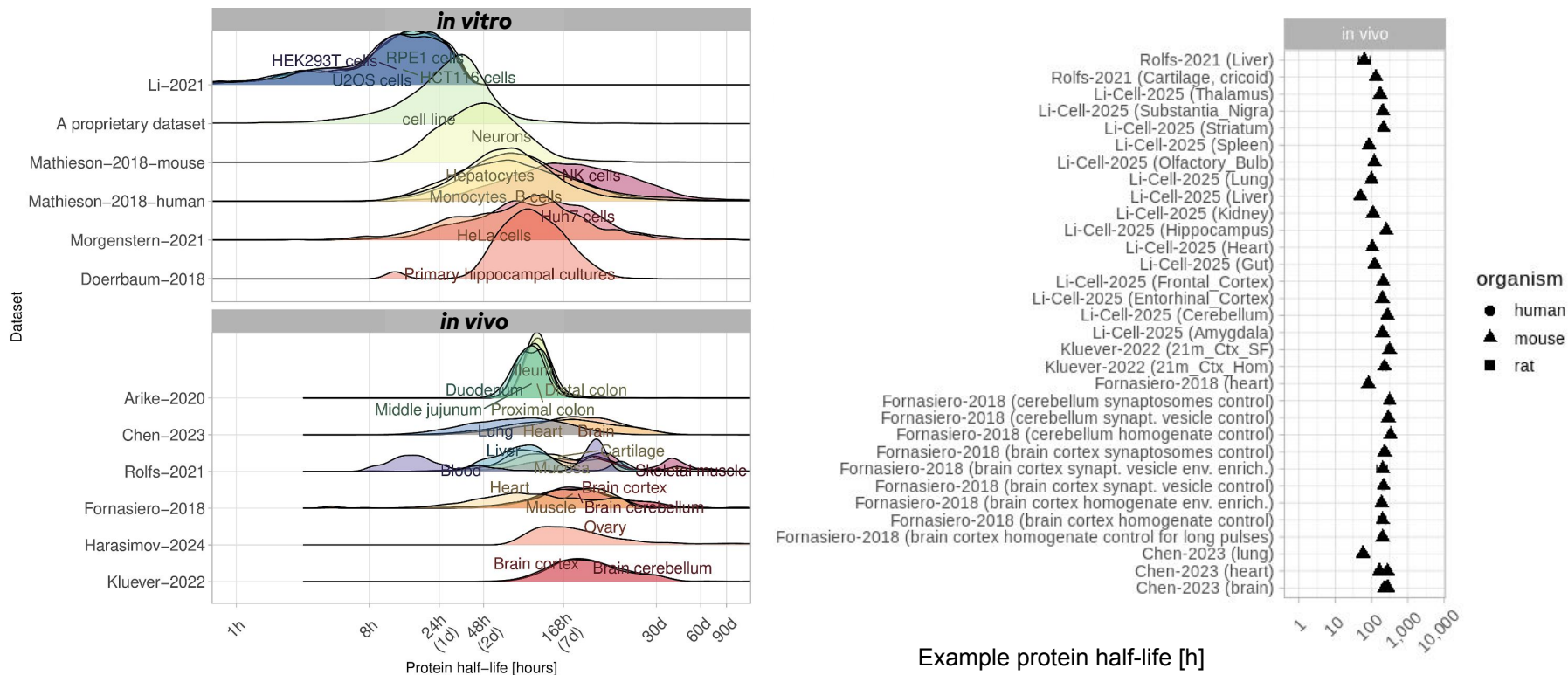
Predictions by open models

With [open-source code and data used for simulation](#)

- A. Higher target synthesis rate increases efficacy while potency remains unchanged.
- B. Higher degradation rate decreases both efficacy and potency.
- C. Keeping the steady-state abundance fixed, increasing both synthesis & degradation rates increase both efficacy and potency.
- D. Higher ligand-target complex elimination rate reduces efficacy while increases potency.
- E. Potency of covalent inhibitors is dictated by k_{deg}/k_{on} : slow turnover and fast on-rate are preferred.



We observe in general longer half-life *in vivo* than *in vitro*, with variations between cell/tissue types

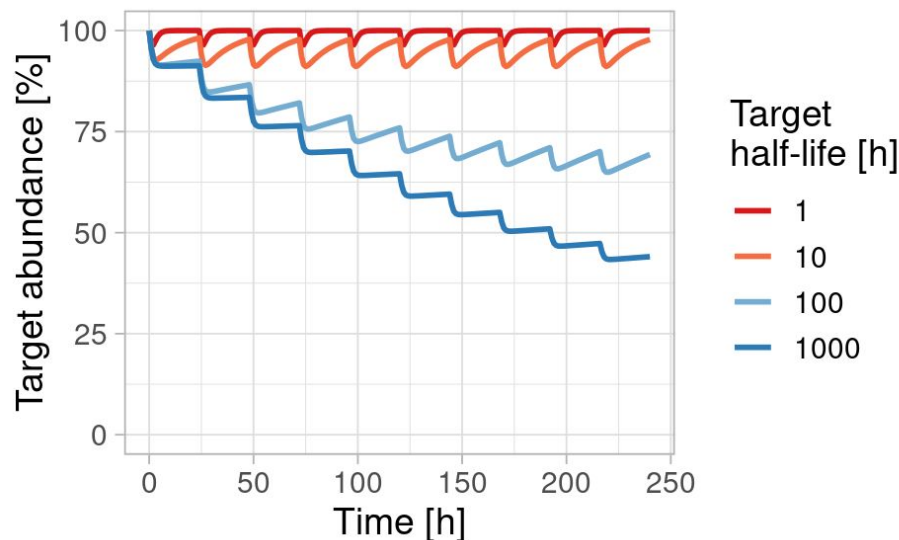


Left: Zhang, Jitao David *et al.*, Medicinal Chemistry Review 2025, [Leveraging Protein Turnover in Drug Discovery: A Practical Guide](#). Right: proprietary data.

Protein half-life is a key parameter of PK/PD models

Modelling and simulation suggests that the PD effect of target degradation by a covalent binder is sensitive to target's turnover: **Short-lived proteins are likely poor targets for covalent inhibitors.**

[Bayer colleagues](#) and [AstraZeneca and U Rice colleagues](#) reported that half-life critically impact PD profiles of targeted protein degraders and molecular glues, too.



Acknowledgement: Martin Ebeling, Kenichi Umehara, Holger Fischer, Roland Schmucki, Andrés Olivares, Uwe Grether, Marcus Bantscheff, Miro Eigenmann, Björn Bartels, Joachim Rudolph, Lothar Lindemann, Sarah Morillo Leonardo, Lizzie Gill and Arne Rufer

Further points for consideration

1. Open Models and the importance of protein turnover does not only affect covalent binders: they are applicable to reversible and irreversible drug-target interactions, as well as to all protein targeting modalities including small molecules, large molecules (for instance antibodies), and PROTACs.
 - a. By taking consideration of the dynamics of RNAs, the Open Models can be extended to RNA-targeting modalities as well as gene therapies.
2. Protein turnover does not only affect drug's potency and duration of response *in vivo*: turnover of enzymes and transporters also affects metabolism and transport.
3. Looking forward, we believe open models, together with experimental data and/or predictions based on modeling and simulation and machine learning/generative models, can help us rationally select modalities. Many experiments are on-going or being planned. We look forward to collaborations.

Conclusions

- Understanding a drug's ADME and PK profiles, PK/PD relationship, safety profile, and biomarkers is critical in translational research.
- We use mechanistic models, statistical and machine-learning models, and causal models to translate findings and make predictions between biological model systems.
- Protein turnover affects drugs' potency, efficacy, ADME properties and likely safety profiles.

Acknowledgement

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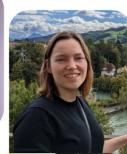


PMDA Summer School 2024 Alumni

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Inken (U Bern)



Arion (U Freiburg)



Kyriakos (U Leipzig)



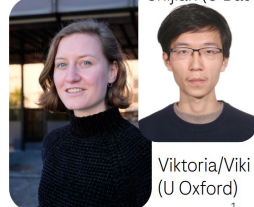
Quentin (ETH Zürich)



Federico (ETH Zürich)

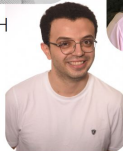


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Eve (ETH Zürich)

Ali (EPFL)



Sara (ETH Zürich)

Michael (U Lausanne)



Cecilia (EPFL)



Viktoria/Viki (U Oxford)



References

1. Figures: [Lumen Learning](#), [Exploring Nature](#), [National Geographics](#), [Platelet cells](#) (Graham Beards, CC-BY-SA 4.0), [Lymphocytes](#) (NicolasGrandjean, CC-BY-SA 3.0), [Adipocytes](#) (Public Domain), [Hepatocytes](#) (CC-BY-NC 2.0), [Neurons and Glia](#) (Public Domain), [Blood](#) (CC 3.0), [Blood Cells](#) (By A. Rad and M. Häggström. CC-BY-SA 3.0 license), [A selective JAK3 inhibitor](#) (London Lab/Weizmann institute)
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Supplementary Information

***In vitro*→*In vivo*→Human is not the only way**

POLICY FORUM

BIOMEDICAL RESEARCH

Discovery research in physiologically maintained deceased

Expanded research opportunities in deceased humans require ongoing ethical inquiry

...this approach
may...mitigate
risks traditionally
borne by
human subjects
during phase 1
clinical trials...

Douglas B. Pet^{1,2}, Brendan Parent³, Neel S. Singhal^{1,2}, Claire D. Clelland^{1,4}

Pet, Douglas B., Brendan Parent, Neel S. Singhal, and Claire D. Clelland. 2025. "Discovery Research in Physiologically Maintained Deceased." *Science* 388 (6746): 473–76. <https://doi.org/10.1126/science.adt3527>.

The Hill function as a typical PD model

The Hill function is one of the mostly useful non-linear functions to model biological systems.

In its general form, H_{max} indicates the maximal value to which the function is asymptotic, n is the shape parameter (known as the Hill's coefficient), and k is the reflection point, often abbreviated as XC_{50} ($X=I, E, C, \dots$), the half-saturation constant.

The Michaelis-Menten model is a special case of the Hill function with $n=1$.

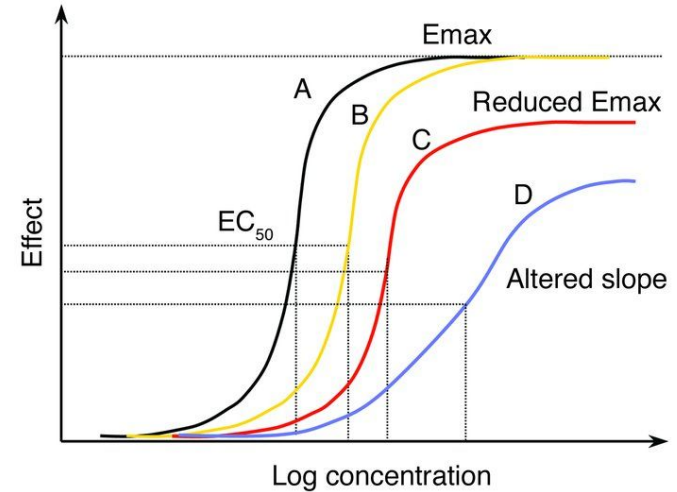
$$H = H_{max} \frac{x^n}{k^n + x^n}$$

General form of the Hill function

$$E = E_{max} \frac{[L]^n}{EC_{50}^n + [L]^n}$$

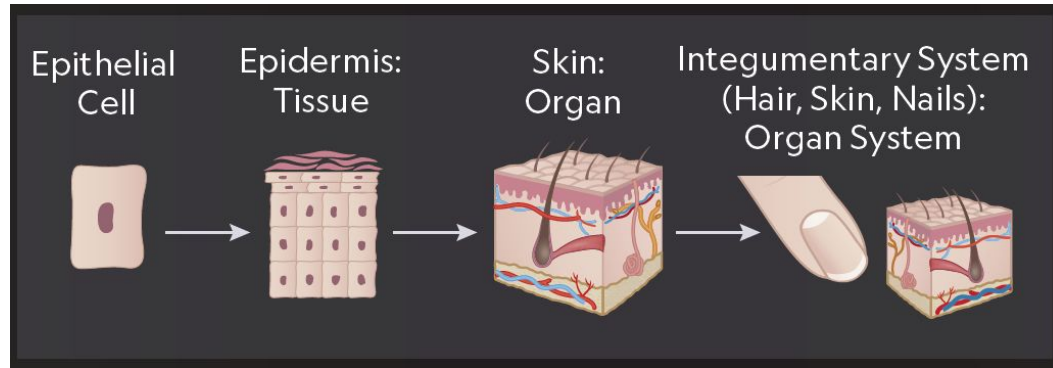
$$= E_{max} \frac{1}{1 + \left(\frac{EC_{50}}{[L]}\right)^n}$$

Modelling dose-dependent effect



White. *J Clin Invest.* 2004;113(8):1084-1092.
<https://doi.org/10.1172/JCI21682>.

Complexity Increases Through a System



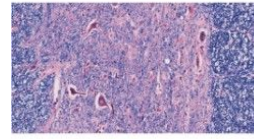
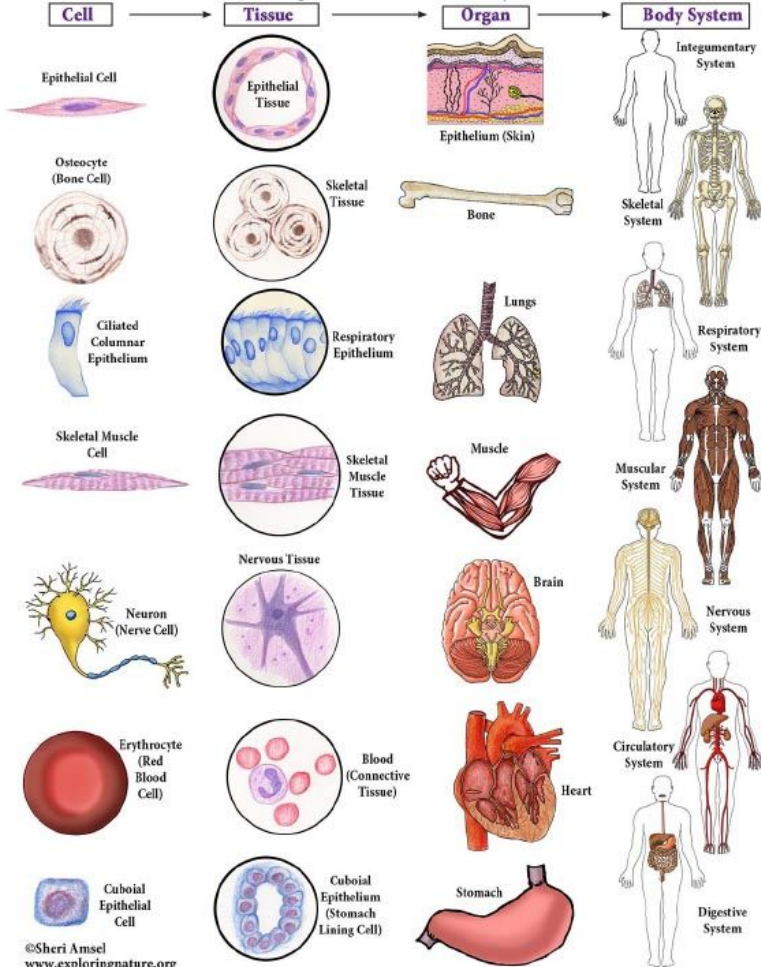
Cells: basic building blocks, variable morphologies and functions

Tissues: groups of specialized cells that communicate and collaborate

Organ: group of tissues to perform specific functions

Organ systems: group of organs and tissues

Organization of the Body

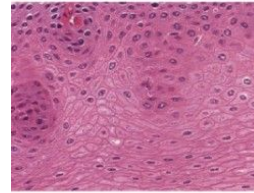


Nervous tissue

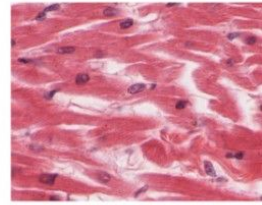
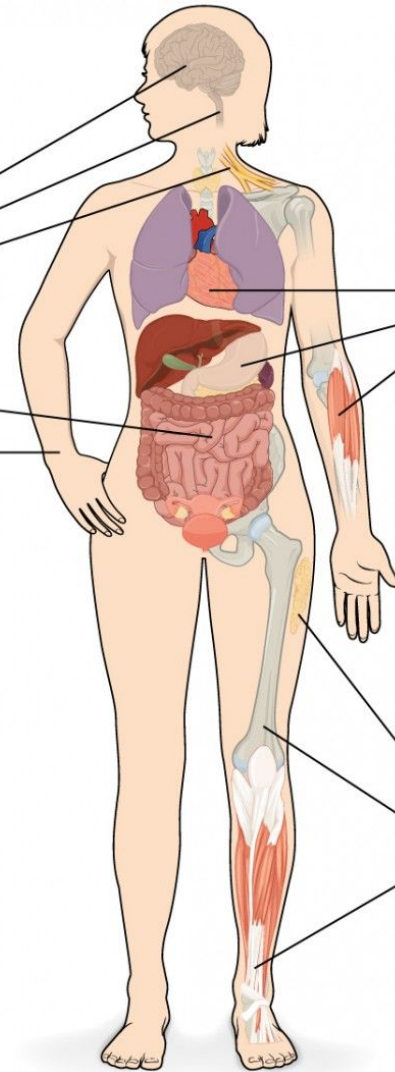
Brain
Spinal cord
Nerves

Epithelial tissue

Lining of GI tract organs
and other hollow organs
Skin surface (epidermis)

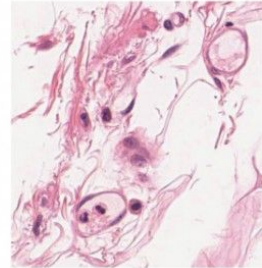


Four major tissue types



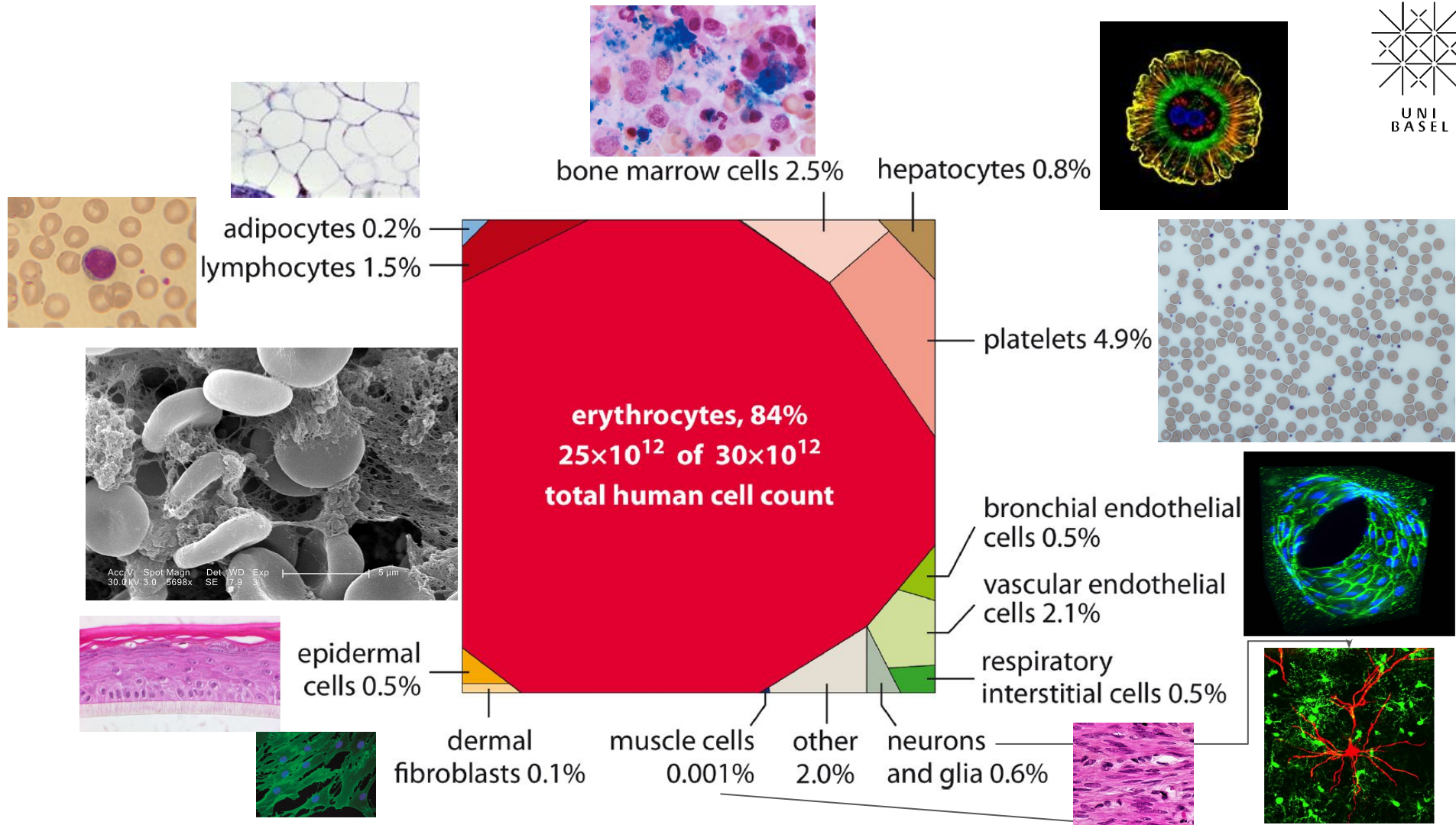
Muscle tissue

Cardiac muscle
Smooth muscle
Skeletal muscle

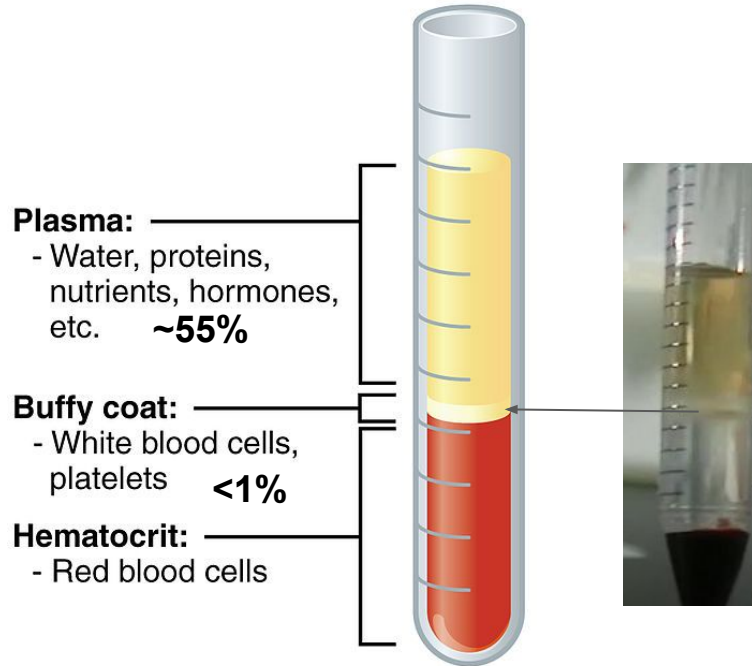


Connective tissue

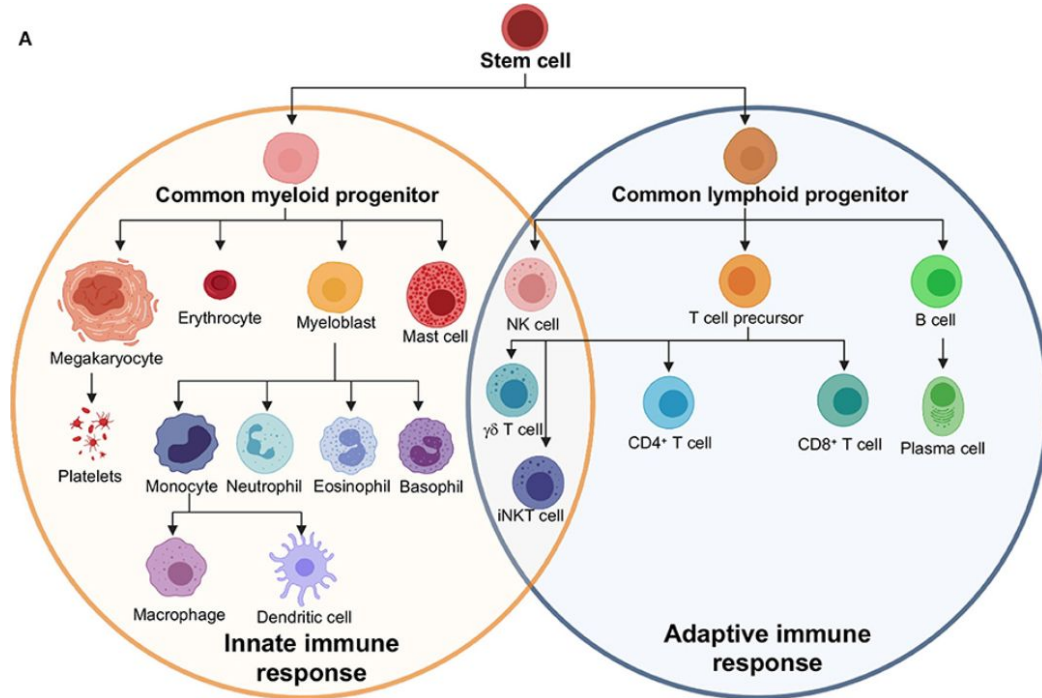
Fat and other soft padding tissue
Bone
Tendon



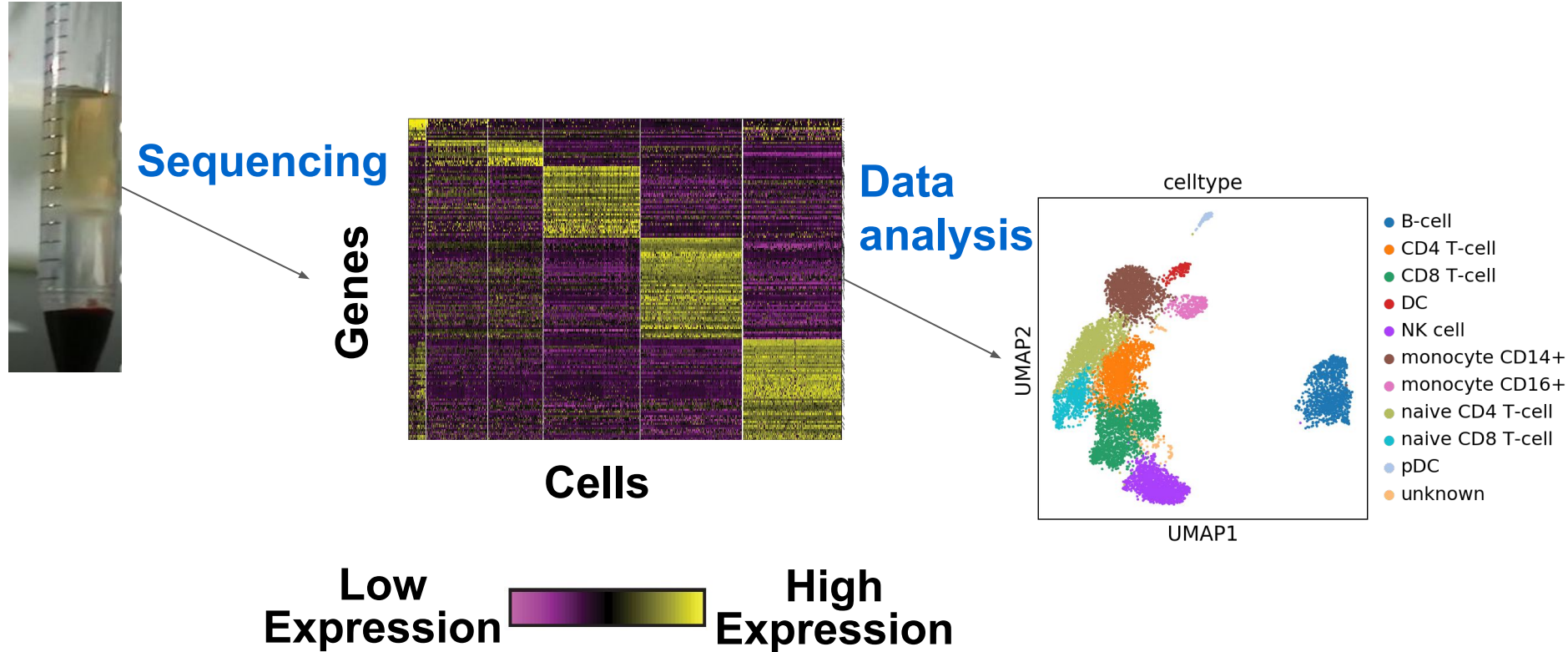
What's in a drop of blood? Ask a doctor or a biologist!



Normal Blood:
♀ 37%–47% hematocrit
♂ 42%–52% hematocrit



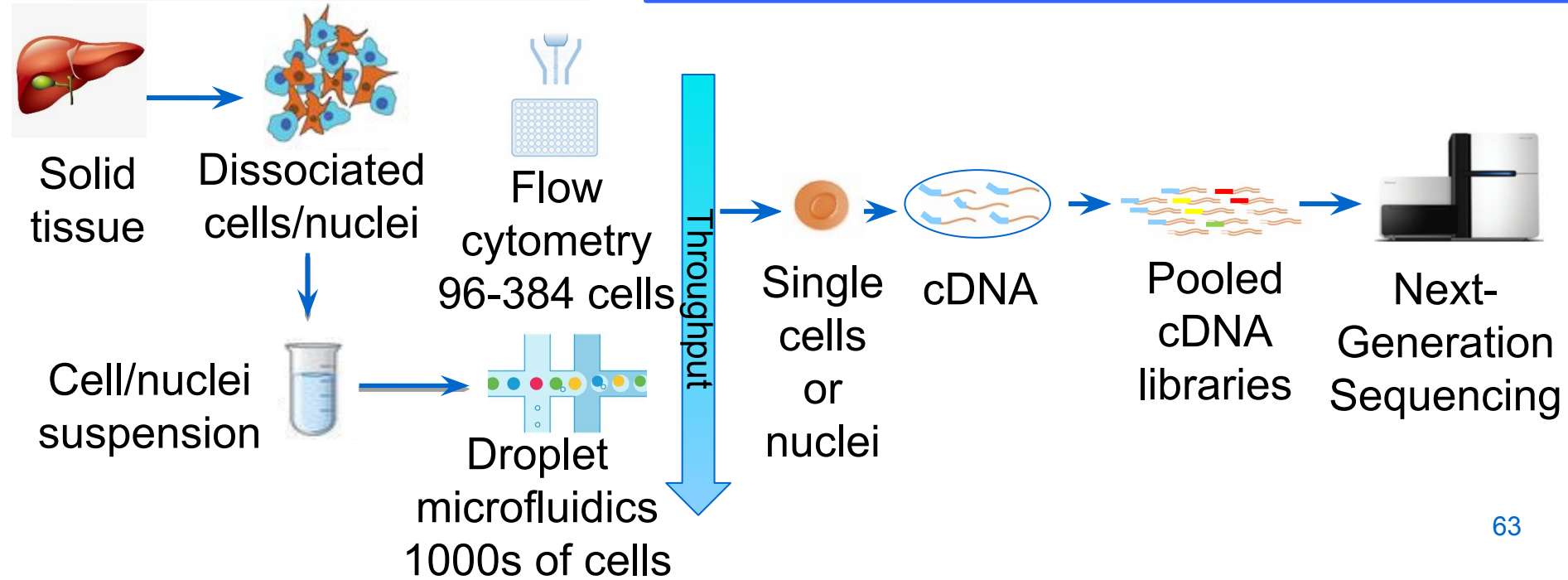
What's in a drop of blood? Count the genes!

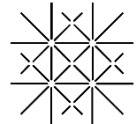


Single-cell sequencing (scSeq) workflow

Tissue dissociation

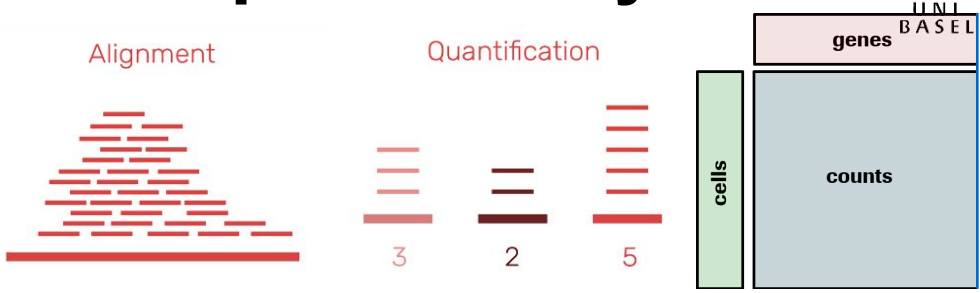
Single cell capture and transcriptome sequencing



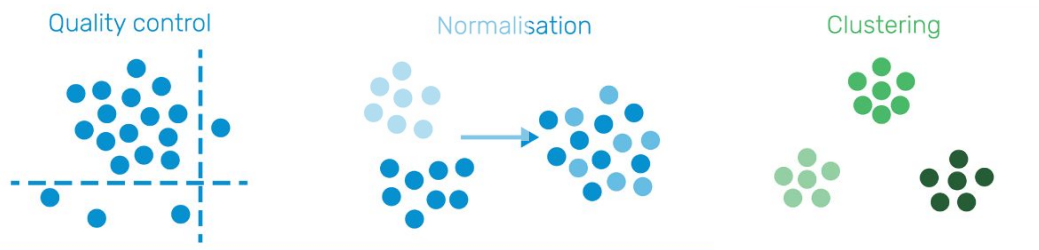


A linearized workflow of scSeq data analysis

From short reads to gene-cell matrix



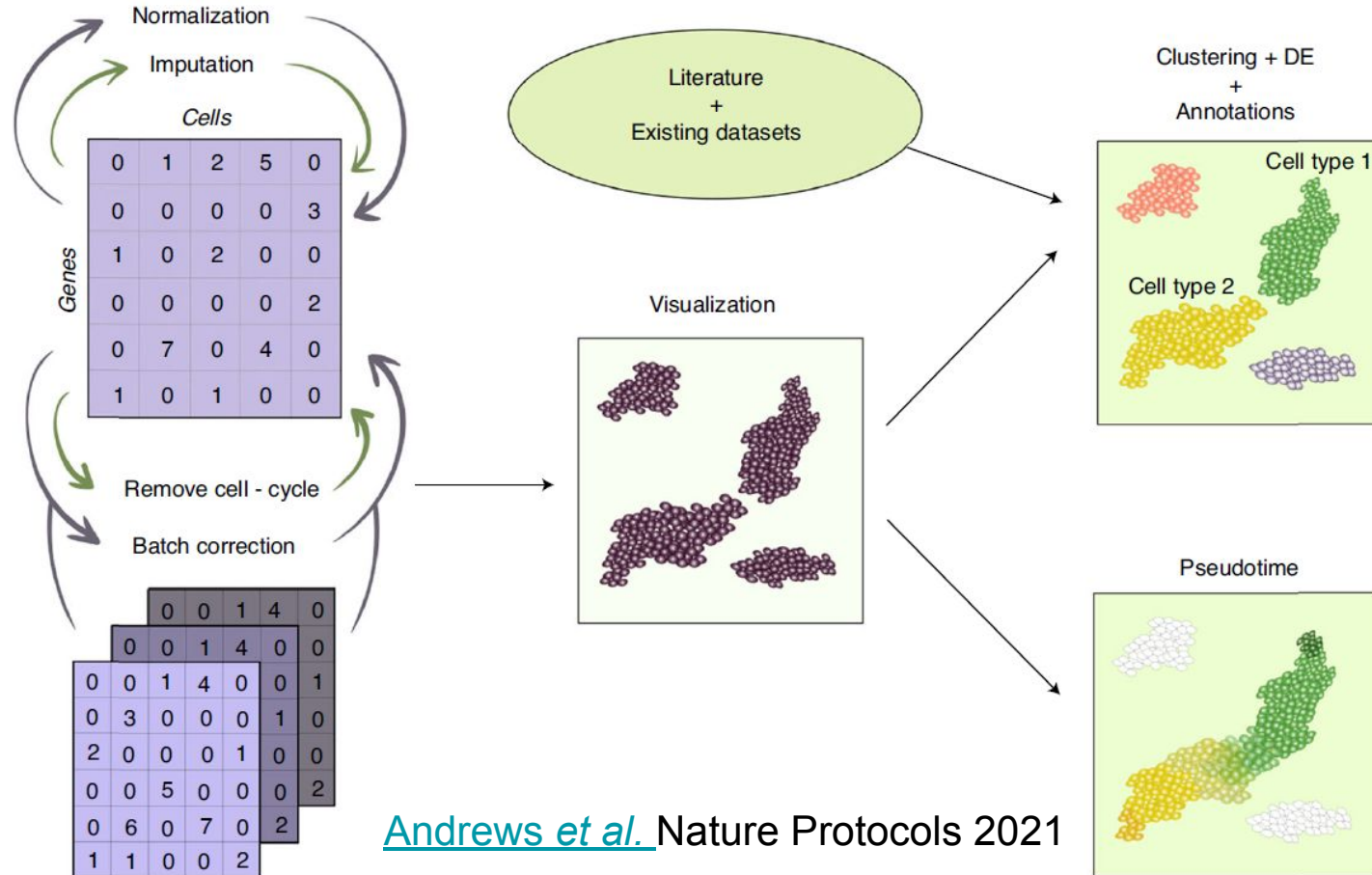
QC, filtering & normalization, dimensionality reduction, and clustering



Downstream analysis

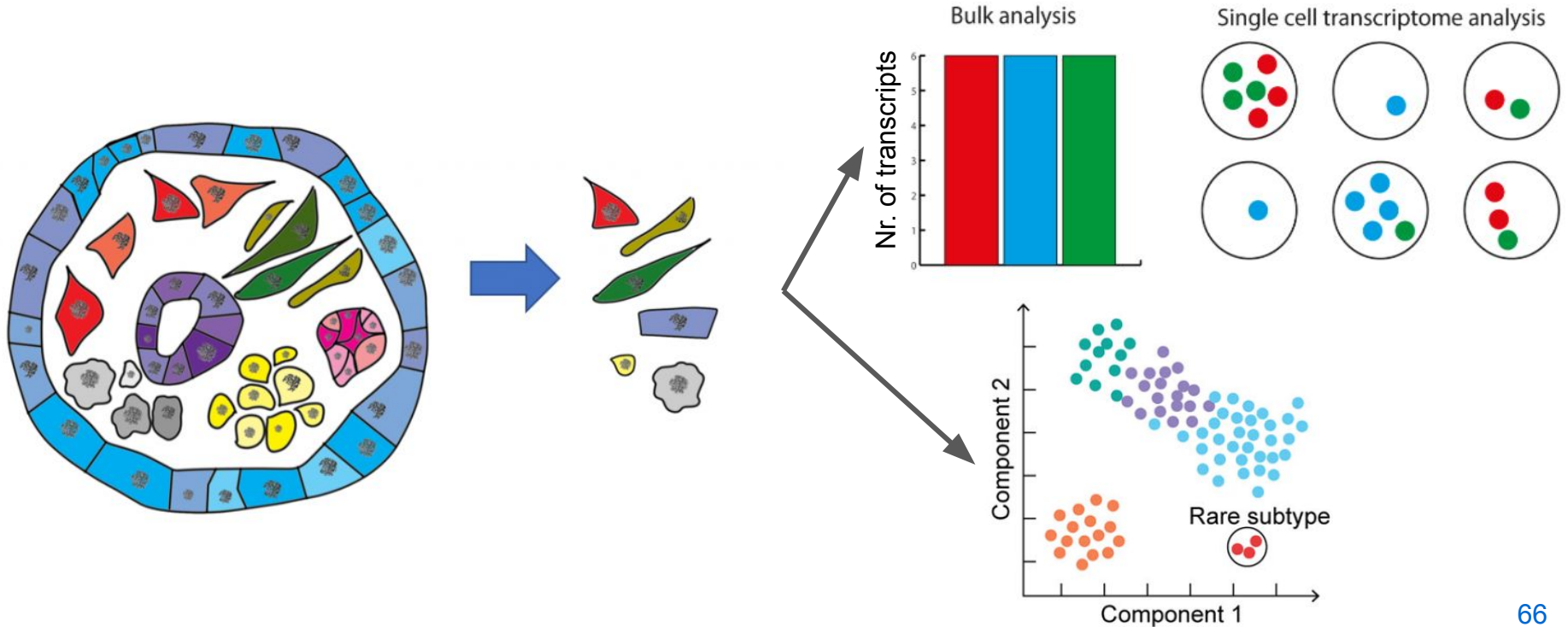


Overview of the computational workflow



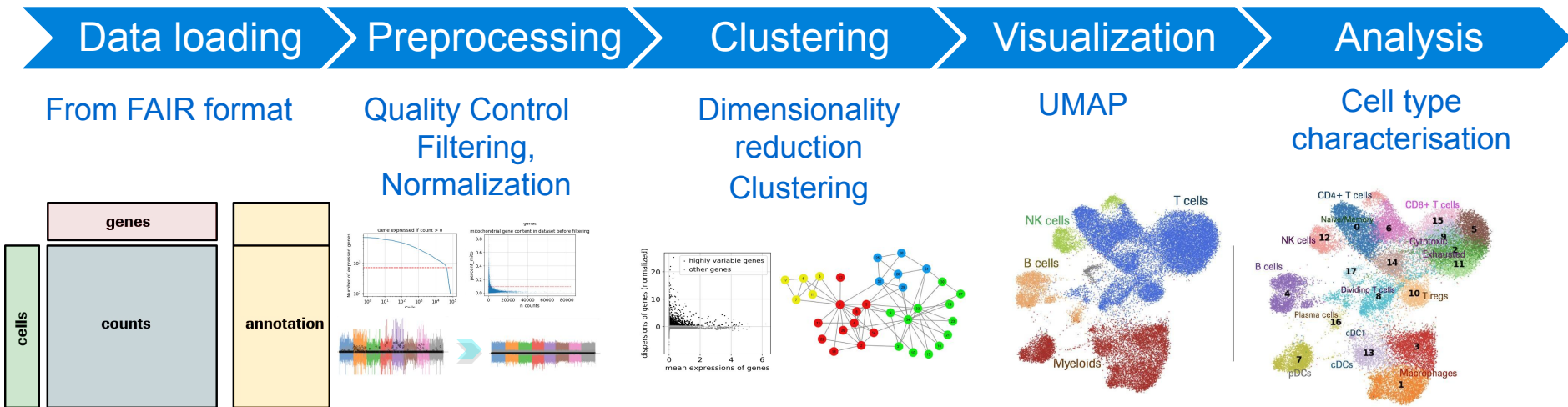
[Andrews et al.](#) Nature Protocols 2021

Single-cell biology benefits both disease understanding and drug discovery



BESCA: An open-source Python package for single-cell gene expression analysis

An automatized standard workflow

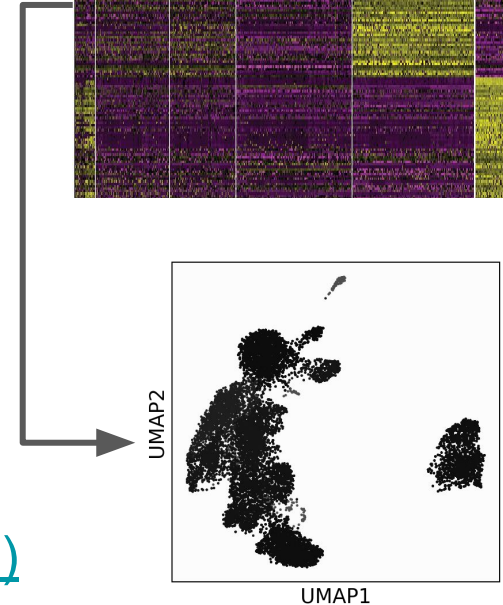
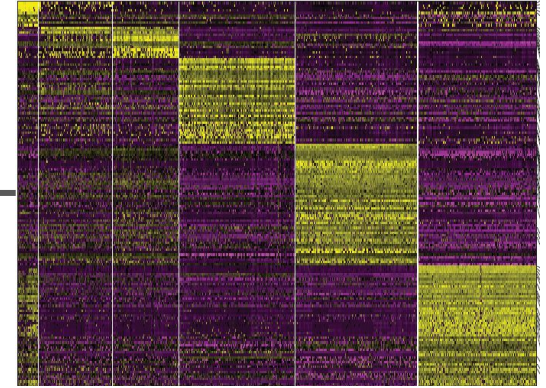
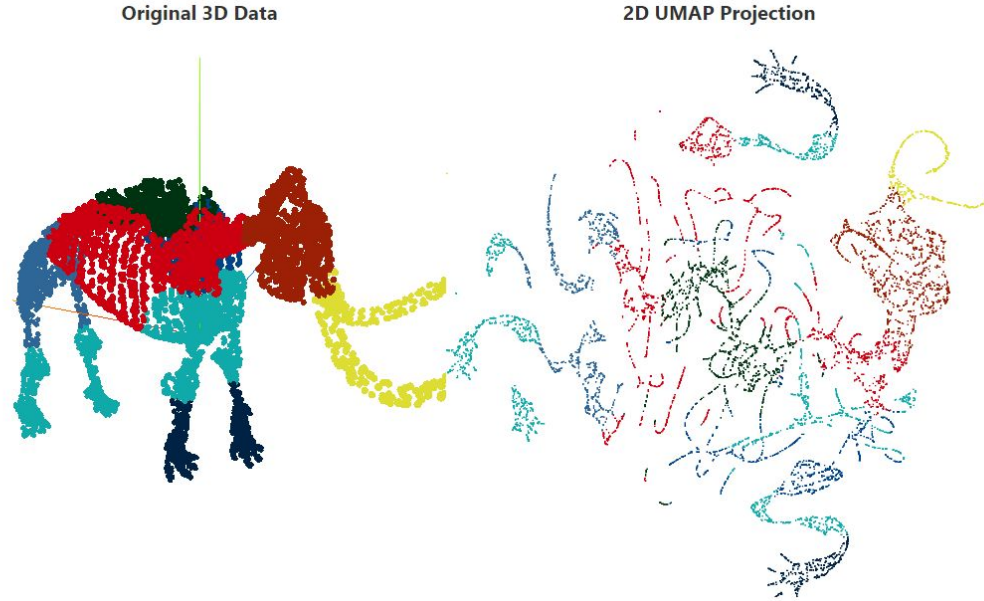


How to represent voxels with pixels?



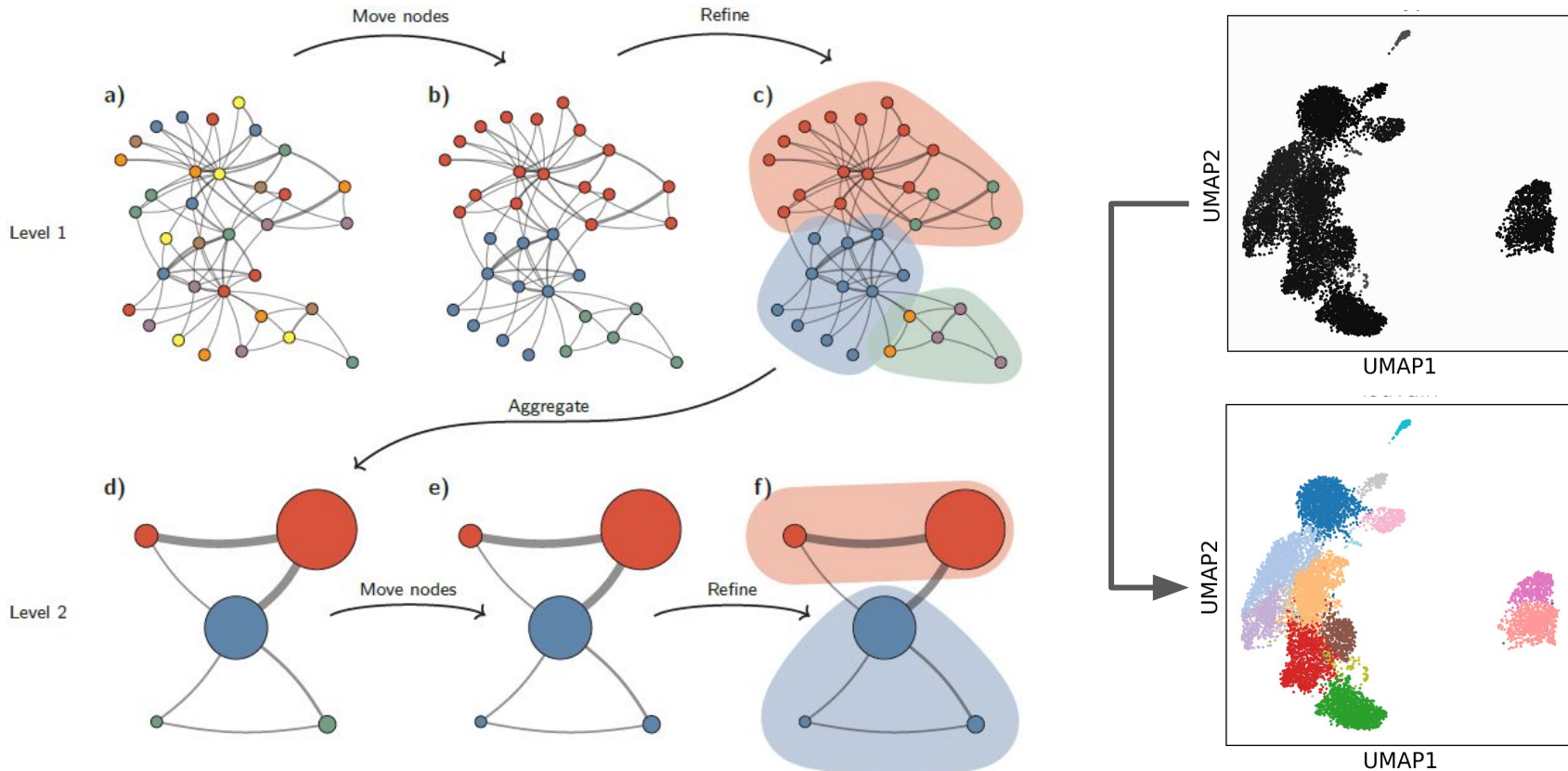
The elephant bull *Tusker* (1992-2023) at Zolli Basel plays with a tree trunk on a post (2022)

Uniform Manifold Approximation and Projection (UMAP) for dimension reduction



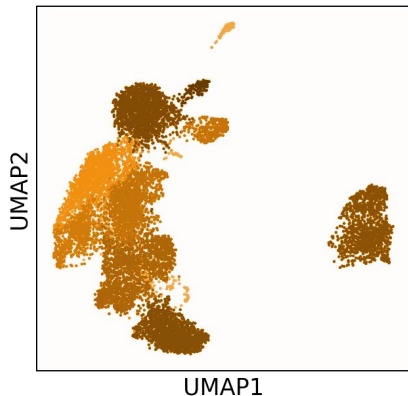
[Understanding UMAP by A. Coenen and A. Pearce](#)
[UMAP by Leland McInnes on SciPy 2018 \(YouTube\)](#)

The Leiden Algorithm for Community Detection

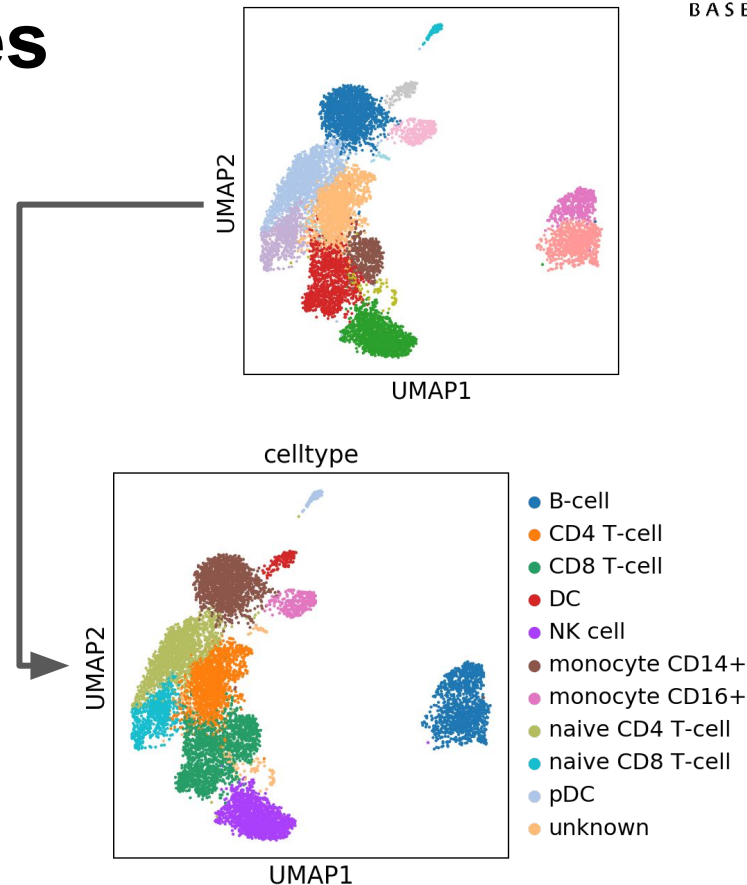


Biological knowledge and visual inspection is used to annotate cell types

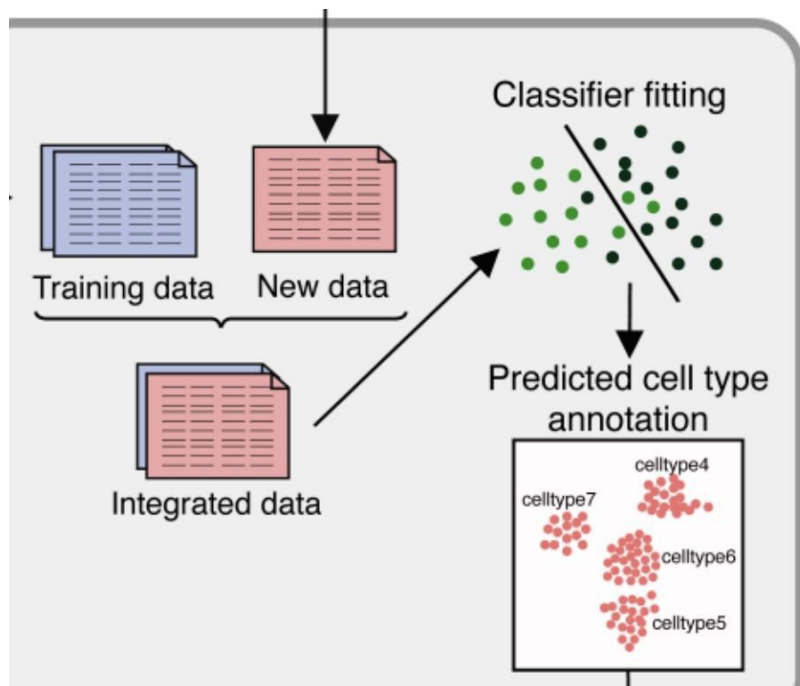
Heatmap
of gene X



lymphocyte	PTPRC								
myeloid	S100A8	S100A9	CST3						
Bcell	CD19	CD79A	MS4A1						
Tcells	CD3E	CD3G	CD3D						
CD4	CD4								
CD8	CD8A	CD8B							
NKcell	NKG7	GNLY	NCAM1						
monocyte	CST3	CSF1R	ITGAM	CD14	FCGR3A	FCGR3B			
macrophage	CD14	IL1B	LYZ	CD163	ITGAX	CD68	CSF1R	FCGR3A	



Cell type annotation with machine learning

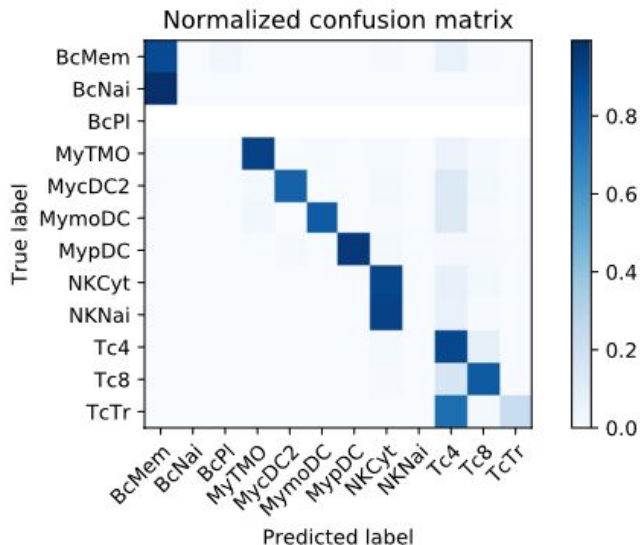
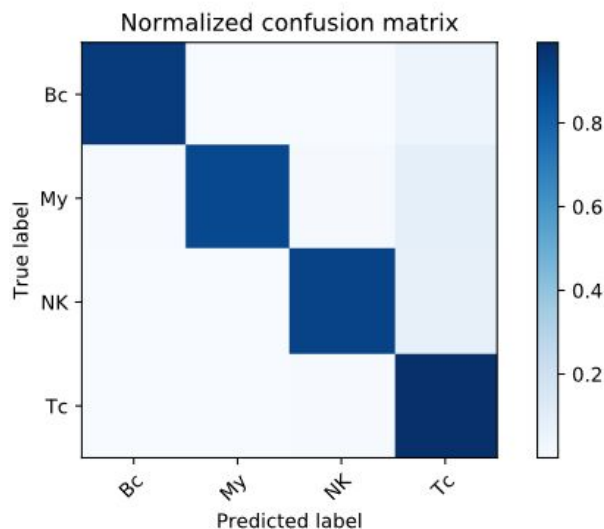
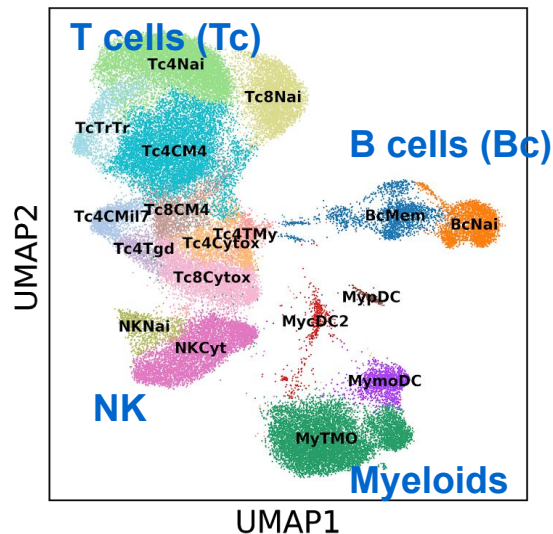


	Pancreas					CellBench		TM	Allen Mouse Brain			PBMC	
SVM _{rejection}	0.99	0.99	0.98	1	0.98	1	1	0.99	1	1	0.98	0.99	0.92
scPred	1	0.98	0.98	1	0.95	1	1	0.97	1	1	0.69	0.96	
SVM	0.98	0.98	0.97	1	0.99	1	1	0.98	1	0.99	0.89	0.95	0.7
singleCellNet	0.97	0.96	0.97	0.99	1	1	1	0.94	1	0.99	0.87	0.88	0.74
ACTINN	0.97	0.98	0.97	1	0.95	1	1	0.97	1	0.99	0.86	0.88	0.74
CaSTLe	0.93	0.94	0.96	0.98	0.96	1	0.99	0.94	1	0.99	0.79	0.84	0.79
scmapcell	0.98	0.98	0.97	1	0.73	1	1	0.98	1	1	0.91	0.73	0.64
LDA	0.94	0.97	0.96	0.99	0.89	1	1	0.95	1	0.99	0.88	0.63	0.66
scmapcluster	0.99	0.95	0.97	1	1	1	1	0.87	1	0.98	0.88	0.73	0.44
RF	0.94	0.94	0.96	0.98	0.85	1	1	0.91	1	0.99	0.73	0.81	0.66
SingleR	0.96	0.97	0.95	0.97	0.99	1	1	0.88	1	0.97	0.86	0.66	0.32
LAmbDA	0.92	0.8	0.95	0.96	0.97	1	1	0.62	1	0.99	0.84		0.4
NMC	0.92	0.91	0.84	0.93	0.99	0.92	0.9	0.69	0.99	0.97	0.81	0.71	0.55
CHETAH	0.91	0.94	0.96	0.97	0.96	1	1	0.83	1	0.96	0.81	0.65	0.11
scVI	0.98	0.56	0.97	0.99	1	1	1	0	1	0.97	0	0.97	0.64
scID	0.75	0.59	0.95	0.85	0.8	1	1	0.42	1	0.95	0.63	0.61	0.42
Cell_BLAST	0.11	0.89	0.79	0.08	0.63	1	0.99	0.97	1	0.99	0.76	0.91	0.74
kNN	0.91	0.95	0.95	0.85	0.03	1	0.98	0.92	1	0.64	0.13	0.45	0.54
SCINA												1*	1*
DigitalCellSorter												0.99*	0.78*
Garnett _{Cy}												0.94*	0.6*
Garnett _{pretrained}												0.98*	0.54*
Moana												0.93*	0.5*
Garnett _{DE}												0.65	0.37
SCINA _{DE}												0.38	0.47
DigitalCellSorter _{DE}												0	0

Median F1-score

0 0.25 0.5 0.75 1

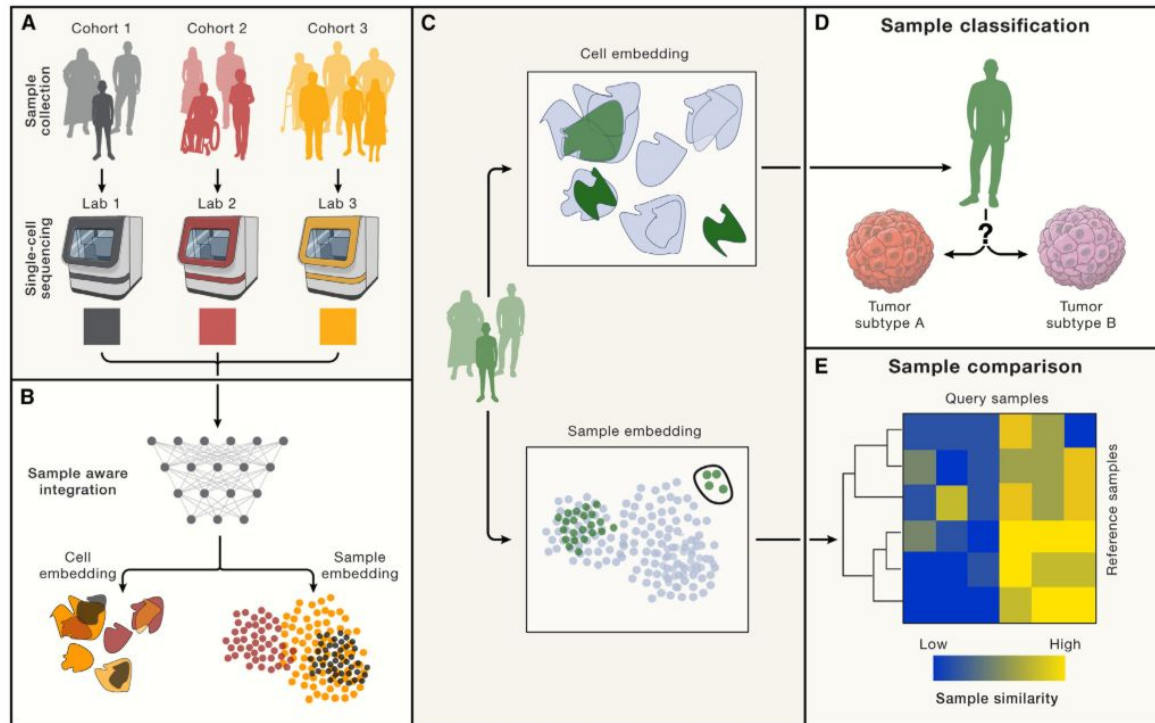
A PBMC example of cell type annotation



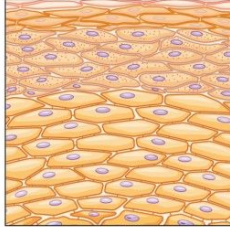
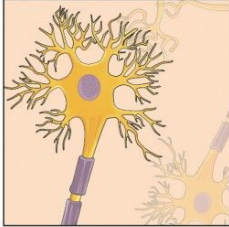
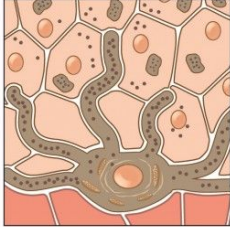
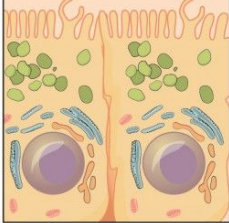
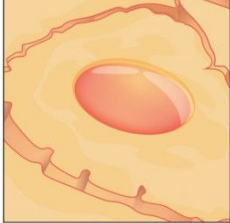
- Broad level cell types, including B cells (Bc), Myeloid (My), NK cells (NK) and T cells (Tc), are successfully predicted.
- Missing and highly similar cell types cause challenges with increased granularity. Essential: reference data quality and knowledge of cell types.

Reference mapping at population scale

- The data platform Chan Zuckerberg CELL by GENE Discover ([CZ CELLxGENE](#)) provides data of 85 million cells as of April 2024 to be explored online.
- Much research and development now devotes to mapping data from different labs to reference datasets in order to annotate cells and samples in a (semi-)automated fashion

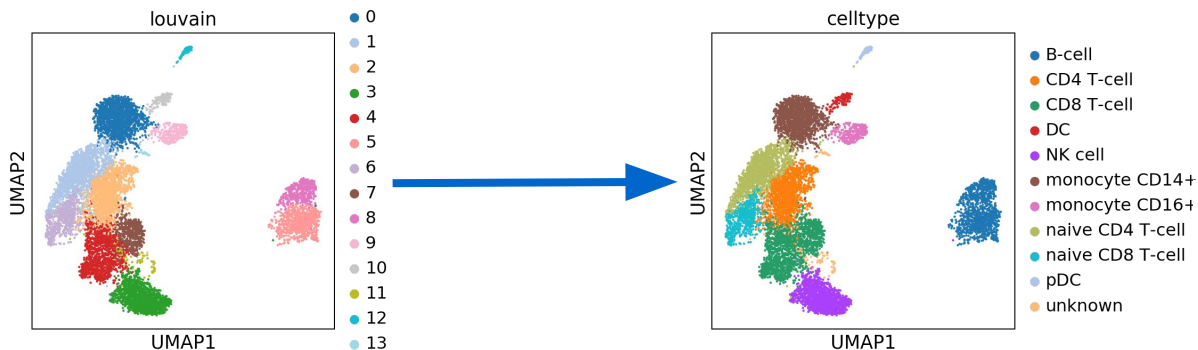


Embryonic origins of tissues

Germ Layer	Gives rise to:
Ectoderm	Epidermis, glands on skin, some cranial bones, pituitary and adrenal medulla, the nervous system, the mouth between cheek and gums, the anus  Skin cells  Neurons  Pigment cell
Mesoderm	Connective tissues proper, bone, cartilage, blood, endothelium of blood vessels, muscle, synovial membranes, serous membranes lining body cavities, kidneys, lining of gonads  Cardiac muscle  Skeletal muscle  Tubule cell of kidney  Red blood cells  Smooth muscle
Endoderm	Lining of airways and digestive system except the mouth and distal part of digestive system (rectum and anal canal); glands (digestive glands, endocrine glands, adrenal cortex)  Lung cell  Thyroid cell  Pancreatic cell

An intern project: Cell type annotation

From unsupervised clustering and cluster based annotation



Luis Wyss
RAAN intern 2019

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5	Label
Training Cell 1	10	50	0	12	4	Celltype A
Training Cell 2	8	45	78	3	23	Celltype B
Training Cell 3	14	55	78	65	55	Celltype B
Training Cell 4	78	12	13	9	58	Celltype A
Training Cell 5	45	23	65	98	11	Celltype C

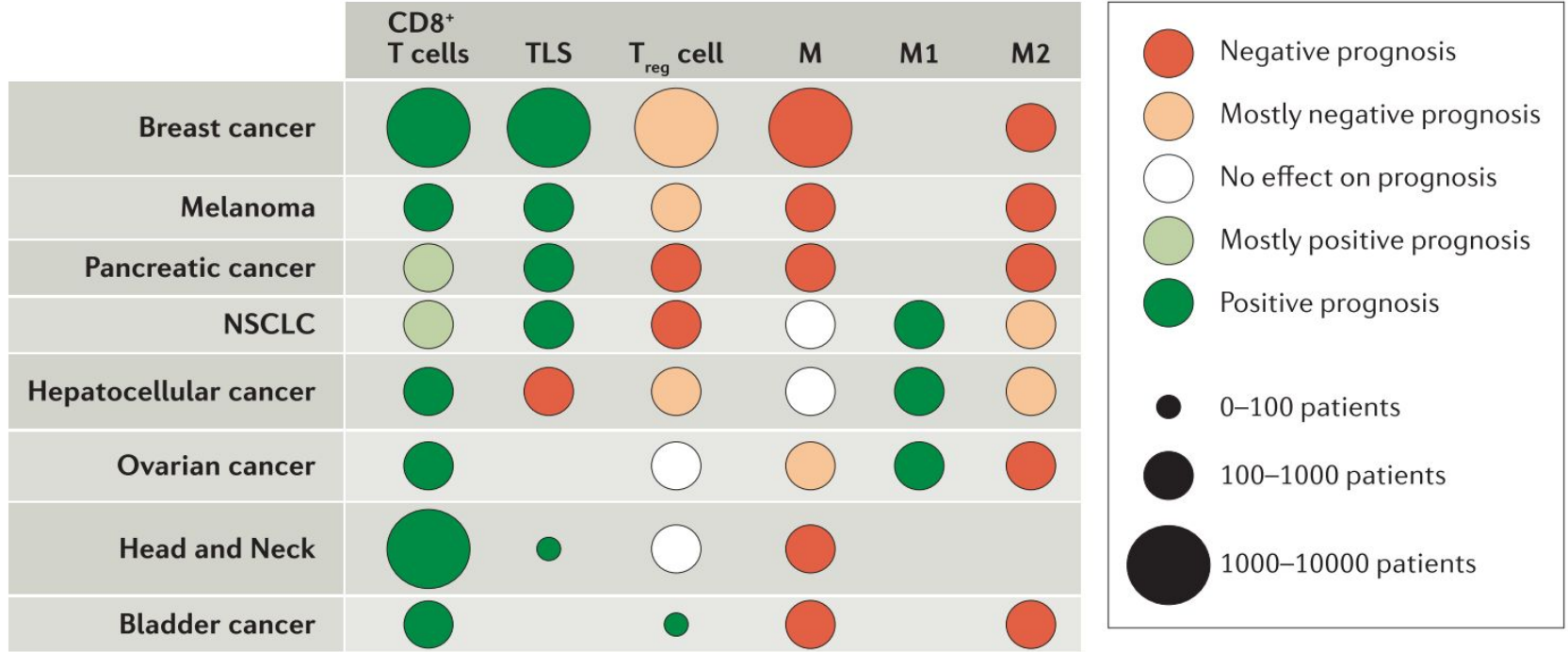
To supervised annotation at single-cell level:

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5
Cell 1	45	45	8	56	3
Cell 2	65	120	78	45	12
Cell 3	79	12	34	65	88
Cell 4	7	59	32	47	62

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5	Prediction
Cell 1	45	45	8	56	3	Celltype A
Cell 2	65	120	78	45	12	Celltype B
Cell 3	79	12	34	65	88	Celltype C
Cell 4	7	59	32	47	62	Celltype B

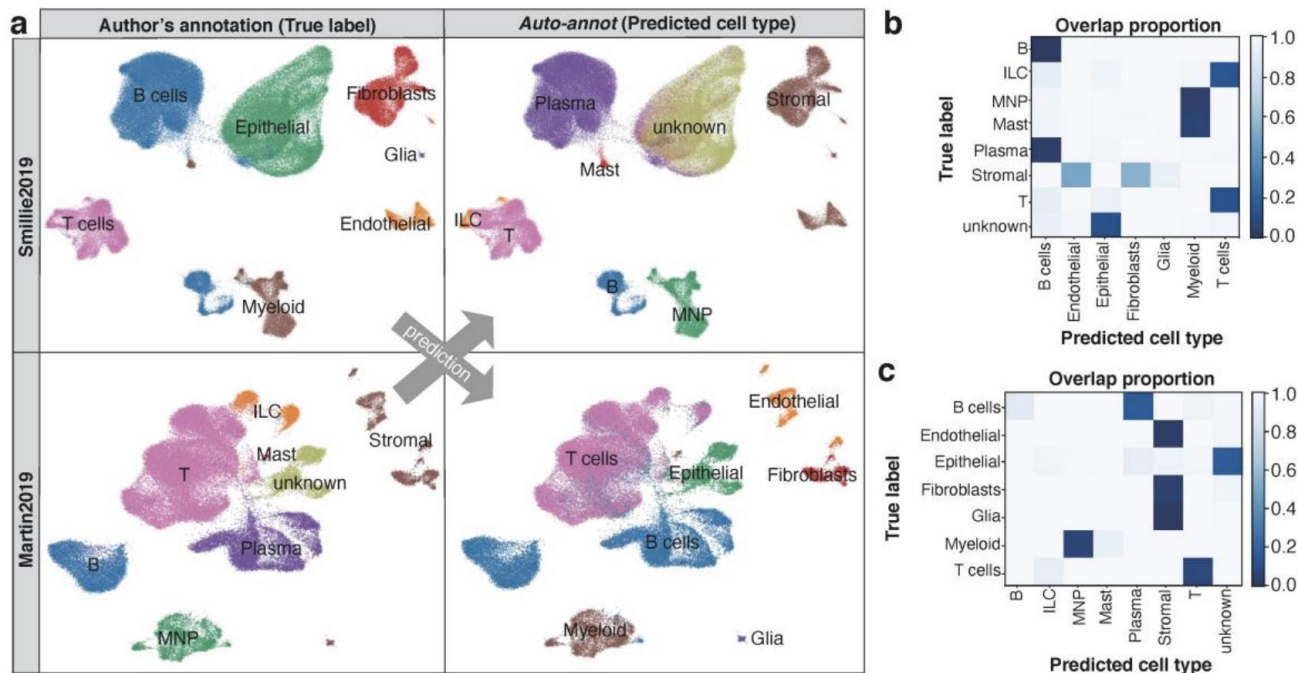
Advantages: (1) automation, (2) annotation independent from clustering, and (3) we can estimate the confidence of prediction

Abundance of immune cells in tumor microenvironments affect outcome



TLS: tertiary lymphoid structures; T_{reg}: regulatory T cells; M: macrophages; M1/M2: subtypes of macrophages

An example of Inflammatory Bowel Disease (IBD)



We observed Inconsistent cell type nomenclature across studies.
Machine learning allows us compare and integrate multiple studies.

We are living ecosystems

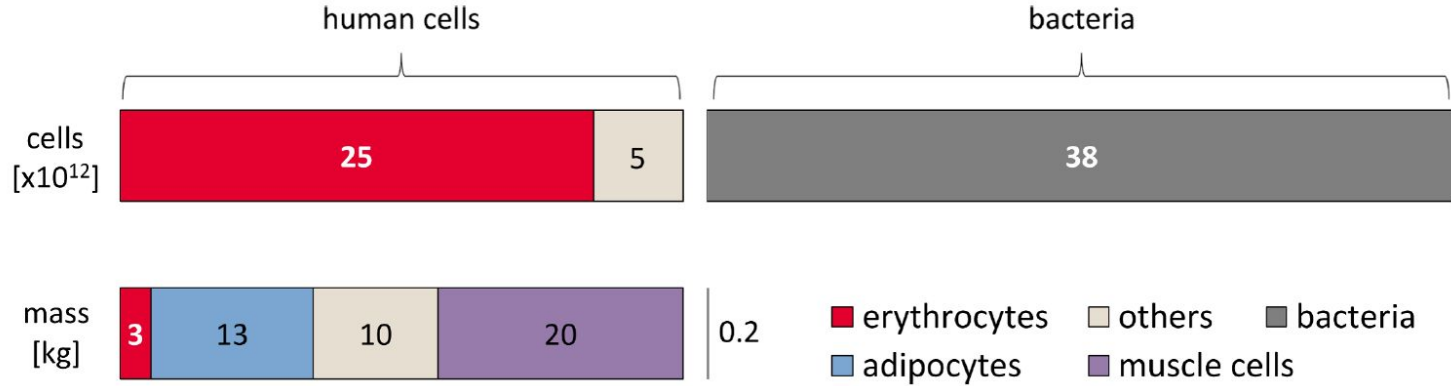
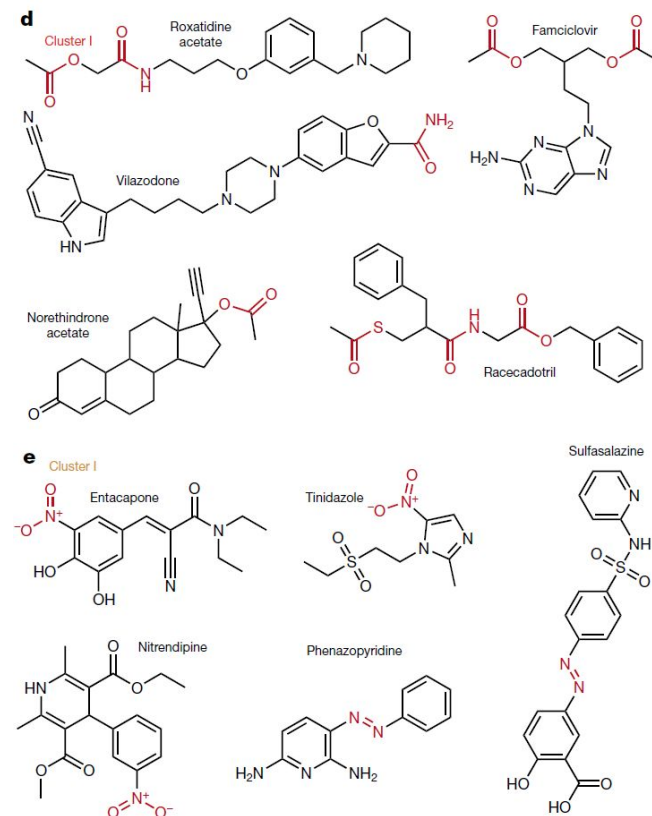
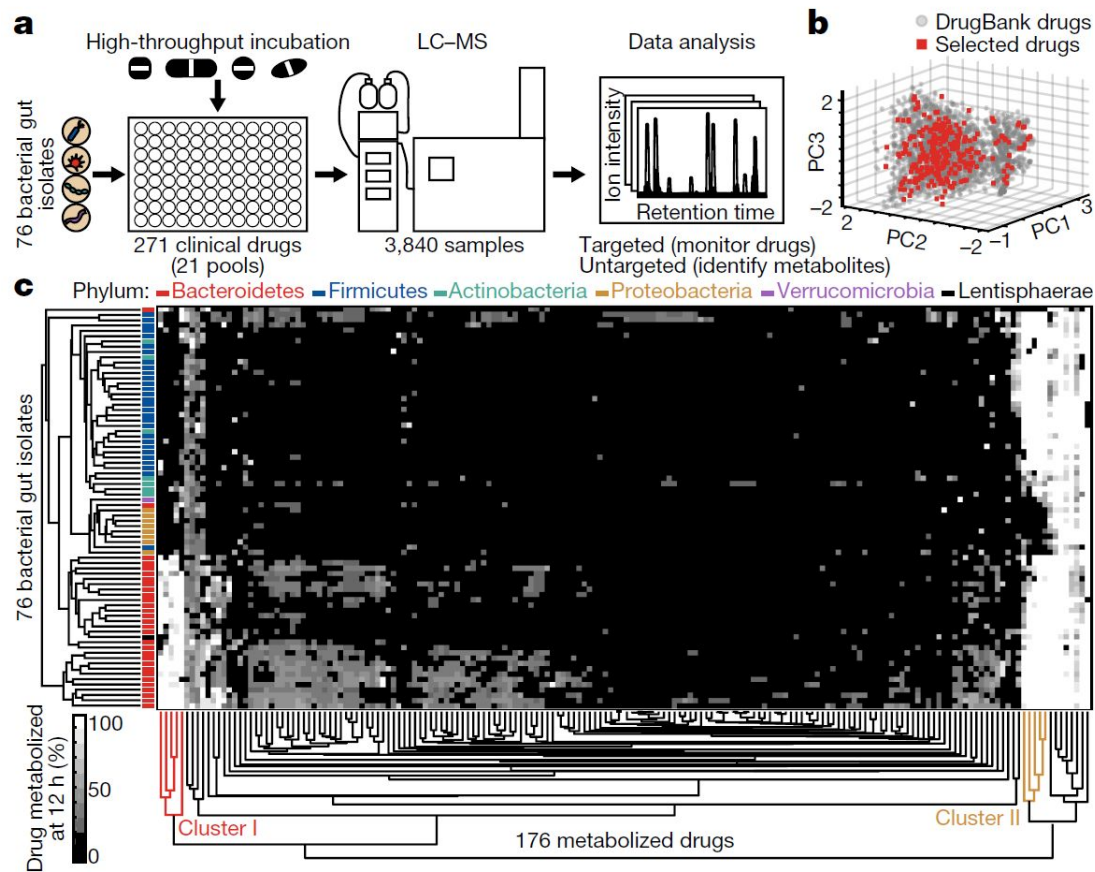


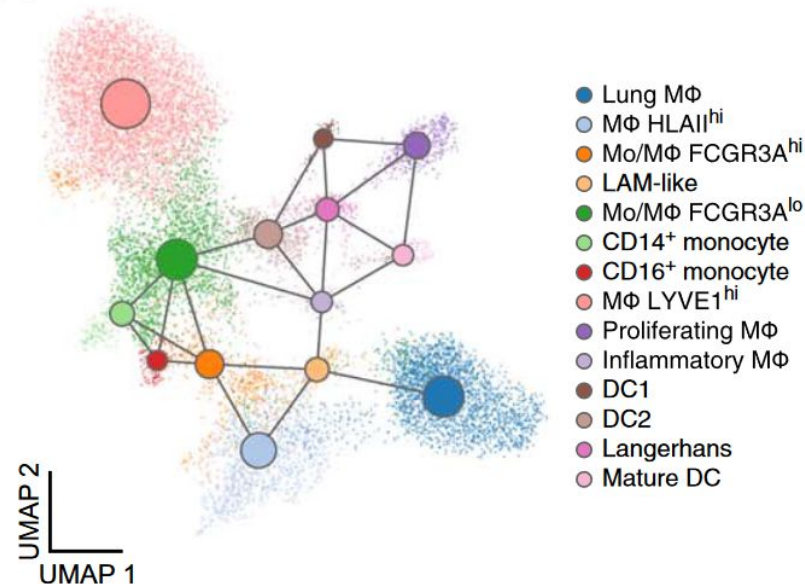
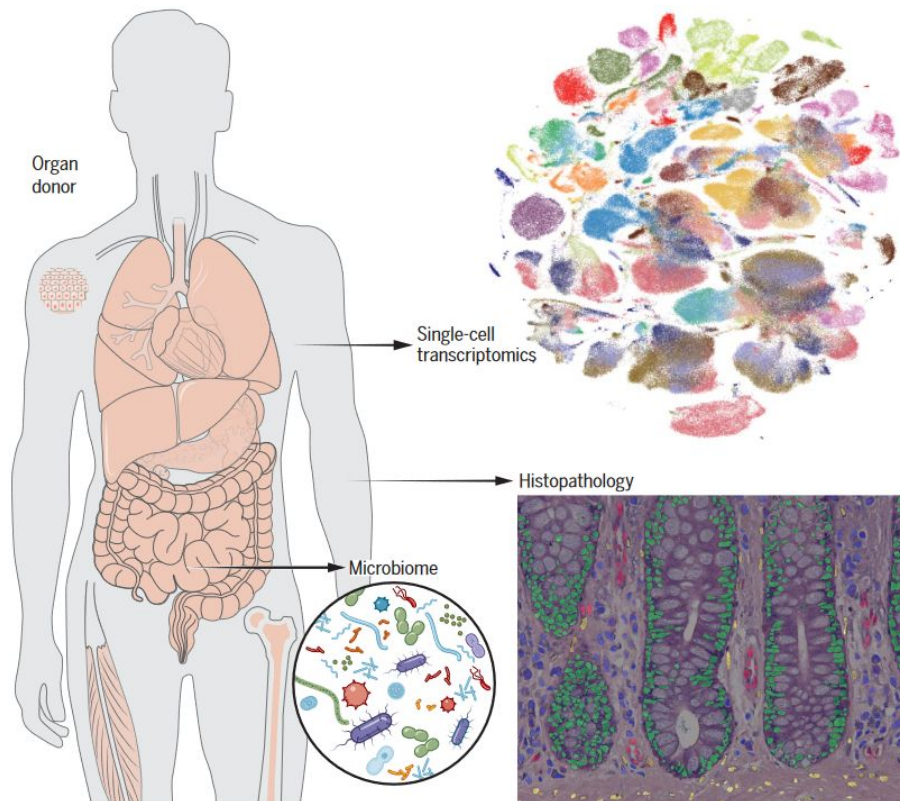
Table 3. B/H ratio for different population. See Table B in [S1 Appendix](#) for full references.

population segment	body weight [kg]	age [y]	blood volume [L]	RBC count [$10^{12}/L$]	colon content [g]	bac. conc. [$10^{11}/g$ wet] ⁽¹⁾	total human cells [10^{12}] ⁽²⁾	total bacteria [10^{12}]	B:H
ref. man	70	20–30	4.9	5.0	420	0.92	30	38	1.3
ref. woman	63		3.9	4.5	480	0.92	21	44	2.2
young infant	4.4	4 weeks	0.4	3.8	48	0.92	1.9	4.4	2.3
infant	9.6	1	0.8	4.5	80	0.92	4	7	1.7
elder	70	66	3.8 ⁽³⁾	4.8	420	0.92	22	38	1.8
obese	140		6.7	5.0 ⁽⁴⁾	610 ⁽⁵⁾	0.92	40	56	1.4

Gut microbiome can metabolize drugs differently



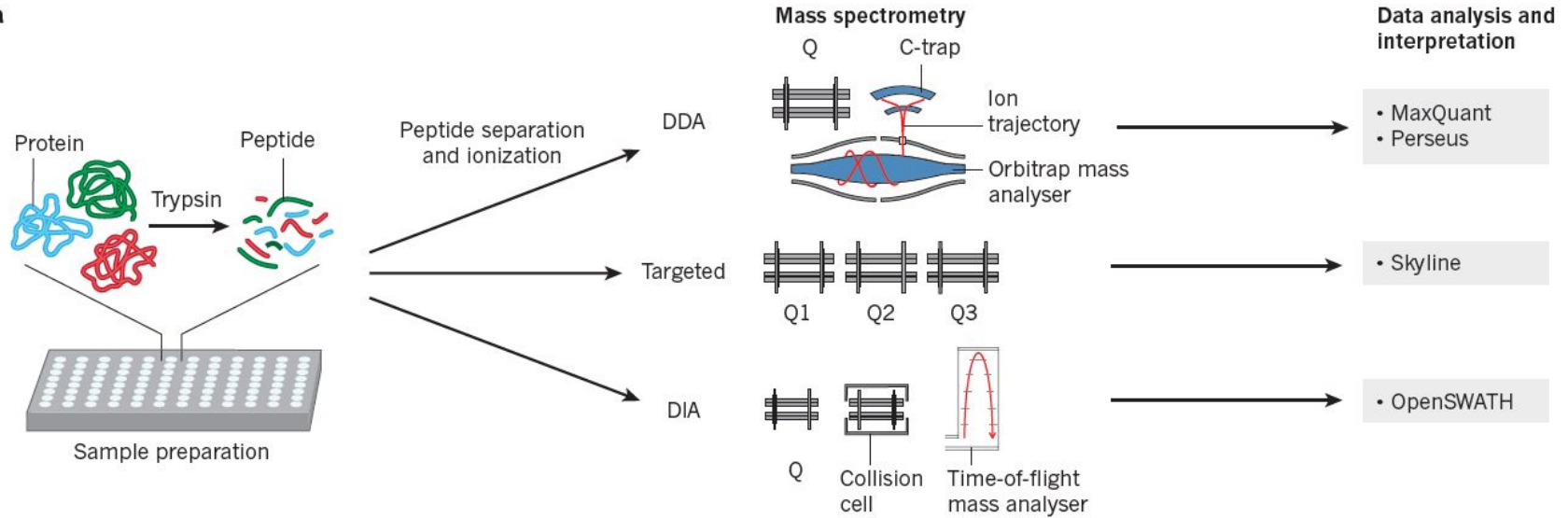
The *Tabula Sapiens* and other community projects offer reference expression data in healthy donors



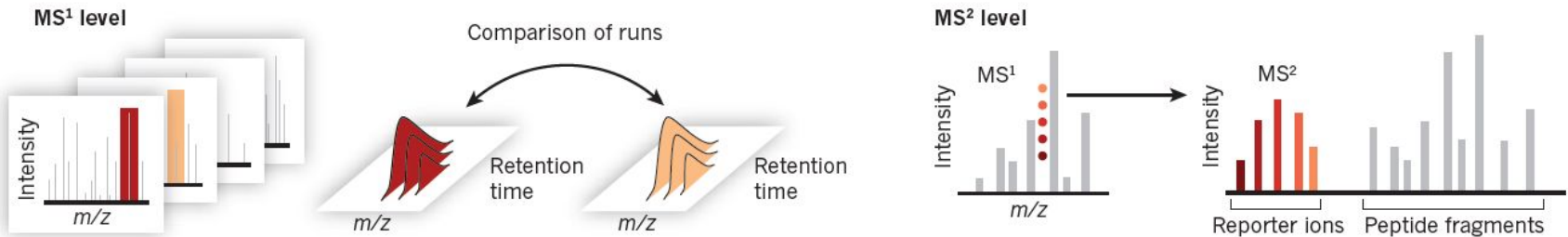
Left: the *Tabula Sapiens*. Right: Myeloid (MΦ=macrophages, Mo/monocytes, LAM=lipid-associated macrophages, DC=dendritic cells) gene expression

Mass-spectrometry based proteomics

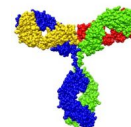
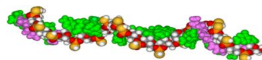
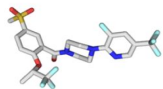
a



b Peptide quantification

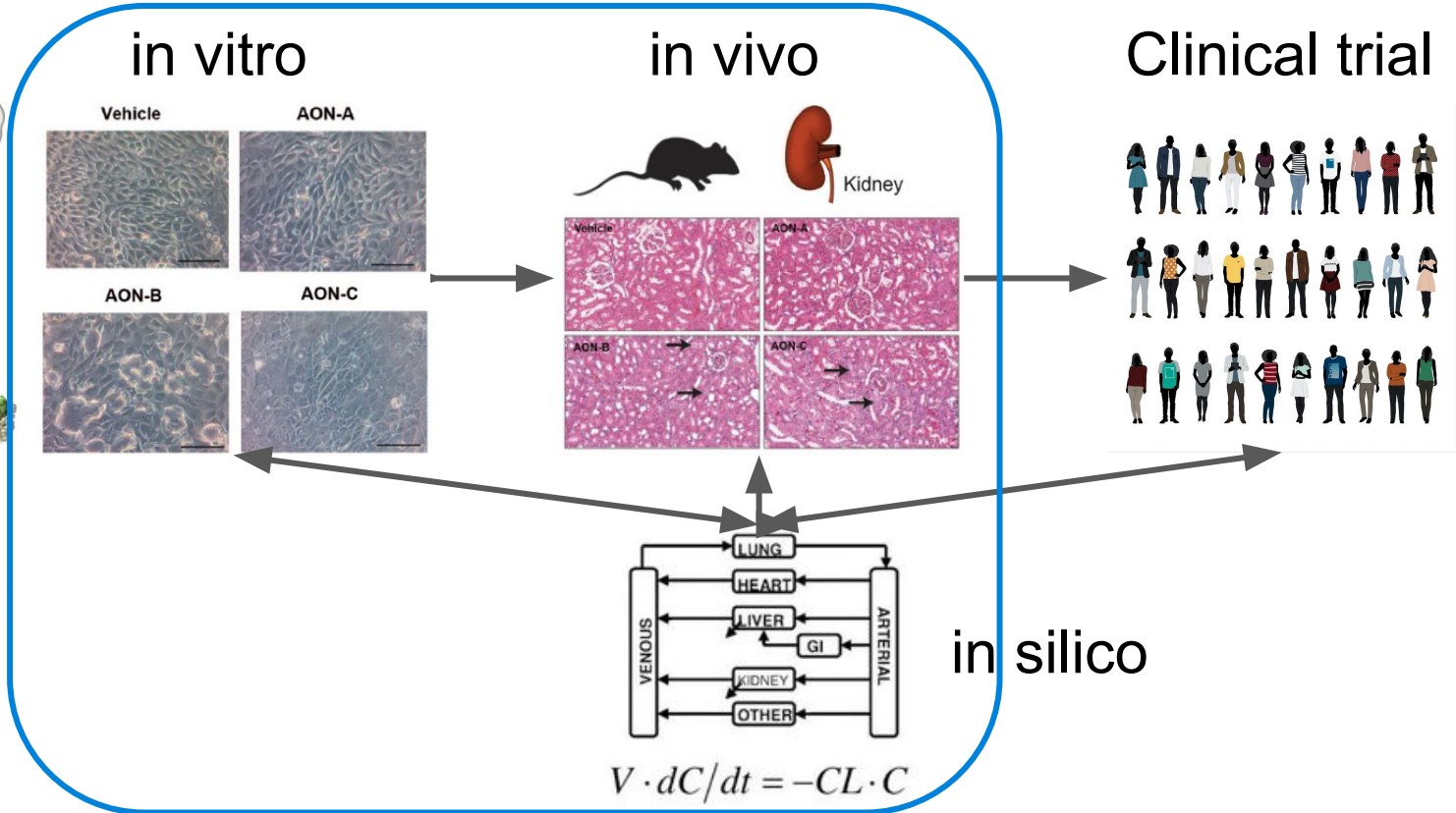


Comparing modalities with regard to safety assessment



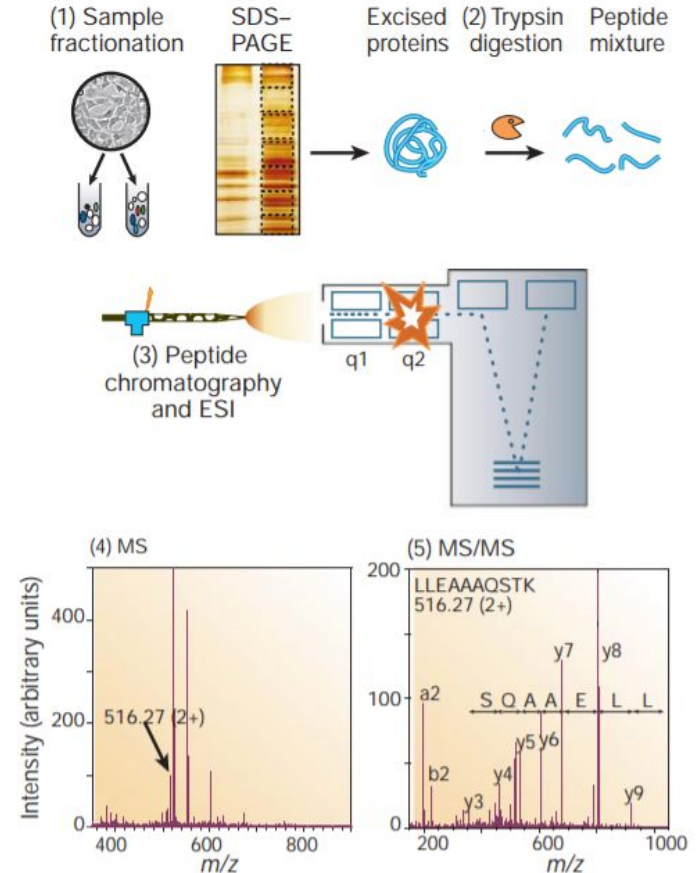
	Small molecules	Single Stranded Oligos	Biologics
Molecular weight	<1000 D	5000-7000 D	> 30000 D
Manufacture	Chemical synthesis	Chemical synthesis	Biologically-derived
Structure	Single entity, high purity	Single entity with 10-15% product-related impurities	Complex, heterogeneous
Chemical-driven toxicity	Yes	Yes	No
Metabolism	Species-specific	Species-independent catabolism by proteolytic degradation	Species-independent catabolism by proteolytic degradation
PK	Generally short $t_{1/2}$	Long (tissue) $t_{1/2}$	Long $t_{1/2}$
Some general aspects	High throughput screening/early safety testing of up to 500 small molecules	Biodistribution with consistent patterns	Fewer, yet complex due to biology/immunology

Proteomics plays an important role in *in vitro* *in vivo* translation



Mass-spectrometry based proteomics

- **SDS-PAGE:** Sodium Dodecyl Sulphate-Polyacrylamide Gel Electrophoresis
- **ESI:** Electrospray ionization
- **q1/q2:** selection/collision/separation cells
- **MS:** Mass spectrometry
- **MS/MS:** tandem mass spectrometry



Proteomics approaches for drug discovery

Affinity purification

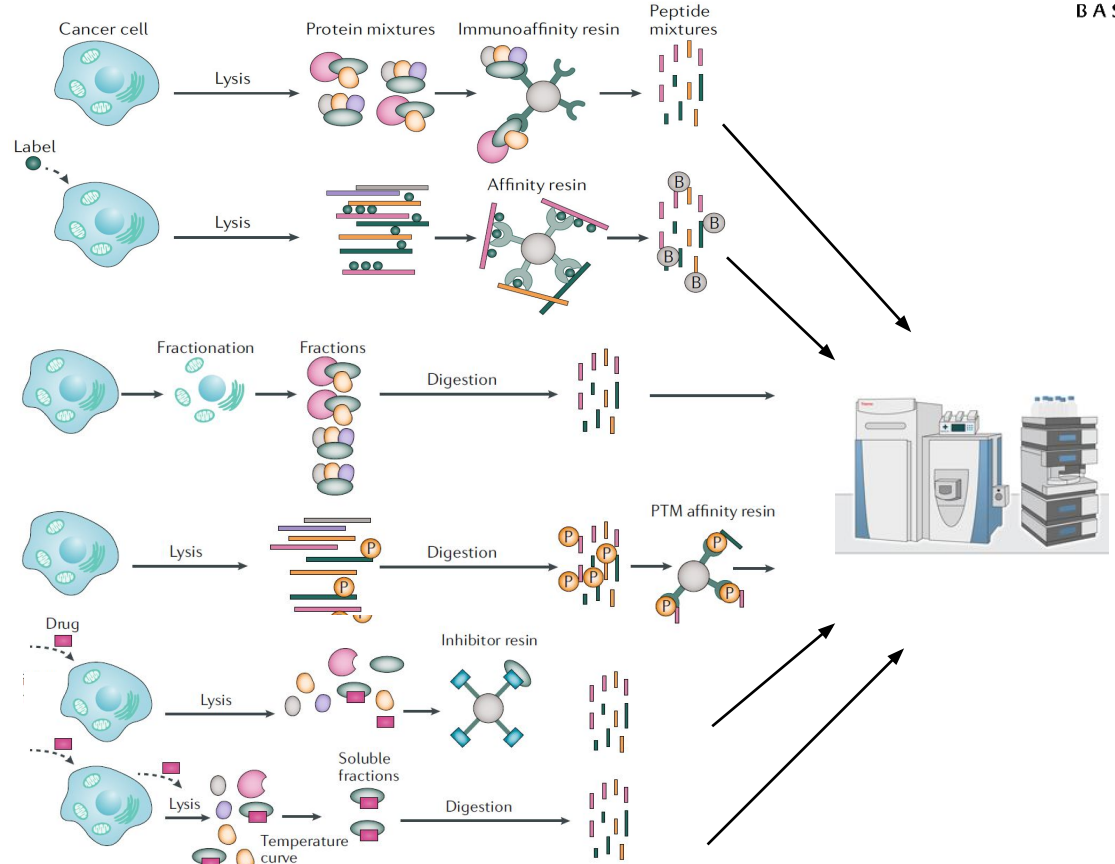
Proximity labelling

Organelle proteome profiling

**Post-translational modification
(PTM) profiling**

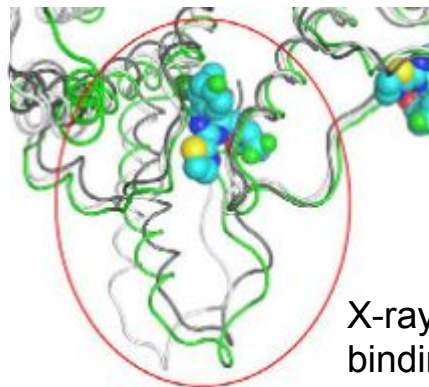
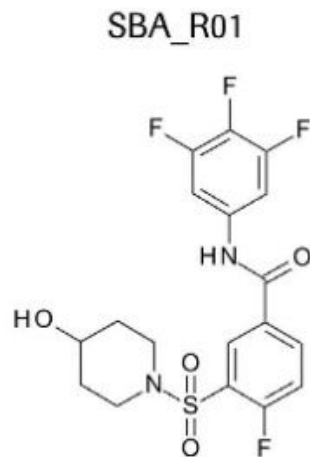
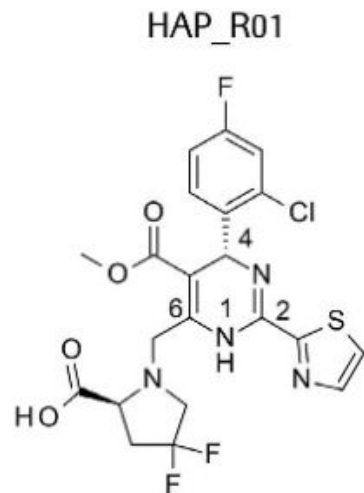
Chemoaffinity enrichment

Thermal proteome profiling

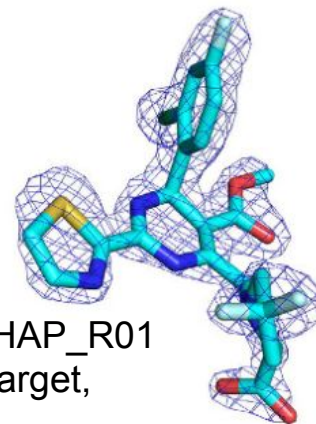


Case 1: Differentiate two compounds that inhibit Hepatitis B Virus with similar mode of action

a

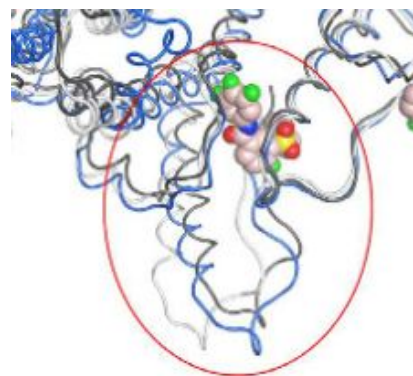


X-ray data of HAP_R01 binding to its target, HBV capsid

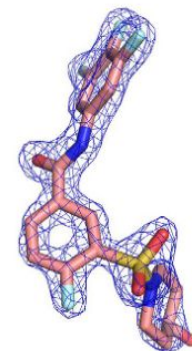


b

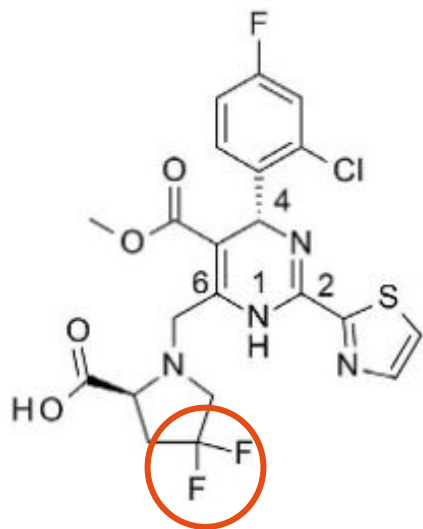
Compound	IC ₅₀ (μM)	HepG2.2.15 EC ₅₀ (μM)	CC ₅₀ (μM)
HAP_R01	0.39 ± 0.13	0.0064 ± 0.0006	34.8 ± 1.8
SBA_R01	1.90 ± 0.22	0.26 ± 0.02	8.05 ± 0.92



SBA_R01

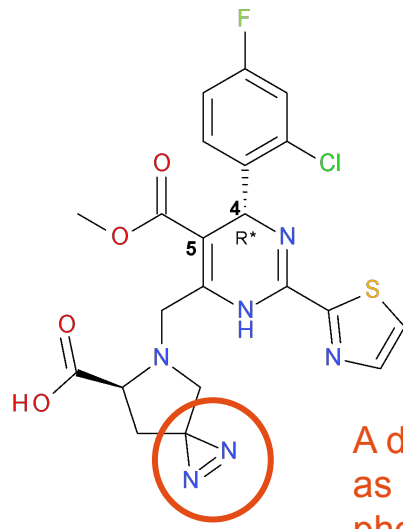


Chemical probes: drug-like molecules to probe its mode of action



Compound IC₅₀ (μM)

HAP_R01 0.39 ± 0.13

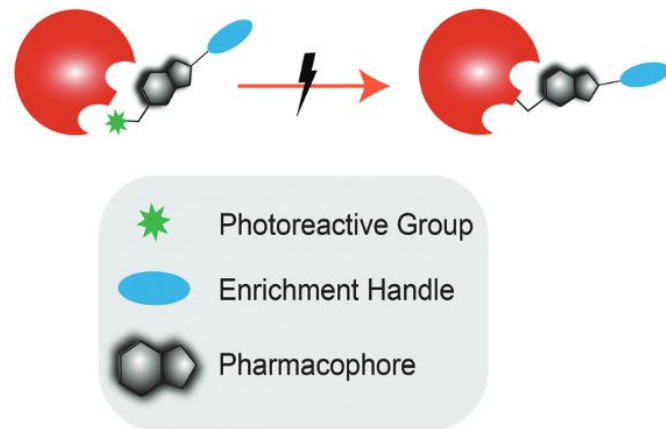


RO-A

EC₅₀: 0.040 μM

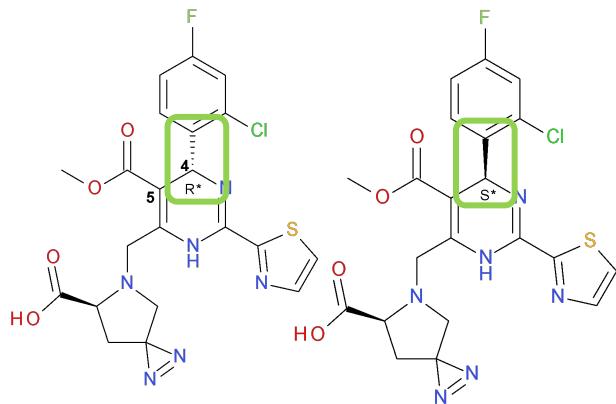
IC₅₀: 0.47 μM

A diazirine group
as the
photoreactive
group



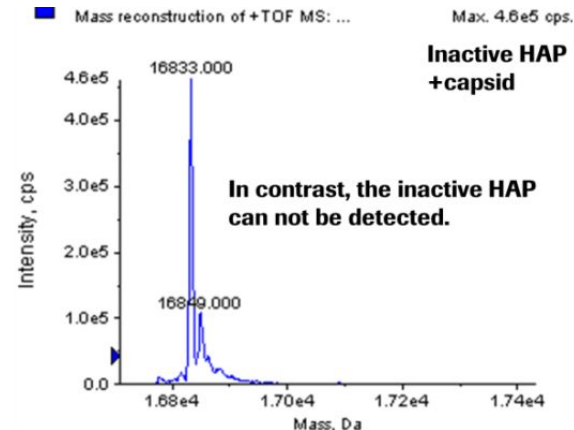
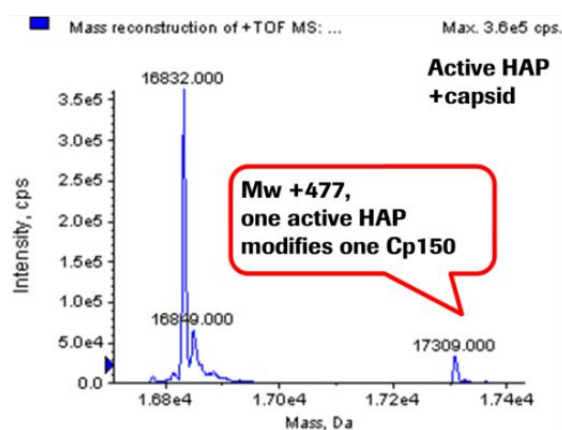
Case 1 solved: Proteomics confirmed target binding and mapped the small molecule binding pocket

+Cp150, UV, MS



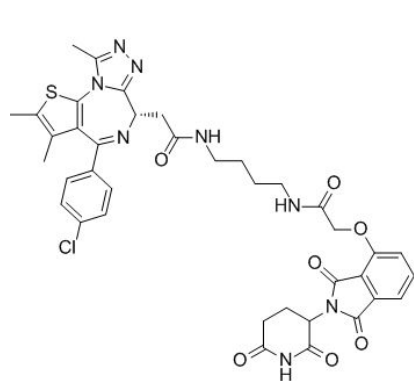
RO-A
EC₅₀: **0.040 μ M**
IC₅₀: **0.47 μ M**

RO-B
EC₅₀: **>1 μ M**
IC₅₀: **>100 μ M**

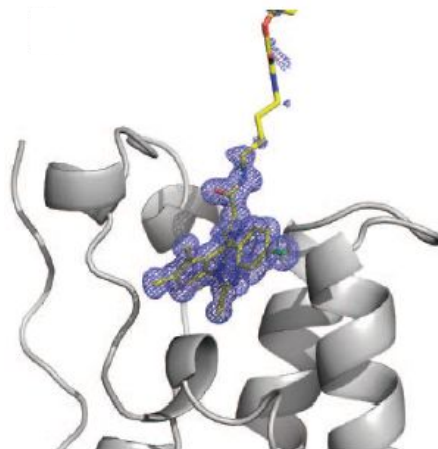
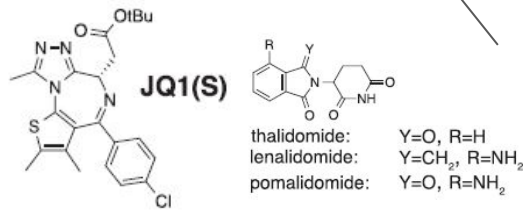


Proteolytic digestion/LC-MS/MS identified labelling site **Y118 (Y=Tyrosine)** of HBV capsid protein. More photoaffinity probes identified labelling sites at **R127 (R=Arginine)** and **Y38**.

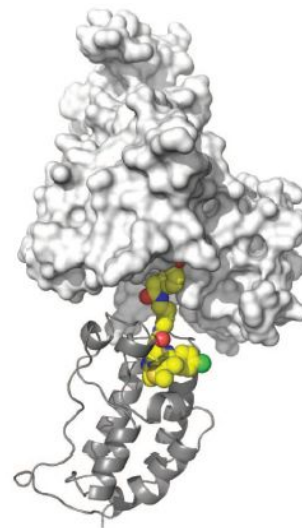
Case 2: Confirmation of selective degradation of protein target *in vivo*



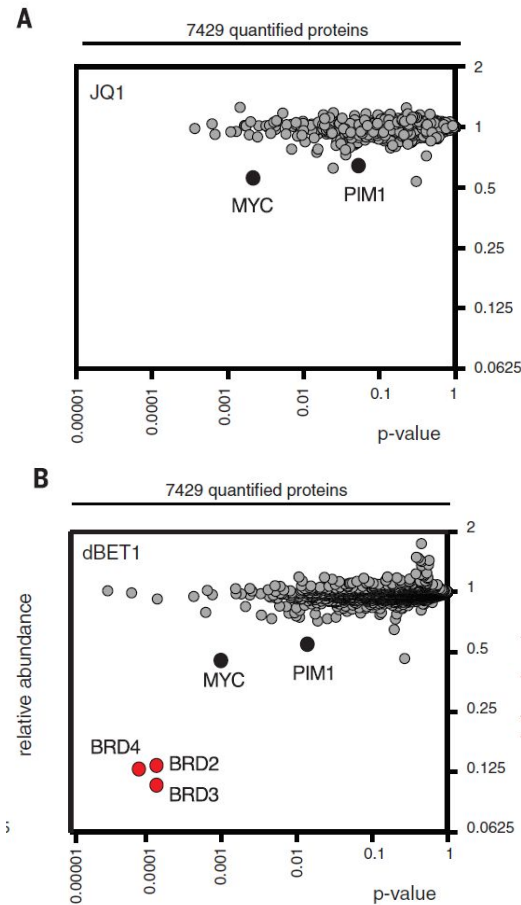
dBET1



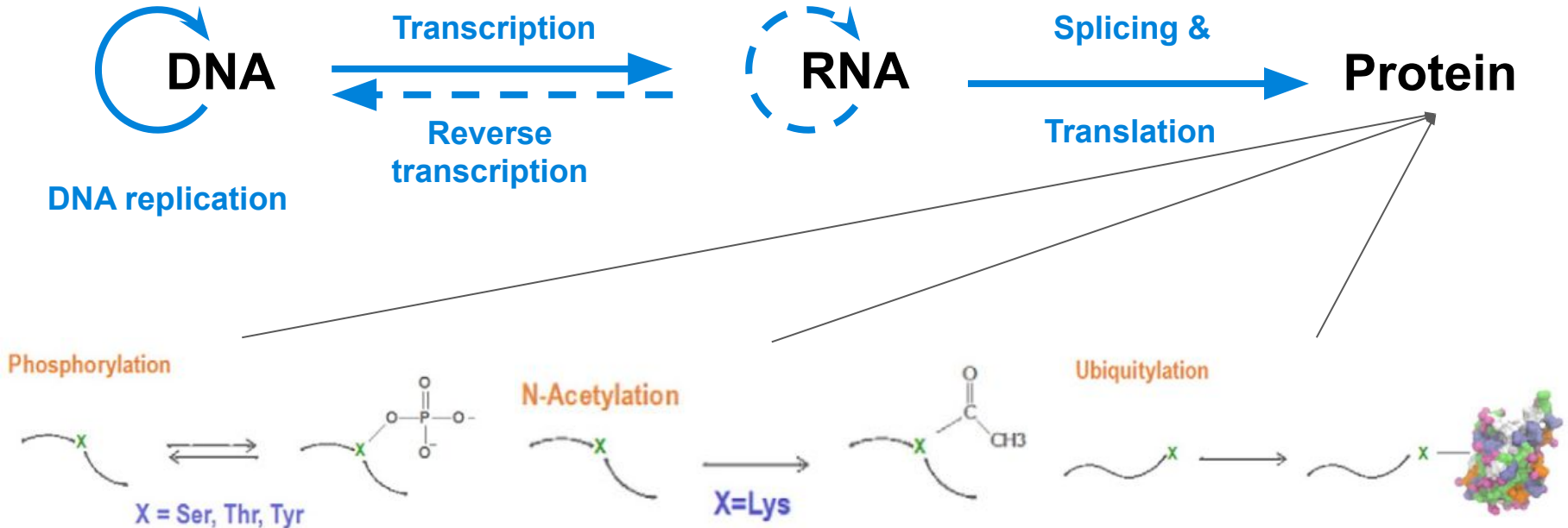
Crystal structure of dBET1 binding to its target BRD4



Docking of dBET1-BRD4 to DDB1-CRBN structure

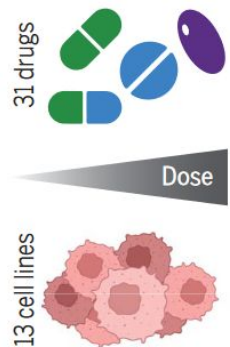


Protein post-translational modifications (PTMs) offer an additional layer of regulation

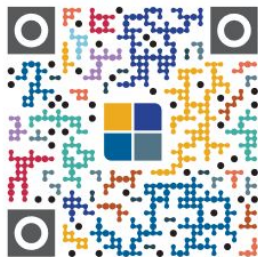


Case 3: Millions of PTM profiles induced by drugs in cancer cell lines

Perturbations

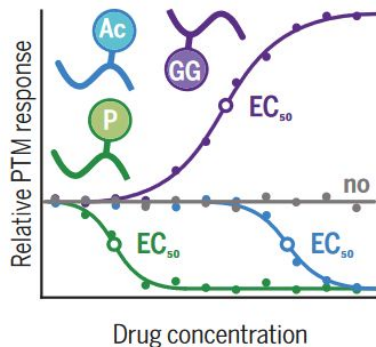


decryptM

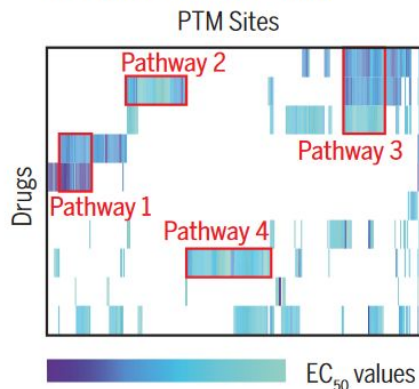


- 1.8 million assays
- online exploration

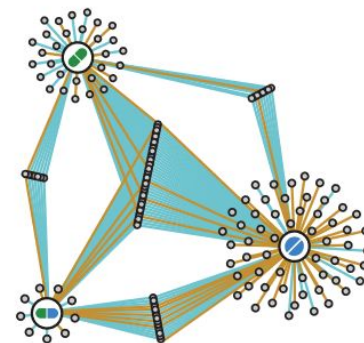
Pathway engagement



PTM functionalization

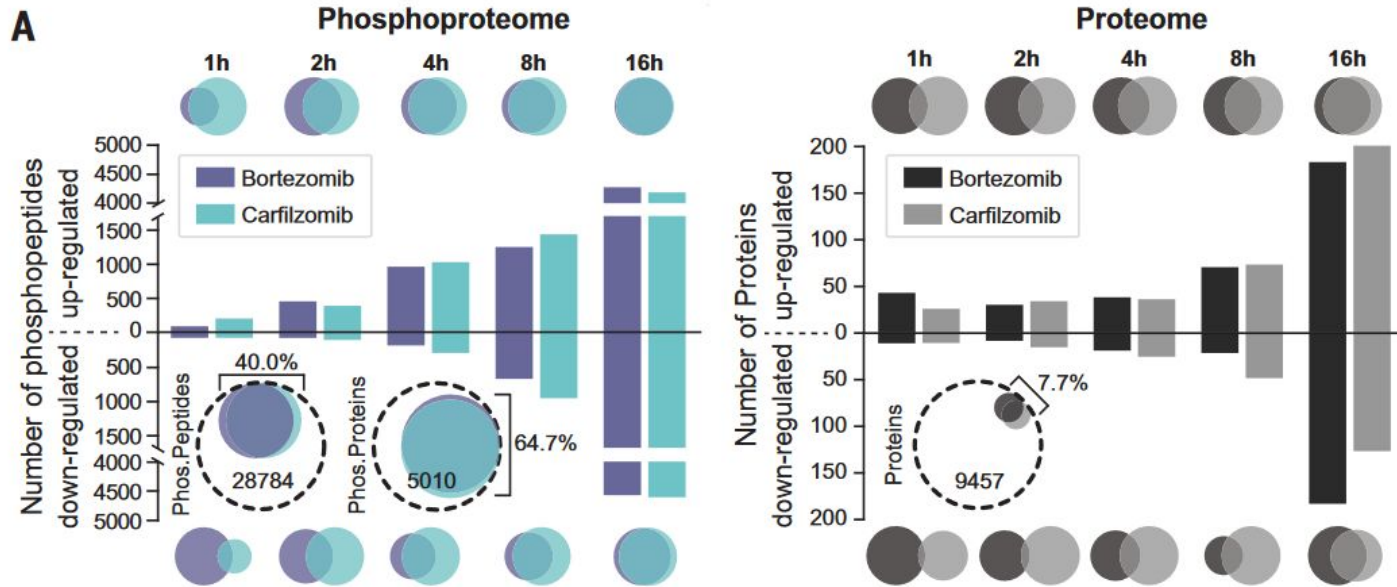


Mechanism of action



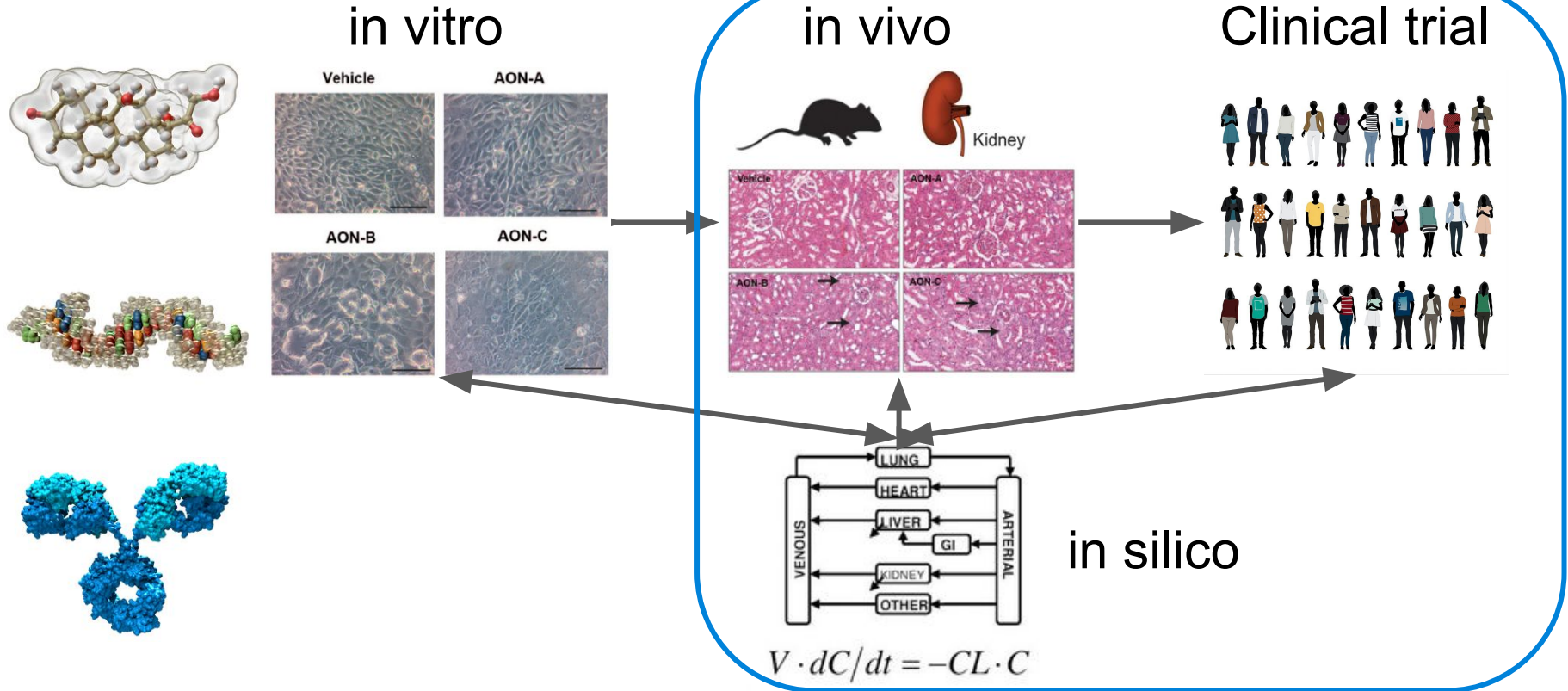
decryptM (Nature 2023): Following the dose-dependent treatment of cancer cells with drugs, quantitative mass spectrometry records dose-response of thousands of posttranslationally modified peptides. EC₅₀: half-maximal effective concentration; Ac, acetylation; GG, ubiquitinylation; P, phosphorylation.

PTM and proteomics characterize MoA of drugs

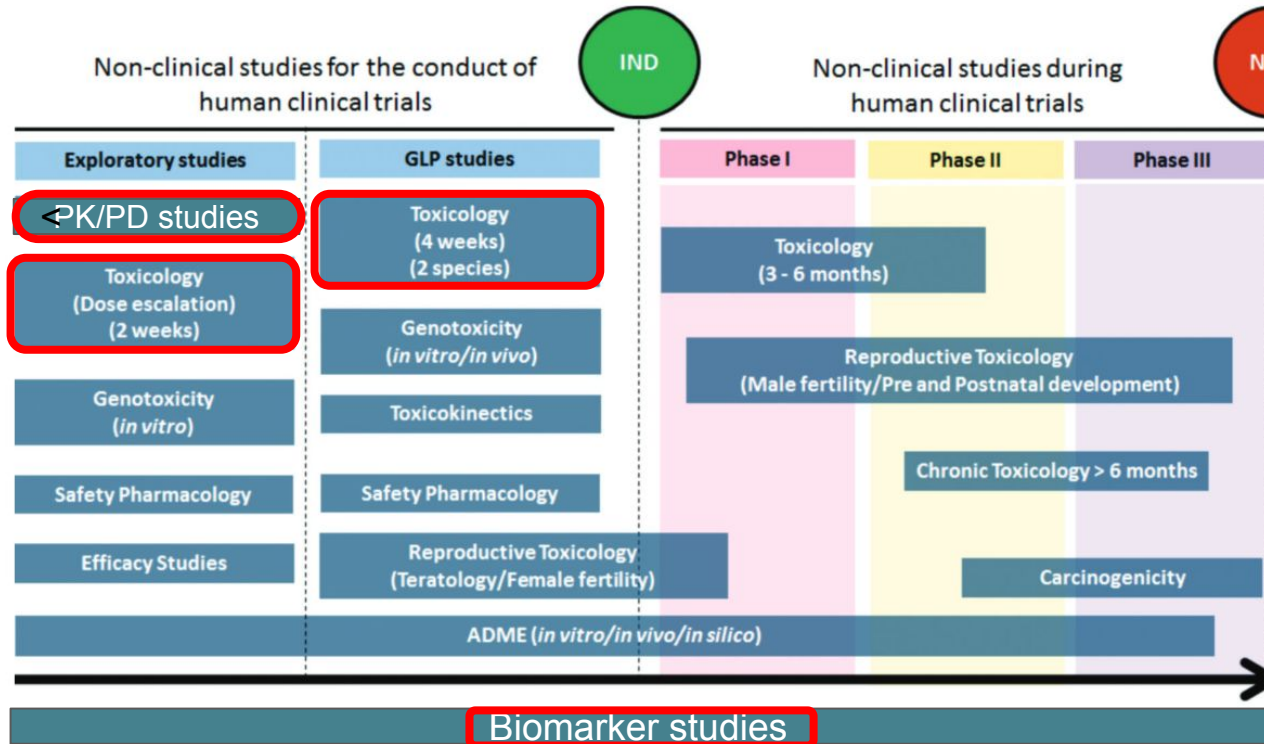


Bortezomib (BTZ) and carfilzomib (CFZ) both treat multiple myeloma by inhibiting the proteasome by reversible covalent (BTZ) or irreversible (CFZ) binding to the protease PSMB5. Time-series data show both the dynamics and the converging signaling.

Dose prediction based on pharmacology and toxicology before entry into human

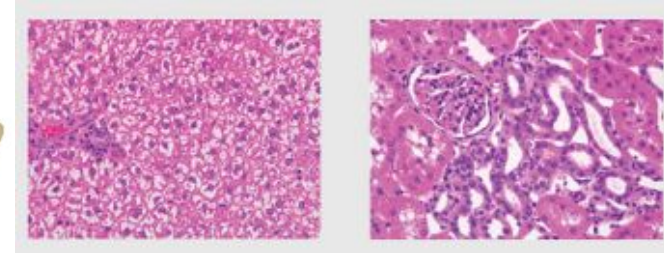
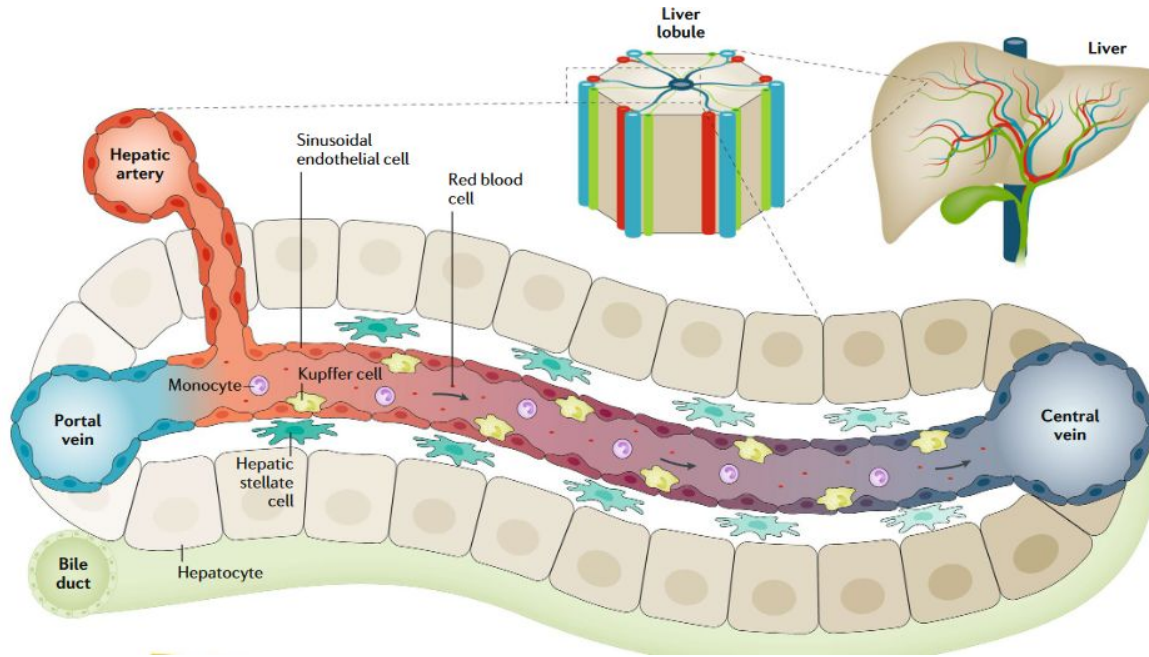


Current practices of non-clinical studies in drug development

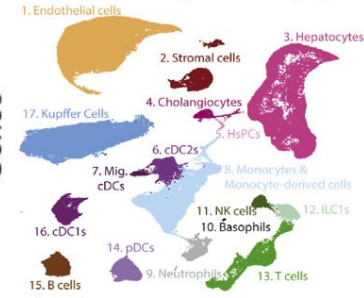
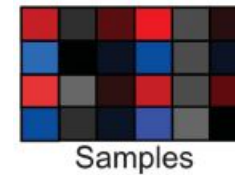


- IND: Investigational New Drug application
- NDA: New Drug Application
- GLP: Good Lab Practice
- Red boxes: Focus areas of this and coming lectures

Current practices of profiling and understanding toxicology: an example with liver



Histopathology

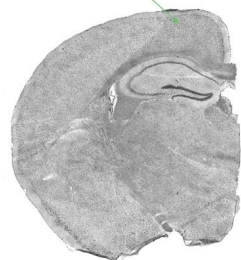


Omics

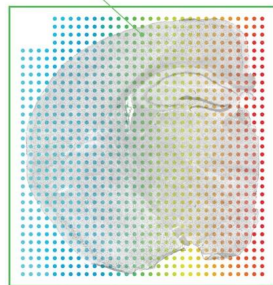
[Liver structure and anatomy \(YouTube Video\)](#)

Spatially resolved omics complement histopathology

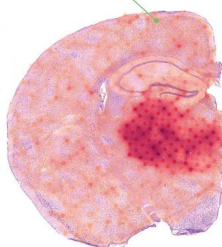
tissue section



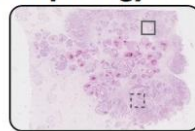
spatial mapping



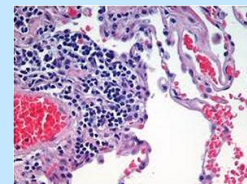
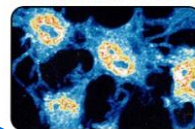
visualization



Morphology level



Molecular level



Spatial Omics

Histopathology

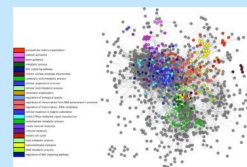
- Morphology

Molecular Biomarkers

- Genetics, Transcriptomics
- Proteomics

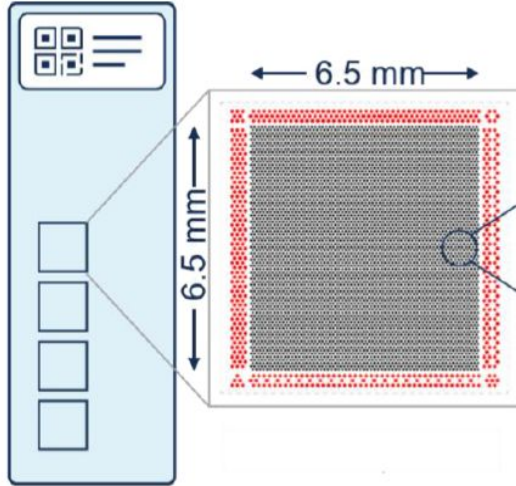
Data insights

- Data analysis & integration



An example: 10x VISIUM Technology

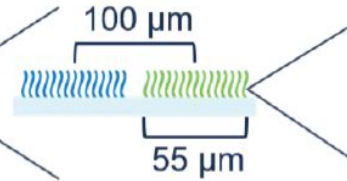
Visium Spatial Gene
Expression Slide



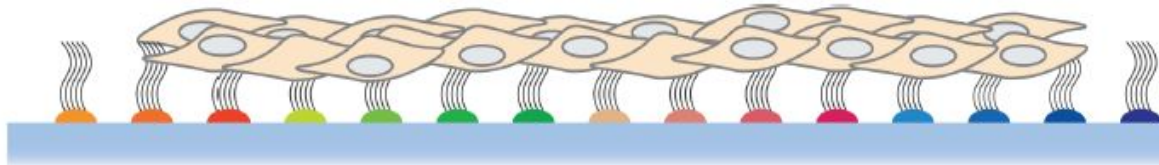
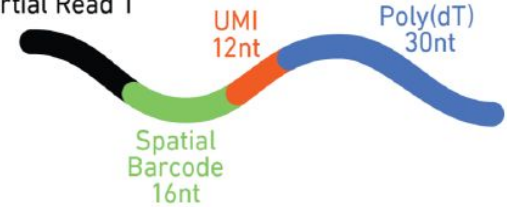
Capture Area with
~5000 Barcoded
Spots



Visium Gene
Expression Barcoded
Spots



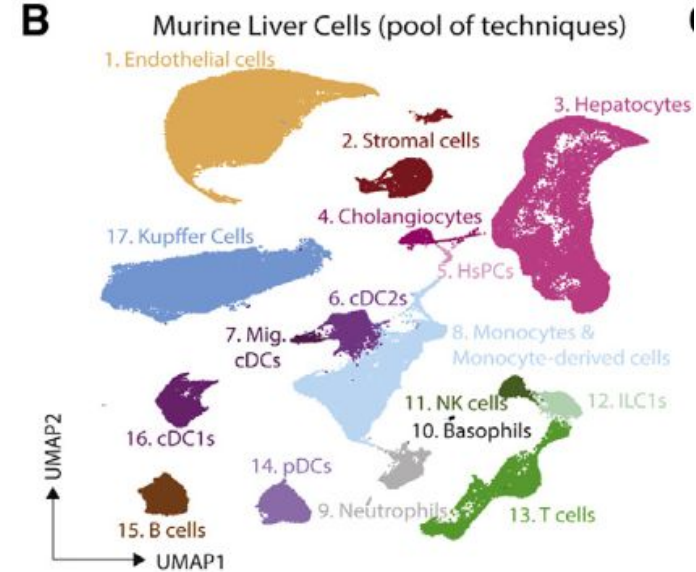
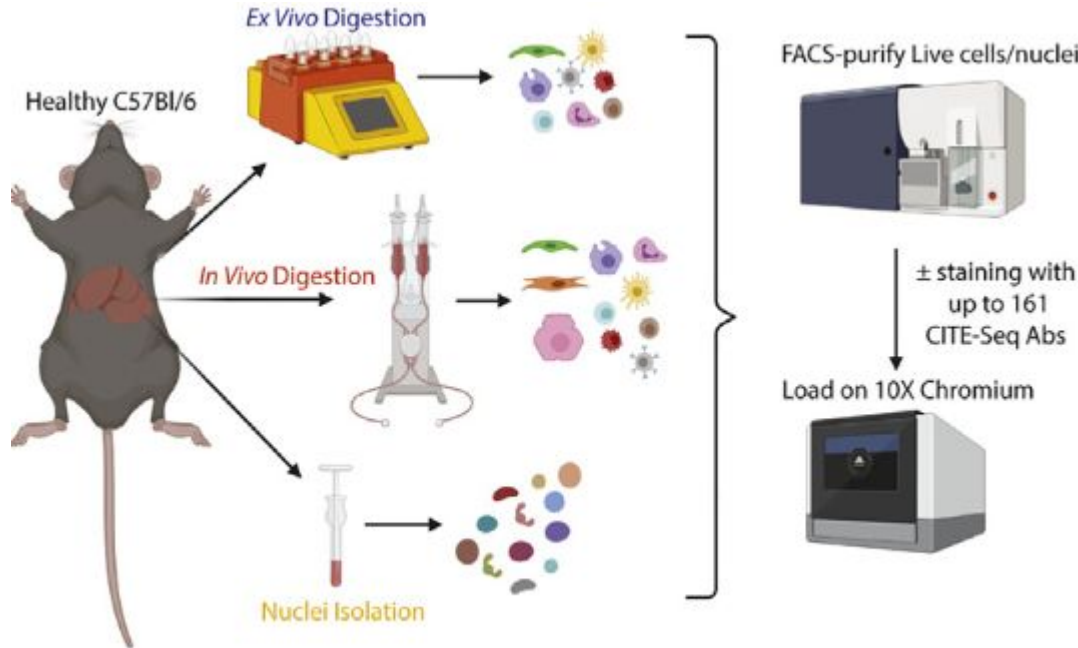
Partial Read 1



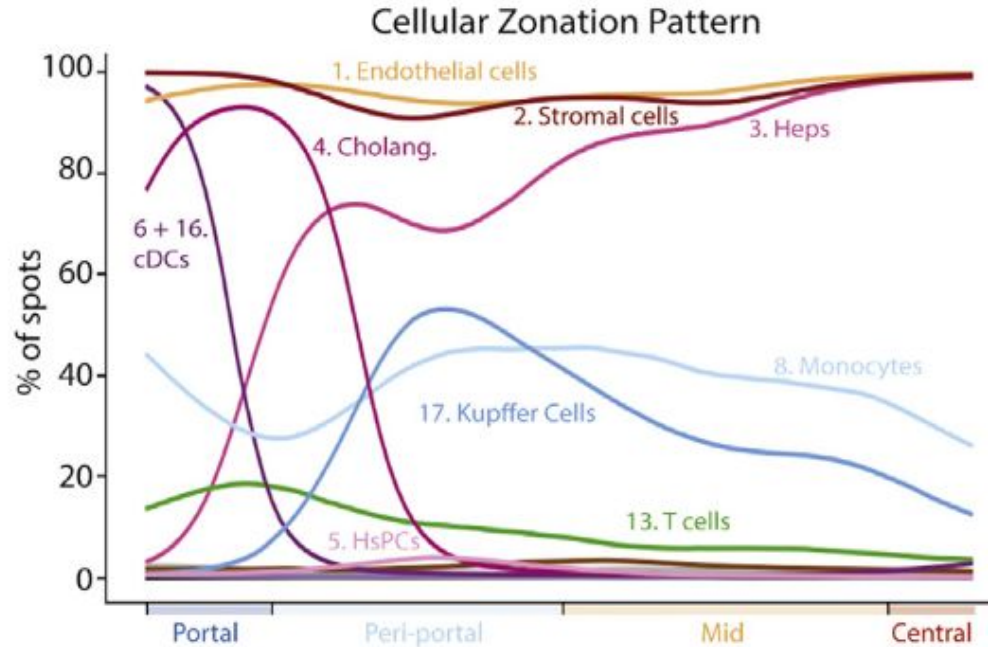
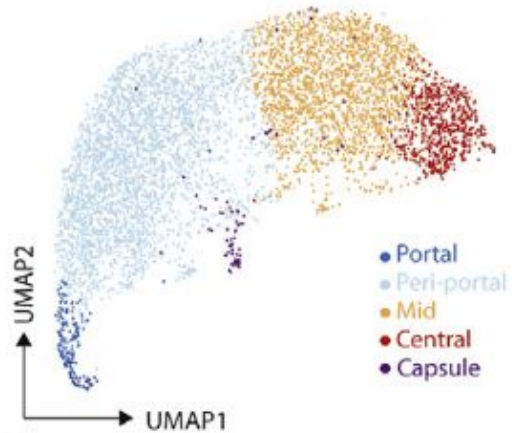
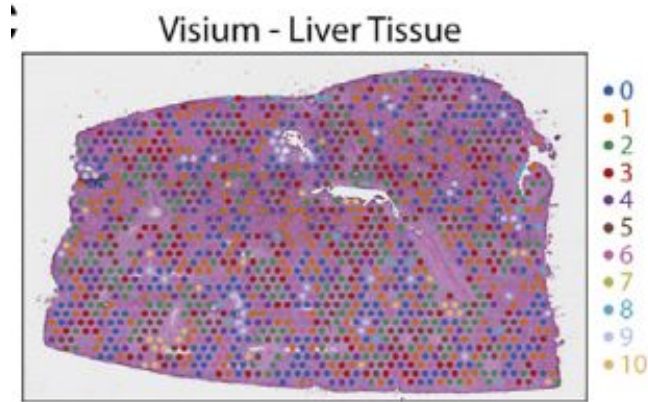
Tissue section

Visium capture area

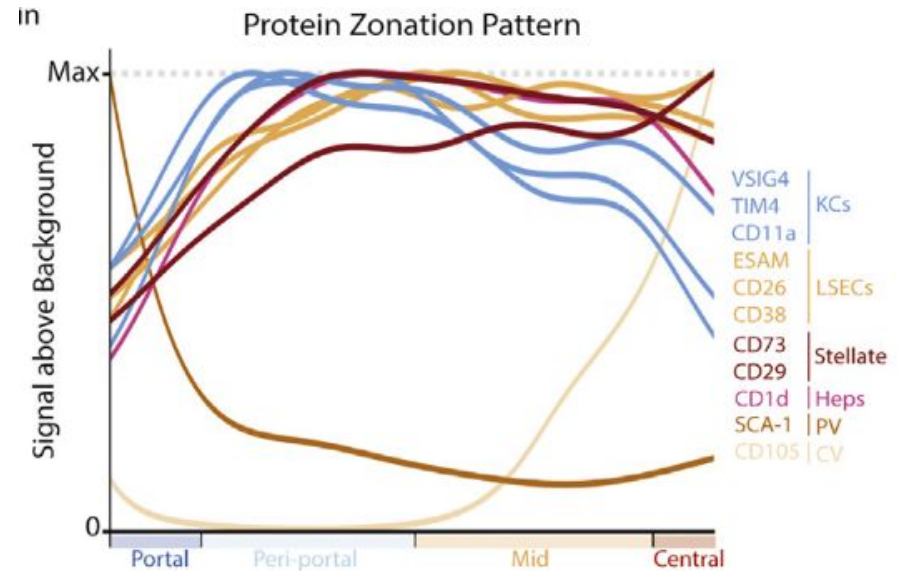
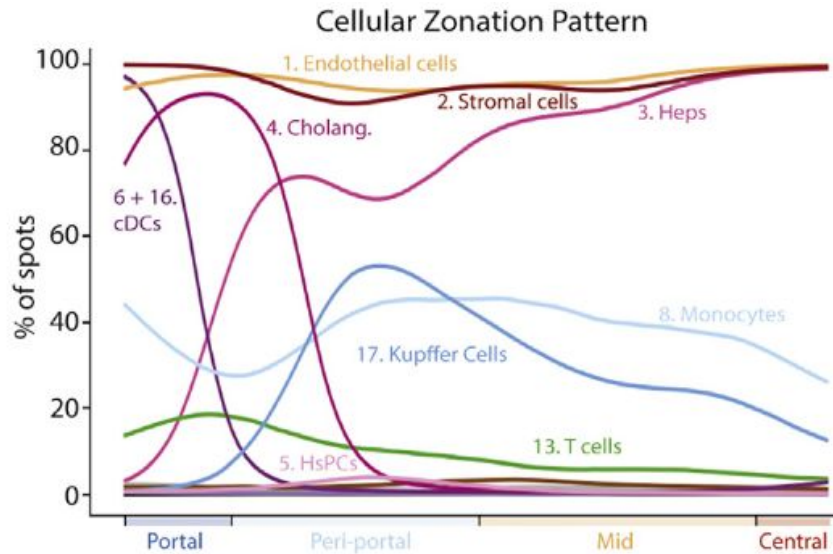
Spatial and single-cell expression of liver cells



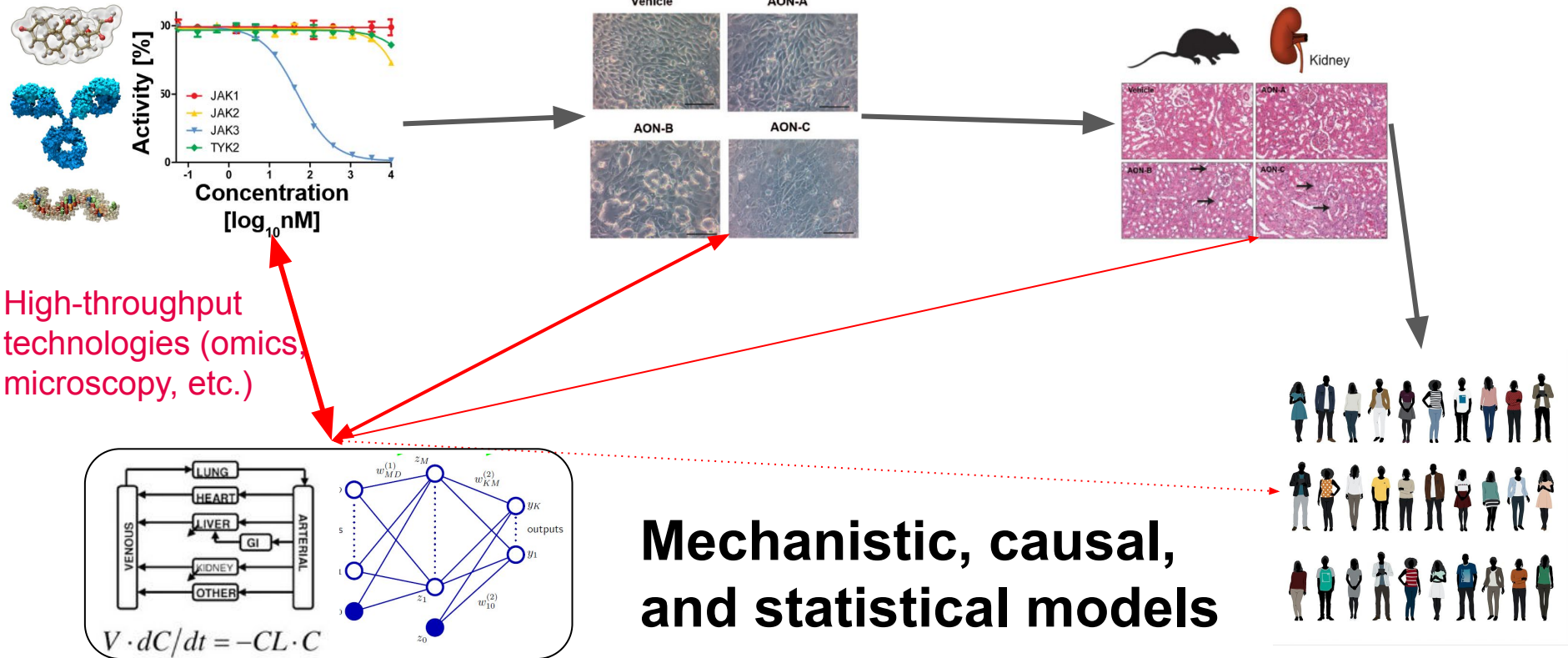
Spatial and single-cell expression of liver cells



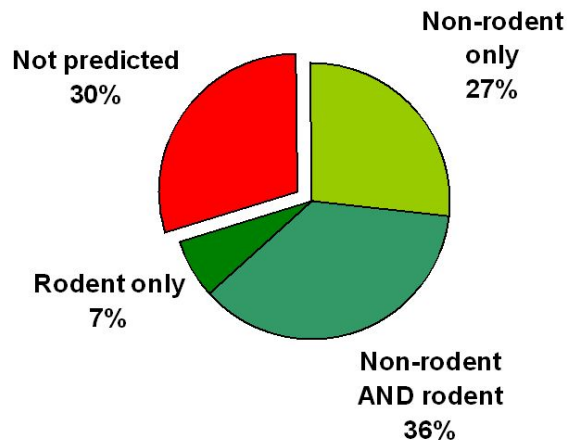
Spatial mRNA and protein expression data empowers digital pathology and biological understanding



Computational methods empower efficacy and toxicity assessment



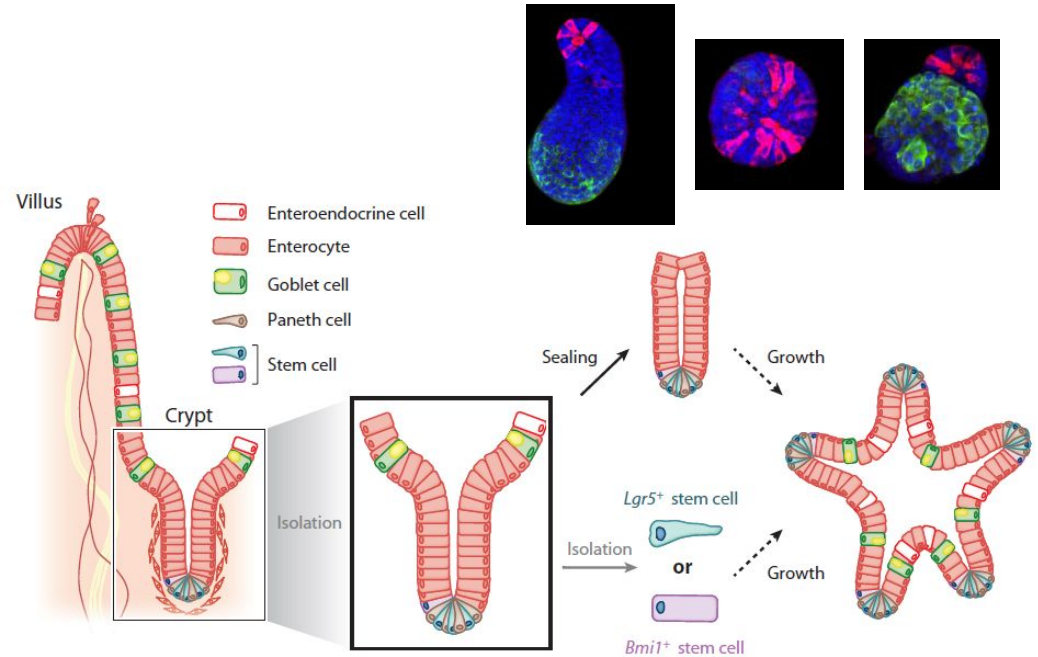
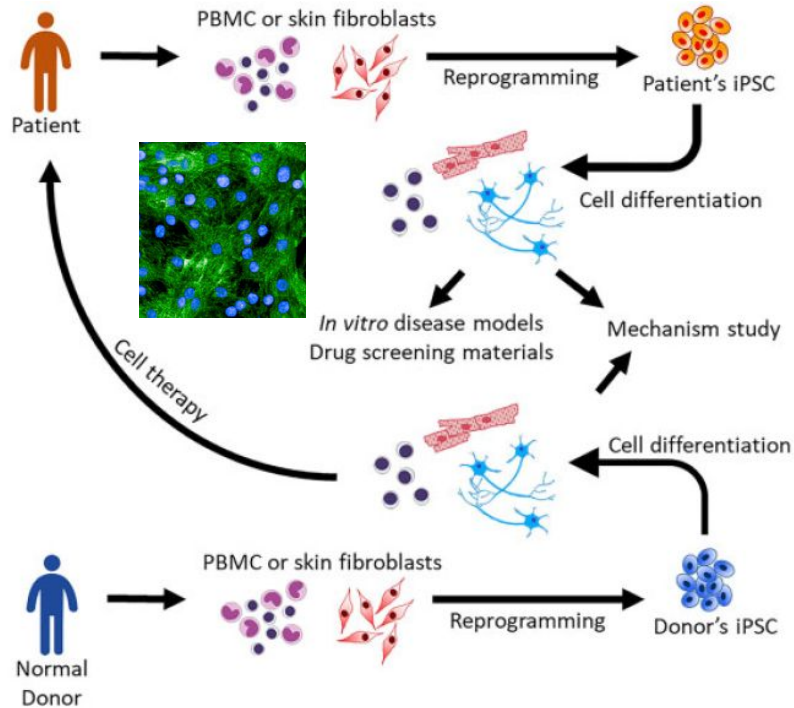
How predictive is animal safety testing for humans? It depends on modality and therapeutic classes.



[Regul Toxicol Pharmacol. 2000;32:56-67](#)

Target organ of ADRs	Small molecule drugs		Large molecule drugs	
	% of ADRs	% of correlation	% of ADRs	% of correlation
Gastrointestinal	21	80	14	19
Neurological	20	34	11	4
Hepatobiliary	11	73	8	21
Hematological	8	75	8	80
Cutaneous	5	56	9	22
Systemic	5	45	8	20
Cardiovascular	4	61	6	0
Ocular	5	64	5	83
Musculoskeletal	3	16	5	0
Metabolic	4	50	3	43
Faucal/oral	4	41	3	38
Urinary	3	61	3	14
Respiratory	1	45	5	32
Infection	0.4	100	6	68
Nasal	1	27	2	33
Application site reaction	1	100	3	81
Others	3	45	1	80

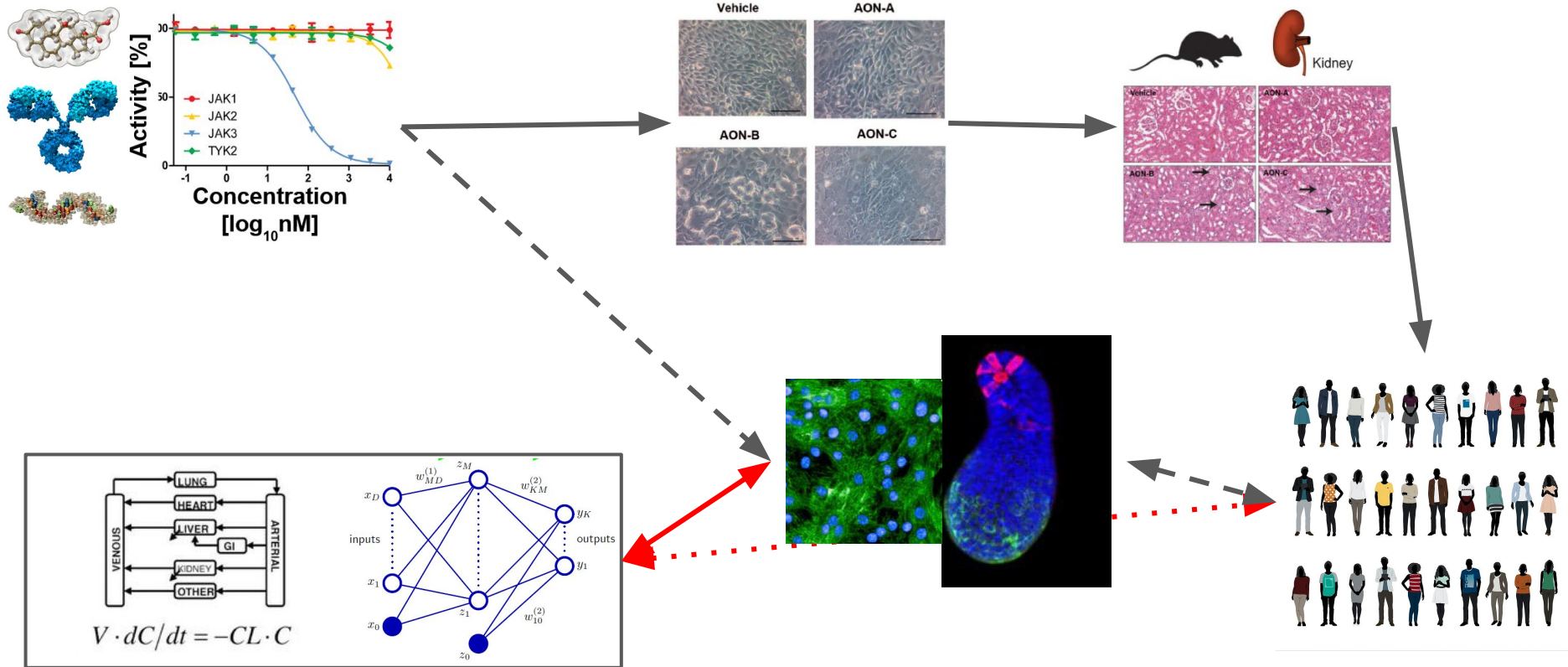
Stem cells and organoids empower efficacy and toxicity assessment



Small-intestinal organoids

Induced pluripotent stem-cells

Computational methods and novel biological models empower efficacy and toxicity assessment



Single-cell biology is important in drug discovery

Disease understanding:
disease-specific cell types
and states



Target identification:
expression pattern in
health and disease across
cell types



Biomarker and patient stratification: which
genes should we measure
in which cell type(s)?



MoA and safety modelling: perturbation
effect at single-cell level

*Does it make sense to develop
covalent drugs for any target?
If not, which targets should be
covalently targeted?*

Proteomics enables the elucidation of protein relations in the protein communities

